YANG–MILLS THEORY AND THE MASS GAP

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Abstract. We prove that four-dimensional lattice Yang–Mills theory with any compact simple gauge group has a mass gap with an explicit lower bound at strong coupling, and that a spectral gap ϖ for resampling cubes of physical size ℓ, uniform in the lattice spacing, gives every nontrivial continuum limit the mass gap $16\varpi/(625e^2\ell)$. We prove this cube gap at strong coupling, for SU(2) up to $\beta_W = \frac{7}{20}$, and reduce it to a decay of correlations below $\frac{1}{2}$ across one slab of width $\ell/2$, which we prove in two dimensions and which remains open in four.

1. Introduction

The Yang–Mills existence and mass gap problem asks for a quantum Yang–Mills theory on $\mathbb{R}^4$ with a compact simple gauge group that satisfies the axioms of constructive quantum field theory, and for a number $\Delta > 0$ such that the spectrum of the Hamiltonian lies in $\{0\} \cup [\Delta, \infty)$ [31]. In the Euclidean formulation the theory is a probability measure on gauge fields whose Schwinger functions satisfy the Osterwalder–Schrader axioms [41, 42]; the mass gap is then exponential decay of the truncated two-point functions of gauge-invariant observables at the rate Δ. The natural regularisation is Wilson's lattice gauge theory [54]: an element Q_e of $G \subset \mathrm{U}(N)$ sits on every link e of the lattice $a\mathbb{Z}^4$, and a configuration has the weight $\exp(N\beta \sum_p \operatorname{Re}\operatorname{Tr} Q_p)$, where Q_p is the ordered product around the plaquette p.[1] Asymptotic freedom forces the continuum limit $a \to 0$ to be taken with $\beta \to \infty$, that is, at weak coupling.

At small β the lattice theory has a unique state, exponential clustering and a transfer-matrix gap, by convergent expansions [43, 45]; Bałaban's renormalisation group [2, 3, 4] gives ultraviolet stability in finite volume at weak coupling (Dimock [18] presents the method for the scalar ϕ^4 model in three dimensions), and Magnen, Rivasseau and Sénéor constructed the theory with an infrared cutoff [39]. No existing result gives a mass gap in physical units. This paper proves a lattice mass gap with explicit constants for every compact simple group, reduces the continuum gap to one estimate on one physical scale, proves that estimate at strong coupling, reduces it further to a decay of correlations across a slab of physical width $\ell/2$, and determines what the methods used here can and cannot reach. It also states precisely what is missing.

1.1. The lattice gap with explicit constants.

Let G be a compact connected Lie group with semisimple Lie algebra, identified with its image under a faithful unitary representation in $\mathrm{U}(N)$, and let $\kappa_G > 0$ be its least Ricci curvature for the bi-invariant metric $\langle X, Y\rangle = -\operatorname{Tr}(XY)$. For a simple group $\kappa_G = h^\vee/(2\ell_\rho)$, with $h^\vee$ the dual Coxeter number and ℓ_ρ the Dynkin index of the representation; for $\mathrm{SU}(N)$, $\kappa_G = N/2$.

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[1] This is the normalisation of [49, 6]. The conventional inverse coupling is $\beta_W = N^2\beta$, and for SU(2), identified with the unit quaternions, $\beta_W = 4\beta$.

Theorem A. *Suppose $N|\beta| < \frac{2}{29}\kappa_G$. For every gauge-invariant local observable the expectations in boxes with free boundary converge, along every exhausting sequence of boxes, to a state ω that is invariant under the lattice symmetries; truncated correlations of Lipschitz observables decay at every rate below*

$$m_G(\beta) = \operatorname{arccosh}\Big(1 + \frac{1}{12}\Big(\frac{\kappa_G}{N|\beta|} - \frac{29}{2}\Big)\Big) > 0,$$

and for $\beta \geq 0$ the state is reflection positive and its transfer matrix has spectral gap $m \geq m_G(\beta)$. For $G = \mathrm{SU}(2)$ the same holds for $|\beta_W| < \frac{1}{3}$ with $m \geq -\log(3\beta_W)$.

Two ingredients carry the proof. The first is a gauge: a field of straight lines on $\mathbb{Z}^4$ that contains exactly one link of every plaquette (Sec. 3). Setting these links to the identity loses no gauge-invariant information, and every remaining link then interacts with exactly twelve others instead of eighteen. The second is a Hessian bound in this gauge, $|\operatorname{Hess} S| \leq \frac{29}{2}$ for the action $S = \sum_p \operatorname{Re}\operatorname{Tr} Q_p$, proved by a divergence identity and a positivity statement for one plaquette (Sec. 5); with the Ricci curvature of G it gives a Bakry–Émery bound [9, 10] with an explicit decay rate. Without the gauge the sharp Hessian bound is 16 [6], so the gauge enlarges the range of every group by the factor $32/29$ and supplies the rate. The SU(2) statement combines the gauge with the sharp Kantorovich modulus $\frac{1}{4}$ of the von Mises–Fisher family [6] in Dobrushin's criterion [19, 20].

1.2. **A continuum mass gap from one cube.** The lattice gap of Theorem A is measured in lattice units and vanishes in physical units as $a \to 0$. The second result isolates a single estimate, posed on one scale, that controls the gap in physical units. On a periodic lattice let $\mathcal{P}_x$ resample the links of the cube of side n at x from their conditional law under the Wilson measure μ. The condition $\mathrm{BP}(n, \varpi)$ asks that

$$\varpi \operatorname{Var}_\mu(f) \leq \frac{1}{n^4} \sum_x \mu((f - \mathcal{P}_x f)^2)$$

for every gauge-invariant f: the dynamics that resamples cubes, each link at rate one, has spectral gap at least ϖ on gauge-invariant functions. The condition involves no gauge fixing and no supremum over boundary conditions.

Theorem B. *Let $a_k \to 0$ and $\beta_k \geq 0$ be couplings, and suppose that there are $\varpi > 0$, $\ell > 0$ and integers $n_k \geq 2$ with $n_k a_k \leq \ell$ such that $\mathrm{BP}(n_k, \varpi)$ holds for the Wilson measure at β_k on all large tori. Then every continuum limit of the corresponding lattice theories has Hamiltonian H with $\sigma(H) \subset \{0\} \cup [\Delta, \infty)$,*

$$\Delta = \frac{16\varpi}{625e^2\ell},$$

and 0 is a simple eigenvalue; if the limit is nontrivial, H has a mass gap of at least Δ.

The proof (Sec. 6) turns the spectral gap into decay of covariances through a finite-speed bound for the cube dynamics, at a rate of order $1/n$ in lattice units, hence of order $1/\ell$ in physical units, and passes this decay through the Osterwalder–Schrader reconstruction.

1.3. **Where the cube condition holds.**

Theorem C. (i) *At $\beta = 0$, $\mathrm{BP}(n, 1)$ holds for every n. For $21N|\beta| < \kappa_G$ and n large, $\mathrm{BP}(n, 1 - \alpha_n)$ holds with $\alpha_n = N|\beta|(18n - 3)/((\kappa_G - 21N|\beta|)n^2) \to 0$; for $\mathrm{SU}(2)$ the range is $|\beta_W| < \frac{5}{18}$, and for $\mathrm{SU}(3)$ it contains $0 \leq \beta_W \leq 0.45$.*

(ii) *For the free scalar field of lattice mass μ in d dimensions the best constant is $\varpi = \frac{\mu^2}{n^d}\langle \mathbf{1}, (-\Delta_D + \mu^2)^{-1}\mathbf{1}\rangle$, with $-\Delta_D$ the Dirichlet Laplacian of the cube, and with $s = n\mu$ (the mass times the physical size of the cube)*

$$\Big(\frac{5}{6}\Big)^d \frac{s^2}{s^2 + 10d} \leq \varpi \leq \frac{s^2}{s^2 + 16d/9}.$$

(iii) *For* $G = \mathrm{SU}(2)$ *and* $|\beta_W| \le \frac{7}{20}$, $\mathrm{BP}(n, \varpi)$ *holds for every* $n \ge 2$ *with* $\varpi = \frac{106}{1933} \cdot 517^{-4}$, *independent of* n *and of the torus, and* $\mathrm{BP}(n, 1 - 259/n)$ *holds for* $n \ge 260$. *For* $0 \le \beta_W \le \frac{9}{25}$ *every torus limit is reflection positive and its transfer matrix has spectral gap at least* $7/(625 \cdot 2957\, e^2)$.

Part (ii) is the model case of the continuum tuning: the constant is uniform in the lattice spacing at fixed $m\ell$, and it vanishes as $m \to 0$. The cube condition is therefore a quantitative statement of mass generation at the scale ℓ. Part (i) is proved in Sec. 7 by the gauge of Theorem A and a weighted Dobrushin criterion for blocks of links. No argument built on single-link updates in this gauge passes $\beta_W = \frac{1}{3}$ for SU(2) (Proposition 7.12). Part (iii) passes that value by resampling *stars*, the eight links at a vertex of one parity (Sec. 8): the perturbation of one plaquette at a vertex enters the law of a star only through the relative variable of two of its links, and no plaquette is internal to a star.

1.4. **Mass generation at one scale.** Statement (D) below asks for the cube condition along the continuum tuning. Section 10 reduces it to a hypothesis of a different kind, closer to physics. Split a box of the lattice, all of whose widths lie between $2d - 1$ and $md - 1$, into two outer parts of width d and a middle of width $d - 1$ in one coordinate, fix an arbitrary configuration outside the box, and let ρ be the maximal correlation between the gauge-invariant observables of the two outer parts under the conditional law. The hypothesis $(\mathrm{M}_{d,m}(c))$ asks that $\rho \le c$ for all such boxes, splittings and boundary conditions. Along the tuning $d \approx \ell/(2a)$, so this is a decay of correlations across a slab of physical width about $\ell/2$, uniform in the boundary condition.

Theorem D proves that $(\mathrm{M}_{d,m}(c))$ with $c < \frac{1}{2}$ and m large in terms of c gives $\mathrm{BP}(2d-1, (1-2c)^4/32)$, hence, along the tuning, statement (D) and the continuum gap $(1-2c)^4/(1250\, e^2\ell)$ in every nontrivial limit. The proof passes through a gap for the dynamics that resamples overlapping patches of a box, obtained from the decorrelation of the slabs by a Markov factorisation, and a finite-size criterion for gauge-invariant functions in the spirit of those of Knabe [34] and of Gosset and Mozgunov [26] for quantum spin systems. The hypothesis also gives exponential decay of gauge-invariant correlations under every boundary condition, at rate $\log(1/c)/d$ per lattice spacing (Proposition 10.10). It holds at strong coupling at fixed d and m, and in two dimensions along the weak-coupling tuning (Theorem G.8). In four dimensions it is not proved.

1.5. **The limits of the methods.** Appendix G proves that several natural routes to (D) cannot work, and locates the threshold of Theorem D.

(a) *The threshold* $\frac{1}{2}$ *is sharp.* For a massless Gaussian chain the maximal correlation of the slab mechanism equals $\frac{1}{2}$ exactly, and the gap of the corresponding block dynamics on M blocks is at most $16/(3M)$; so no decorrelation constant $c \ge \frac{1}{2}$ can give a gap uniform in the size (Proposition G.4).

(b) *Two dimensions.* For SU(2) in two dimensions the middle slab of a box integrates out, after an axial gauge, to a convolution power of the heat kernel, and $(\mathrm{M}_{d,m}(c))$ holds with $c = 20\, e^{-3g^2(da)^2/4}$, which tends to zero as the physical size of the box grows (Theorem G.8). The same condition for strips one link wide fails along the tuning: a pinned strip is a Brownian bridge on SU(2), and the antipodal boundary condition breaks a global symmetry, with correlation close to one (Proposition G.14).

(c) *Curvature.* The Bakry–Émery constant of the Wilson measure without gauge fixing is at most $N/2 - 16N|\beta|$, so the Hessian bound 16 of [6] is sharp (Theorem G.16); in the gauge of Theorem A the constant $\frac{29}{2}$ cannot be replaced by anything below $\frac{38}{3}$, and no Bakry–Émery argument for the gauge-fixed measure reaches $|\beta_W| = \frac{3}{19}$ (Theorem G.17). Every smooth plaquette action, such as an effective action at a coarse scale, obeys a bound of the same kind (Proposition G.18).

(d) *Entropy, covariances and fixed blocks.* Block entropy factorisation implies the cube condition and so cannot be easier (Proposition G.19); the link covariances of every gauge-invariant measure are $\frac{1}{4}I$ and carry no information on the coupling (Proposition G.20); and a single-link spectral gap, or any gap for blocks of fixed lattice size, with constant C_k satisfying $C_k a_k \to 0$, forces every continuum limit to be trivial (Proposition G.21).

(e) *A gap in the limit is not enough.* If the lattice gaps m_k tend to zero, a choice of lattice spacings proportional to m_k produces a nontrivial continuum limit with a mass gap; this limit is kinematic and is not identified with Yang–Mills theory (Theorem G.24).

The content of (D) and (E) is that the gap and the nontriviality hold along the tuning of asymptotic freedom.

1.6. **What remains.** Theorems A and C(i), (iii) are strong-coupling theorems. The Clay problem needs the opposite end of the coupling range: the continuum limit lives at $\beta \to \infty$, which the ranges of these theorems do not reach. Two statements would complete the proof of a continuum mass gap, and neither is proved here:

(D) $\mathrm{BP}(n_k, \varpi)$ with $n_k a_k \le \ell$ along the asymptotic-freedom tuning $\beta_k \to \infty$, uniformly in k;

(E) convergence of the renormalised Schwinger functions to a nontrivial limit satisfying the Osterwalder–Schrader axioms.

Given (D) and (E), Theorem B gives the mass gap $\Delta \ge 16\varpi/(625e^2\ell)$, and by Theorem D statement (D) follows from $(\mathrm{M}_{d_k,m}(c))$ along the tuning with $c < \frac{1}{2}$. Section 11 explains why (D) is the heart of the matter: by Theorem C(ii) it cannot hold without a dynamically generated mass, it fails at fixed cube size at weak coupling, and it fails in the Coulomb phase of the abelian theory, so its proof has to use the non-abelian structure of G at every scale up to ℓ.

1.7. **Related work.** Strong-coupling clustering goes back to Osterwalder and Seiler [43] and Seiler [45]; probabilistic treatments include the survey of Chatterjee [13], the Langevin approach of Shen, Zhu and Zhu [49], whose Bakry–Émery argument we sharpen, the area law in enlarged regions of Cao, Nissim and Sheffield [14], and correlation decay for finite gauge groups at weak coupling by Adhikari and Cao [1]. In two dimensions the continuum measure is known [25, 21, 35]. The Kantorovich modulus of the SU(2) single-link law and the sharp Hessian bound without gauge fixing are proved in [6], and an earlier and much longer account of the one-cube reduction is [7].

One of us (I. J. S.) has formalised in Lean 4 with Mathlib [46], without unproved statements and with only the standard axioms, several ingredients of the strong-coupling analysis on periodic lattices: the Haar probability on $\mathrm{SU}(N)$, an infinite-volume Gibbs state obtained as a limit of torus states, site reflection positivity and the transfer matrix, the Holley–Stroock comparison, Dobrushin's covariance bounds, and the passage from decay of correlations to a transfer-matrix gap (one direction of Proposition 2.1). For the periodic state it proves a transfer-matrix gap for every N at $\beta_W < 5.9 \cdot 10^{-5}$, and explicit single-link heat-bath gaps on boxes of side fifteen for SU(2) at $\beta_W \le 0.166$ and for SU(3) at $\beta_W \le 0.38$ (the coupling there is $\beta_W/2$). It also proves, by a Poincaré inequality and a light-cone argument different from Lemma 6.2, that an ungauged one-cube influence condition gives a lattice gap of order $1/n$; this is a variant of the lattice step, Theorem 6.3 and Corollary 6.4, at a smaller rate. The continuum step, the gauge fixing and the curvature bounds are not part of that development, which shares with this paper the open hypothesis (D).

2. Lattice gauge theory and the transfer matrix

Write $e_0, \dots, e_3$ for the unit vectors of $\mathbb{Z}^4$, (x, μ) for the link from x to $x + e_\mu$, and $Q_{(x,\mu)}^{-1}$ for the variable of the reversed link. A box Λ is a product of four finite intervals, with its set $E(\Lambda)$ of links and its set $P(\Lambda)$ of plaquettes. For a plaquette at x in the plane $\mu < \nu$ with links $e_1 = (x, \mu)$, $e_2 = (x + e_\mu, \nu)$, $e_3 = (x + e_\nu, \mu)$, $e_4 = (x, \nu)$, put $Q_p = Q_{e_1} Q_{e_2} Q_{e_3}^* Q_{e_4}^*$.

With dQ the Haar probability measure of G, the Wilson measure of a box with free boundary is

$$W_\Lambda(dQ) = \frac{1}{Z_\Lambda} \exp\Big(N\beta \sum_{p \in P(\Lambda)} \operatorname{Re} \operatorname{Tr} Q_p\Big) \prod_{e \in E(\Lambda)} dQ_e,$$

and the Wilson measure μ_L of the torus $\Lambda_L = (\mathbb{Z}/L\mathbb{Z})^4$ is defined in the same way with all links and plaquettes of Λ_L. Both are invariant under the gauge transformations $Q_{(x,y)} \mapsto g_x Q_{(x,y)} g_y^{-1}$. A function of finitely many links is a local observable, and it is gauge invariant if it is invariant under these transformations. For $\beta \geq 0$ the measures are reflection positive for reflections in lattice hyperplanes through sites and in hyperplanes bisecting links, on boxes symmetric under the reflection and on tori of even side [43].

Let ω be a state on the gauge-invariant local observables of $\mathbb{Z}^4$ that is invariant under translations and reflection positive in this sense. The Osterwalder–Schrader construction [41, 42, 43] gives a Hilbert space $\mathcal{H}$ with a unit vector Ω, a map $F \mapsto \Psi_F$ from the observables at nonnegative times to $\mathcal{H}$ with dense range, and a self-adjoint transfer matrix $0 \leq \mathbb{T} \leq 1$ with $\mathbb{T}\Omega = \Omega$ and

$$\langle \Psi_F, \mathbb{T}^t \Psi_H \rangle = \omega\big(\vartheta(\bar{F})\, \tau_t H\big), \qquad t \in \mathbb{Z}_{\geq 0},$$

where ϑ is the time reflection and τ_t the time translation; Ψ_F is the class of F for the positive semidefinite form $\omega(\vartheta(\bar{F})H)$, and $\|\Psi_F\|^2 \leq \omega(|F|^2)$. The *spectral gap* of $\mathbb{T}$ is m if its spectrum on $\Omega^\perp$ lies in $[0, e^{-m}]$.

Proposition 2.1. *Let $m > 0$ and let $\mathcal{F}$ be a class of gauge-invariant local observables, supported at nonnegative times, that is dense in $L^2(\omega)$ on each bounded time support. If for every $F \in \mathcal{F}$ there is $C_F < \infty$ with*

$$\omega(\vartheta(\bar{F})\tau_t F) - |\omega(F)|^2 \leq C_F e^{-mt} \qquad (t \geq 0),$$

then $\mathbb{T}$ has spectral gap at least m.

Proof. Put $\Psi'_F = \Psi_F - \langle \Omega, \Psi_F \rangle \Omega$, so that the left side equals $\langle \Psi'_F, \mathbb{T}^t \Psi'_F \rangle = \int \lambda^t \, d\sigma_F(\lambda)$, with σ_F the spectral measure of $\mathbb{T}$ at Ψ'_F. For $\delta > 0$ this is at least $\sigma_F((e^{-m+\delta}, 1]) e^{(-m+\delta)t}$, which exceeds $C_F e^{-mt}$ for large t unless $\sigma_F((e^{-m+\delta}, 1]) = 0$. So the spectral projection of $(e^{-m}, 1]$ annihilates every Ψ'_F. These vectors are dense in $\Omega^\perp$, because $\|\Psi_F\|^2 \leq \omega(|F|^2)$ and the class is dense, so the projection vanishes on $\Omega^\perp$. □

The decay of a single two-point function does not suffice: it places only one spectral measure in $[0, e^{-m}]$, and the density of the class is what gives the gap.

3. The line field

A *direction field* is a map $\sigma : \mathbb{Z}^4 \to \{0, 1, 2, 3\}$; its links are $F = \{(x, \sigma(x)) : x \in \mathbb{Z}^4\}$. At each site put $\pi_x(\mu) = \sigma(x + e_\mu)$. The plaquette at x in the plane $\{\mu, \nu\}$ contains

$$[\sigma(x) \in \{\mu, \nu\}] + [\pi_x(\mu) = \nu] + [\pi_x(\nu) = \mu] \tag{3.1}$$

links of F.

Lemma 3.1. *Every plaquette contains exactly one link of F if and only if, at every site x, π_x fixes $\sigma(x)$ and permutes the other three directions cyclically.*

Proof. Fix x, let $s = \sigma(x)$ and let R be the other three directions. For the plane $\{s, \nu\}$ the count (3.1) is $1 + [\pi_x(s) = \nu] + [\pi_x(\nu) = s]$, which equals one for all $\nu \in R$ exactly when π_x fixes s and maps R into R. For a plane $\{\mu, \nu\} \subset R$ the count is $[\pi_x(\mu) = \nu] + [\pi_x(\nu) = \mu]$. Draw an arrow $\mu \to \pi_x(\mu)$ whenever $\pi_x(\mu) \in R \setminus \{\mu\}$; each vertex has at most one outgoing arrow, and the condition asks for exactly one arrow on each of the three pairs. Three arrows on three vertices with out-degree at most one give out-degree one everywhere, and such a tournament is a directed three-cycle. □

From now on σ is the field of period two of Table 2 in Appendix A; in each row π_x fixes $\sigma(x)$ and is a three-cycle on the other directions, so every plaquette contains exactly one link of F.[2] Call the links outside F *active* and write A for their set.

Lemma 3.2.

(i) *F is a disjoint union of bi-infinite straight lines, one through every site; hence $F \cap E(\Lambda)$ is a forest for every box Λ.*

(ii) *An active link $e = (x, \mu)$ lies in six plaquettes. In four of them the link of F shares an end point with e; in the other two it is opposite to e.*

(iii) *Every active link shares a plaquette with exactly twelve other active links, two in each of its plaquettes, and two links share at most one plaquette.*

Proof. (i) Since π_x fixes $\sigma(x)$, $\sigma(x + e_{\sigma(x)}) = \sigma(x)$, and since σ has period two, also $\sigma(x - e_{\sigma(x)}) = \sigma(x)$. If $(x - e_\nu, \nu) \in F$, straightness at $x - e_\nu$ gives $\sigma(x) = \nu$, so the incoming link of F at x is unique and continues the outgoing one. A line meets a box in a segment.

(ii) A link of F sharing an end point with e is the outgoing or incoming link of F at x, of direction $s = \sigma(x) \neq \mu$, or at $x + e_\mu$, of direction $t = \pi_x(\mu) \in R \setminus \{\mu\}$. They lie in the two plaquettes of e in the plane $\{\mu, s\}$ and the two in the plane $\{\mu, t\}$, four distinct plaquettes as $s \neq t$. In the two plaquettes of the remaining plane $\{\mu, \rho\}$ the unique link of F touches neither end of e, so it is the opposite link.

(iii) Each plaquette of e contains e, one link of F and two other links. □

4. Gauge fixing and the free-boundary state

For a function O of the links let O^F be O with every link of F set to the identity.

Lemma 4.1 (gauge fixing)**.** *Let μ_Λ^F be the probability measure on $G^{A \cap E(\Lambda)}$ with density proportional to $\exp(N\beta \sum_{p \in P(\Lambda)} \operatorname{Re} \operatorname{Tr} Q_p^F)$. For every gauge-invariant O on $G^{E(\Lambda)}$, $W_\Lambda(O) = \mu_\Lambda^F(O^F)$.*

Proof. Each segment of $F \cap E(\Lambda)$ is a path $x_0, \dots, x_k$. Given the variables on F, put $g_{x_0} = 1$ and $g_{x_{i+1}} = g_{x_i} Q_{(x_i, x_{i+1})}$ along every segment. The transformed field equals the identity on $F \cap E(\Lambda)$, and for fixed values on F the map from the active variables to the transformed ones is a product of translations, which preserve Haar measure. Gauge invariance of the action and of O gives the claim. □

No holonomy remains because $F \cap E(\Lambda)$ is a forest. The measures μ_Λ^F are finite-volume versions of the specification γ^F on G^A in which γ_e^F resamples an active Q_e from the law with density proportional to $\exp(N\beta \sum_{p \ni e} \operatorname{Re} \operatorname{Tr} Q_p^F)$ given the other active links. Fix a bi-invariant metric δ on G of diameter D and define the Kantorovich coefficients

$$C_{ee'} = \sup \frac{W_1\big(\gamma_e^F(\cdot \mid Q), \gamma_e^F(\cdot \mid Q')\big)}{\delta(Q_{e'}, Q'_{e'})},$$

the supremum over configurations that differ only at e'; W_1 is the Kantorovich distance [51], and $C_{ee'} = 0$ unless e and e' share a plaquette.

Lemma 4.2 (comparison)**.** *Suppose every row and column of $(C_{ee'})$ sums to at most $\alpha < 1$, and let $\lambda > 0$ with $\theta = \alpha e^\lambda < 1$. Let $S \subset A$ and let μ_1, μ_2 be probability measures on G^A invariant under γ_e^F for $e \notin S$. Let $T \subset A \setminus S$ be finite, $w_b = e^{-\lambda \operatorname{dist}(b,T)}$ with the distance of link midpoints, and suppose $\sum_{z \in S} w_z < \infty$. There is a coupling of μ_1 and μ_2 with $\varrho(b) = \mathbb{E}\, \delta(\sigma_b, \sigma'_b)$ satisfying*

$$\sum_{b \notin S} w_b \varrho(b) \leq \frac{\theta D}{1 - \theta} \sum_{z \in S} w_z,$$

[2] A field with one link of F at every site and one in every plaquette can exist only in four dimensions: a site carries $d(d-1)/2$ plaquettes at which it is the base point and a link lies in $2(d-1)$ plaquettes, and the two counts agree only for $d = 4$.

and $\mu_1 = \mu_2$ if $S = \emptyset$.

Proof. The couplings form a weakly compact set on which $\Psi(\pi) = \sum_b w_b \mathbb{E}_\pi \delta(\sigma_b, \sigma'_b)$ is continuous; let π^* minimise it. For $e \notin S$ replace (σ_e, σ'_e) by a sample of an optimal coupling of $\gamma_e^F(\cdot \mid \sigma)$ and $\gamma_e^F(\cdot \mid \sigma')$, chosen measurably [51, Cor. 5.22]. Both μ_i are invariant, so this is again a coupling, and changing the links that enter γ_e^F one at a time gives $\varrho(e) \le \sum_{e'} C_{ee'} \varrho(e')$ by minimality. Multiply by w_e and sum over $e \notin S$; where $C_{ee'} \neq 0$, $w_e \le e^\lambda w_{e'}$, so $\sum_{e \notin S} w_e \varrho(e) \le \theta[\sum_{e' \notin S} w_{e'} \varrho(e') + D \sum_{z \in S} w_z]$. If $S = \emptyset$ this forces $\varrho \equiv 0$. □

Conditioning on the links of the support of H [33, Ch. 3] and applying the lemma to the two conditional measures gives, for the Gibbs measure μ of γ^F, a local F' that is L-Lipschitz in each link and a bounded H with finite support disjoint from that of F',

$$|\operatorname{Cov}_\mu(F', H)| \le \|H\|_\infty L \frac{\theta D}{1-\theta} |\operatorname{supp} H| \, e^{-\lambda \operatorname{dist}(\operatorname{supp} F', \operatorname{supp} H)}. \tag{4.1}$$

Proposition 4.3. *Suppose every row and column of $(C_{ee'})$ sums to at most $\alpha < 1$. Then γ^F has a unique Gibbs measure μ_F; for every gauge-invariant local O the limit $\omega(O) = \lim W_\Lambda(O)$ exists along every sequence of boxes exhausting $\mathbb{Z}^4$ and equals $\mu_F(O^F)$; ω is invariant under translations and the hypercubic group; truncated correlations of Lipschitz gauge-invariant observables decay at every rate below $-\log\alpha$; and for $\beta \ge 0$, ω is reflection positive and its transfer matrix has spectral gap at least $-\log\alpha$.*

Proof. Gibbs measures exist by compactness and coincide by Lemma 4.2 with $S = \emptyset$. Extend μ_Λ^F by Haar measure outside Λ and let S be the set of active links with a plaquette outside $P(\Lambda)$; off S the extended measure is invariant under γ_e^F. Lemma 4.2 with T the support of O^F bounds $|\mu_\Lambda^F(O^F) - \mu_F(O^F)|$ by $C_T R^3 e^{-\lambda R}$, R the distance from T to the complement of Λ, for Lipschitz O, and uniform approximation handles continuous O; Lemma 4.1 identifies $\mu_\Lambda^F(O^F)$ with $W_\Lambda(O)$. Since $W_\Lambda(\tau_v O) = W_{\Lambda - v}(O)$ and the limit does not depend on the exhausting sequence, ω is invariant; the line field is a device for computing W_Λ and is never moved. Reflection positivity of W_Λ on symmetric boxes passes to the limit. The decay is (4.1), since $\operatorname{Cov}_\omega(O, O') = \operatorname{Cov}_{\mu_F}(O^F, O'^F)$. Lipschitz gauge-invariant functions are dense in the gauge-invariant part of $L^2(\omega)$ on every support, being Lipschitz functions on the compact quotient by the gauge group, so Proposition 2.1 gives the gap. □

Proof of Theorem A for SU(2). Identify SU(2) with the unit quaternions and use the chordal metric. For an active link e and a plaquette $p \ni e$ let s_p be the product of the other three variables, ordered so that $\operatorname{Re}(Q_e s_p) = \frac{1}{2} \operatorname{Tr} Q_p$, with links of F equal to one. Since $N\beta \operatorname{Re}\operatorname{Tr} Q_p = 4\beta \langle Q_e, \bar{s}_p \rangle$ in $\mathbb{R}^4$, the conditional law of Q_e is the von Mises–Fisher law $\nu_b(dx) \propto e^{\langle x, b\rangle} dx$ on S^3 with $b = 4\beta \sum_{p \ni e} \bar{s}_p$. Moving a partner e' by δ moves one staple by an isometric image of that displacement, hence b by $4|\beta|\delta$, and the sharp quarter (Theorem C.4 in Appendix C), $W_1(\nu_b, \nu_{b'}) \le \frac{1}{4}|b - b'|$, gives $C_{ee'} \le |\beta|$. By Lemma 3.2(iii) every row and column has twelve nonzero entries, so $\alpha \le 12|\beta| = 3|\beta_W|$, and Proposition 4.3 applies for $|\beta_W| < \frac{1}{3}$. □

The value $\frac{1}{3}$ is the end of this argument: at $b = 0$ the derivative of ν_b has Kantorovich norm exactly $\frac{1}{4}$, the six staples of an active link can be arranged with sum zero while one of them moves freely, and so every row sum of the matrix is exactly $3|\beta_W|$. Without the gauge every link has eighteen partners and the same argument stops at $\beta_W = \frac{2}{9}$ [6, Prop. 13].

5. Curvature in the gauge

In this section $G \subset \mathrm{U}(N)$ is closed and connected with Lie algebra $\mathfrak{g}$ and the inner product $\langle X, Y \rangle = -\operatorname{Tr}(XY)$. A tangent vector at a field Q is $v = (X_e Q_e)$ with $X_e \in \mathfrak{g}$, and $|v|^2 = \sum_e |X_e|^2$. For the plaquette p at x in the plane $\mu < \nu$, with links $e_1, \dots, e_4$ as in Sec. 2, transport the vectors to x,

$$Y_1 = X_{e_1}, \quad Y_2 = Q_{e_1} X_{e_2} Q_{e_1}^*, \quad Y_3 = Q_{e_4} X_{e_3} Q_{e_4}^*, \quad Y_4 = X_{e_4},$$

and write $W = Q_p$. Along the geodesics $Q_e(t) = e^{tX_e}Q_e$ the plaquette variable is $Q_p(t) = e^{tY_1}e^{tY_2}We^{-tY_3}e^{-tY_4}$, and differentiating twice gives the second derivative of $\operatorname{Re}\operatorname{Tr} Q_p$,

(5.1)
$$f_p = \sum_{i=1}^{4} \operatorname{Re}\operatorname{Tr}(Y_i^2 W) + 2\operatorname{Re}\operatorname{Tr}(Y_1Y_2W) + 2\operatorname{Re}\operatorname{Tr}(Y_3Y_4W) - 2\sum_{i\in\{1,2\},\, j\in\{3,4\}} \operatorname{Re}\operatorname{Tr}(Y_iWY_j),$$

so that $\operatorname{Hess} S_0(v,v) = \sum_p f_p$ for $S_0 = \sum_p \operatorname{Re}\operatorname{Tr} Q_p$. The *corner form* of p is

(5.2)
$$c_p = 2\langle Y_1, Y_4\rangle - 2\langle Y_1, Y_2\rangle + 2\langle Y_2, WY_3W^*\rangle - 2\langle Y_3, Y_4\rangle;$$

it collects the inner products of the vectors of two links that meet at a right angle at a corner of p.

Lemma 5.1 (divergence identity)**.** *On $\mathbb{Z}^4$, or on a box with free boundary, for every field Q and every square-summable X,*

$$\sum_y \Big[|D_yX|^2 + \sum_\mu \big|X_{y,\mu} + Q^*_{y-e_\mu,\mu}X_{y-e_\mu,\mu}Q_{y-e_\mu,\mu}\big|^2\Big] = 4|X|^2 + \sum_p c_p,$$

where $D_yX = \sum_\mu[X_{y,\mu} - Q^*_{y-e_\mu,\mu}X_{y-e_\mu,\mu}Q_{y-e_\mu,\mu}]$ is the covariant divergence and terms with a link outside the box are omitted.

Proof. At a site y let o_μ be the vector of the outgoing link of direction μ and i_μ the transported vector of the incoming one, so that $D_yX = \sum_\mu(o_\mu - i_\mu)$. Every link is outgoing at one end and incoming at the other, so the squares $|o_\mu|^2 + |i_\mu|^2$, which occur once in $|D_yX|^2$ and once in $|o_\mu + i_\mu|^2$, add up to $4|X|^2$. The collinear cross term $-2\langle o_\mu, i_\mu\rangle$ of $|D_yX|^2$ cancels the term $+2\langle o_\mu, i_\mu\rangle$ of $|o_\mu + i_\mu|^2$. Every remaining cross term of $|D_yX|^2$ pairs two perpendicular ends at y, and such a pair is a corner of exactly one plaquette, with the sign and transport of (5.2); in a box two perpendicular ends at y span a plaquette of the box. □

Lemma 5.2 (one plaquette)**.** *Let $k \in \{1,2,3,4\}$, let $k \pm 1$ (mod 4) be the positions adjacent to k and $k+2$ the opposite one. For every N, every $W \in \mathrm{U}(N)$ and all $Y_1, \dots, Y_4 \in \mathfrak{u}(N)$ with $Y_k = 0$,*

$$\pm f_p + \tfrac{9}{10}c_p \le \tfrac{21}{10}(|Y_{k-1}|^2 + |Y_{k+1}|^2) + \tfrac{5}{4}|Y_{k+2}|^2.$$

Proof. Replacing W by $-W$ changes the sign of f_p and fixes c_p, so the sign + suffices, and conjugation by a unitary matrix preserves both sides, so we may take $W = \operatorname{diag}(w_a)$. The difference $\mathcal{F}$ of the two sides then splits into a block for each diagonal position a, in the real variables $y_i = -i(Y_i)_{aa}$, and a block for each pair $a \neq b$, in the complex variables $z_i = (Y_i)_{ab}$, since $(Y_i)_{ba} = -\bar{z}_i$. Appendix B shows that every block is positive definite. □

Theorem 5.3 (Hessian bound)**.** *For every closed subgroup $G \subset \mathrm{U}(N)$, every box or $\mathbb{Z}^4$, every field Q and every tangent vector v that vanishes on the links of F,*

$$|\operatorname{Hess} S_0(v,v)| \le \tfrac{29}{2}|v|^2.$$

Proof. Every plaquette has exactly one link of F, on which X vanishes, and by Lemma 3.2(ii) each active link is adjacent to that link in four of its plaquettes and opposite to it in two. Summing Lemma 5.2 over the plaquettes,

$$\sum_p(\pm f_p + \tfrac{9}{10}c_p) \le (4\cdot\tfrac{21}{10} + 2\cdot\tfrac{5}{4})|X|^2 = \tfrac{109}{10}|X|^2.$$

By Lemma 5.1, $\frac{9}{10}\sum_p c_p = \frac{9}{10}\Sigma - \frac{36}{10}|X|^2$ with $\Sigma \ge 0$ the left side of that lemma. Hence $\pm\operatorname{Hess} S_0(v,v) \le (\frac{109}{10} + \frac{36}{10})|X|^2 = \frac{29}{2}|X|^2$; links near the boundary of a box lie in fewer plaquettes and receive less. Both lemmas concern vectors in $\mathfrak{u}(N)$, so the bound holds on every closed subgroup, whose geodesics $e^{tX}Q_e$ with $X \in \mathfrak{g}$ are geodesics of $\mathrm{U}(N)$. □

The factor $\frac{9}{10}$ lets the divergence identity pay for the corners at the price $\frac{36}{10}$ while the blocks of one plaquette stay positive. Without the gauge the sharp bound is 16 [6].

TABLE 1. The curvature $\kappa_G = h^\vee/(2\ell_\rho)$ in a faithful representation of least dimension N, and the range of Theorem A for $\beta_W = N^2\beta$.

Group	ρ	$h^\vee$	ℓ_ρ	κ_G	$\beta_W <$
$\mathrm{SU}(N)$, $N \geq 2$	$\mathbb{C}^N$	N	1	$\frac{N}{2}$	$\frac{N^2}{29}$
$\mathrm{SO}(N)$, $N \geq 5$	$\mathbb{R}^N$	$N-2$	2	$\frac{N-2}{4}$	$\frac{N(N-2)}{58}$
$\mathrm{Sp}(n)$, $n \geq 2$	$\mathbb{C}^{2n}$	$n+1$	1	$\frac{n+1}{2}$	$\frac{2n(n+1)}{29}$
G_2	7	4	2	1	$\frac{14}{29}$
F_4	26	9	6	$\frac{3}{4}$	$\frac{39}{29}$
E_6	27	12	6	1	$\frac{54}{29}$
E_7	56	18	12	$\frac{3}{4}$	$\frac{84}{29}$
E_8	248	30	60	$\frac{1}{4}$	$\frac{124}{29}$

Lemma 5.4 (Ricci curvature)**.** *Let $G \subset \mathrm{U}(N)$ be closed and connected with simple Lie algebra, and let $\mathfrak{t} \subset \mathfrak{g}$ be a maximal abelian subalgebra. Write the eigenvalues of $\operatorname{ad} H$ on $\mathfrak{g} \otimes \mathbb{C}$ and of H on $\mathbb{C}^N$, $H \in \mathfrak{t}$, as $i\alpha(H)$ and $i\mu(H)$, over the roots α and the weights μ counted with multiplicity. Then $\mathrm{Ric}_G = \kappa_G\langle\cdot,\cdot\rangle$ with*

$$\kappa_G = \frac{\sum_\alpha \alpha(H)^2}{4\sum_\mu \mu(H)^2} = \frac{h^\vee}{2\ell_\rho} \qquad (0 \neq H \in \mathfrak{t}).$$

Proof. For a bi-invariant metric $R(X,Y)Z = -\frac{1}{4}[[X,Y],Z]$ on left-invariant fields [37, Sect. 7], so for an orthonormal basis (E_i), $\mathrm{Ric}(X,X) = \frac{1}{4}\sum_i |[X,E_i]|^2 = -\frac{1}{4}\operatorname{Tr}(\operatorname{ad} X)^2$. The Killing form and $\langle\cdot,\cdot\rangle$ are invariant and $\mathfrak{g}$ is simple, so the Killing form is $-4\kappa\langle\cdot,\cdot\rangle$ for a constant κ, and on $\mathfrak{t}$ the two forms are $-\sum_\alpha \alpha(H)^2$ and $\sum_\mu \mu(H)^2$. For H the coroot of the highest root, $\sum_\alpha \alpha(H)^2 = 4h^\vee$ and $\sum_\mu \mu(H)^2 = 2\ell_\rho$ [32, Ch. 6]. □

For two links $e \neq e'$ of a plaquette p write $\operatorname{Hess}\operatorname{Re}\operatorname{Tr} Q_p(v,v) = \sum_{e,e'\in p} H^p_{ee'}(X_e, X_{e'})$ with $H^p_{ee'}(a,b) = H^p_{e'e}(b,a)$. By (5.1) a mixed block is, after isometric transport, $\pm\operatorname{Re}\operatorname{Tr}(abW)$ or $\pm\operatorname{Re}\operatorname{Tr}(aWb)$, so

$$|H^p_{ee'}(a,b)| \leq |a|\,|b|. \tag{5.3}$$

5.1. **Decay with an explicit rate.** Let A' be a finite set of active links and P' a finite set of plaquettes whose active links lie in A'. Call $S = \sum_{p\in P'} c_p \operatorname{Re}\operatorname{Tr} Q^F_p$, $0 \leq c_p \leq 1$, *admissible* if $|\operatorname{Hess} S(v,v)| \leq \frac{29}{2}|v|^2$ for v tangent to $G^{A'}$. By Theorem 5.3 the action $S_\Lambda = \sum_{p\in P(\Lambda)} \operatorname{Re}\operatorname{Tr} Q^F_p$ of a box is admissible, and so is every convex combination of admissible actions; by (5.3) and Lemma 3.2(iii) the mixed blocks of $\operatorname{Hess} S$ have norm at most one and each link has at most twelve partners. Let μ have density proportional to $e^{N\beta S}$ on $G^{A'}$, let $L = \Delta + N\beta\langle\nabla S, \nabla\cdot\rangle$ be its Langevin generator, self-adjoint in $L^2(\mu)$, and $P_t = e^{tL}$. For weights $w > 0$ on A' put $\Gamma^w(f) = \sum_e w_e|\nabla_e f|^2$, with ∇_e the gradient in the factor of e; Γ is the case $w \equiv 1$.

Lemma 5.5. *Let G have semisimple Lie algebra, $\lambda \geq 0$, $w_e = e^{\lambda d_e}$ with $|d_e - d_{e'}| \leq 1$ for partners, and*

$$K_\lambda = \kappa_G - N|\beta|\Big(\tfrac{29}{2} + 12\big(\cosh\tfrac{\lambda}{2} - 1\big)\Big).$$

Then $\Gamma^w(P_t f) \leq e^{-2K_\lambda t} P_t \Gamma^w(f)$ for smooth f and $t \geq 0$.

Proof. Put $V = -N\beta S$. On the product $G^{A'}$ the projection to one factor is parallel and the Ricci tensor is block diagonal, so Bochner's formula applied factor by factor gives

$$\tfrac{1}{2}L\Gamma^w(g) - \Gamma^w(g, Lg) = \sum_e w_e\big(|\nabla\nabla_e g|^2 + \mathrm{Ric}_G(\nabla_e g, \nabla_e g)\big) + \sum_{e,e'} w_e \operatorname{Hess} V_{ee'}(\nabla_e g, \nabla_{e'} g).$$

With $u_e = \nabla_e g$, $v_e = \sqrt{w_e} u_e$ and the symmetry of $\operatorname{Hess} V$, the last sum is

$$\operatorname{Hess} V(v,v) + \sum_{e \neq e'} \Big(\cosh \tfrac{\lambda(d_e - d_{e'})}{2} - 1 \Big) \operatorname{Hess} V_{ee'}(v_e, v_{e'}),$$

since $(w_e + w_{e'})/(2\sqrt{w_e w_{e'}}) = \cosh \frac{\lambda(d_e - d_{e'})}{2}$. The first term is at least $-\frac{29}{2} N|\beta||v|^2$; in the second only partners contribute, each with coefficient at most $\cosh \frac{\lambda}{2} - 1$ and a block of norm at most $N|\beta|$, and with twelve partners it is at least $-12(\cosh \frac{\lambda}{2} - 1) N|\beta||v|^2$. As $\mathrm{Ric}_G \geq \kappa_G$ and $|v|^2 = \Gamma^w(g)$, the left side is at least $K_\lambda \Gamma^w(g)$. For $\varphi(s) = P_s \Gamma^w(P_{t-s} f)$ this gives $\varphi' \geq 2K_\lambda \varphi$, hence $\varphi(0) \leq e^{-2K_\lambda t} \varphi(t)$. □

Lemma 5.6. *Suppose $K_\lambda > 0$. Let f, g be smooth functions of the links of A_f and B_g, and let D be the graph distance from A_f to B_g in the graph on A' whose edges join partners. Then*

$$|\operatorname{Cov}_\mu(f,g)| \leq \frac{e^{-\lambda D/2}}{K_\lambda} \|\nabla f\|_\infty \sum_{e \in B_g} \|\nabla_e g\|_\infty.$$

Proof. Lemma 5.5 with $\lambda = 0$ shows that $P_t f \to \mu(f)$ uniformly, so $\operatorname{Cov}_\mu(f,g) = \int_0^\infty \mu(\Gamma(P_t f, g))\, dt$. Let d_e be the smaller of D and the graph distance from e to A_f; then $w = 1$ on A_f, $w = e^{\lambda D}$ on B_g, and $\Gamma^w(f) \leq \|\nabla f\|_\infty^2$. For $e \in B_g$, Lemma 5.5 gives $|\nabla_e P_t f|^2 \leq e^{-\lambda D} e^{-2K_\lambda t} \|\nabla f\|_\infty^2$, and integration in t gives the bound. □

Proof of Theorem A. Let λ^* solve $K_{\lambda^*} = 0$, so that $\lambda^*/2 = m_G(\beta)$, and let $0 < \lambda < \lambda^*$. Lemma 5.6 with $S = S_\Lambda$ gives $|\operatorname{Cov}_{\mu_\Lambda^F}(f,g)| \leq C_{f,g} e^{-\lambda D/2}$ uniformly in the box. For boxes $\Lambda \subset \Lambda'$ interpolate $S_s = (1-s)S_\Lambda + s S_{\Lambda'}$, which is admissible; for smooth f of the active links of Λ,

$$\frac{d}{ds} \mu_s(f) = N\beta \sum_{p \in P(\Lambda') \setminus P(\Lambda)} \operatorname{Cov}_{\mu_s}(f, \operatorname{Re}\operatorname{Tr} Q_p^F),$$

with $|\nabla_e \operatorname{Re}\operatorname{Tr} Q_p| \leq \sqrt{N}$. A plaquette outside $P(\Lambda)$ is at graph distance at least $R-1$ from the support of f, R the distance to the complement of Λ, and at most $C(k+1)^4$ plaquettes lie at graph distance k, so $|\mu_{\Lambda'}^F(f) - \mu_\Lambda^F(f)| \leq C_f R^4 e^{-\lambda R/2}$. The limit exists along every increasing sequence of boxes, two sequences are compared through their intersections, and continuous gauge-invariant observables are uniform limits of smooth ones $e^{\tau\Delta} O$, the heat semigroup commuting with translations of G. With Lemma 4.1 the limit ω exists; invariance and reflection positivity follow as in Proposition 4.3. Since $\mathrm{Ric}_G \geq 0$, $e^{\tau\Delta} O$ has the Lipschitz constant of O [10, Sect. 3.3], so the decay passes to Lipschitz observables; two links of a plaquette have times differing by at most one, so observables at time separation t are at graph distance at least $t - c$, and truncated correlations decay at every rate below $m_G(\beta)$. Proposition 2.1 gives the gap. □

The rate behaves like $\log \frac{\kappa_G}{6N|\beta|}$ as $\beta \to 0$. The proof also gives a Poincaré inequality with constant $1/(\kappa_G - \frac{29}{2} N|\beta|)$ for every μ_Λ^F, uniformly in the box [10, Cor. 5.7.2].

6. One cube and the continuum gap

Setting. Let $G \subset \mathrm{U}(N)$ be compact and nontrivial and φ a positive continuous class function on G, such as $e^{N\beta \operatorname{Re}\operatorname{Tr}(\cdot)}$. On the torus $\Lambda_L = (\mathbb{Z}/L\mathbb{Z})^4$, L even and $L \geq 2n+4$, with links E and plaquettes P, let

$$\mu(dU) = \frac{1}{Z} \prod_{p \in P} \varphi(U_p) \prod_{e \in E} dU_e.$$

A link (y, ν) has base point y; differences of base points are taken in $(-L/2, L/2]^4$ and $|\cdot|_\infty$ is the corresponding distance. For $x \in \Lambda_L$ let V_x be the set of links with base point in $x + [0,n)^4$; every link lies in exactly n^4 of the L^4 cubes V_x. For $V \subset E$ let ∂V be the

links outside V that share a plaquette with a link of V; the conditional law $\gamma_V^0(\cdot \mid U)$ of U_V given the other links has density proportional to $\prod_{p\cap V\neq\emptyset}\varphi(U_p)$ and depends only on $U_{\partial V}$. Let $\mathcal{P}_x f(U) = \int f\, d\gamma_{V_x}^0(\cdot \mid U)$, the conditional expectation given the links outside V_x, an orthogonal projection on $L^2(\mu)$, and let $\mathcal{A}$ be the closed subspace of gauge-invariant functions. By gauge covariance of $\gamma_{V_x}^0$, $\mathcal{P}_x$ maps $\mathcal{A}$ into itself.

Definition 6.1. For $n \geq 2$ and $\varpi > 0$, the condition $\mathrm{BP}(n,\varpi)$ for μ is

$$\varpi \operatorname{Var}_\mu(f) \leq \frac{1}{n^4}\sum_{x\in\Lambda_L} \mu\big((f-\mathcal{P}_x f)^2\big) \qquad \text{for every real } f \in \mathcal{A}.$$

Put $q_n = (2+1/n)^4$, so that $16 < q_n \leq \frac{625}{16}$ for $n \geq 2$, and let $\mathcal{K} = L^{-4}\sum_x \mathcal{P}_x$ be the chain that resamples one uniformly chosen cube; for $Y \subset \Lambda_L$ let $\mathcal{K}_Y = L^{-4}(\sum_{x\in Y}\mathcal{P}_x + (L^4 - |Y|)I)$ be the same chain with the moves outside Y suppressed. Write $y \rightsquigarrow z$ if $\partial V_z \cap V_y \neq \emptyset$. A link of ∂V_z has its base point in $z + [-1,n]^4$, so $y \rightsquigarrow z$ implies $|z-y|_\infty \leq n$, and for each z at most $(2n+1)^4 = q_n n^4$ points y have $y \rightsquigarrow z$. For finite $S \subset E$ let $Y_1(S) = \{y : V_y \cap S \neq \emptyset\}$ and $Y_{j+1}(S) = Y_j(S) \cup \{y : y \rightsquigarrow z \text{ for some } z \in Y_j(S)\}$, $Y_0(S) = \emptyset$.

Lemma 6.2 (finite speed)**.** *Let f be a bounded function of the links of a finite set S, let $m_0 \geq 0$ and $Y = Y_{m_0}(S)$. Then $\mathcal{K}_Y^t f$ depends only on the links of $S \cup \bigcup_{y\in Y}\partial V_y$, and for every integer $t \geq 0$*

$$\big\|\mathcal{K}^t f - \mathcal{K}_Y^t f\big\|_\infty \leq \frac{2|S|}{q_n}\,\|f\|_\infty \sum_{m>m_0}\frac{(q_n\tau)^m}{m!}, \qquad \tau = \frac{tn^4}{L^4}.$$

Proof. If h depends only on the links of W, then $\mathcal{P}_x h = h$ when $V_x \cap W = \emptyset$, and otherwise $\mathcal{P}_x h$ depends only on the links of $(W \setminus V_x) \cup \partial V_x$; induction gives the first claim. For the second run two chains from the same configuration with the same uniformly chosen cubes $x_1, x_2, \ldots$. If $x_s \notin Y$ the first chain resamples V_{x_s} and the second does nothing; if $x_s \in Y$ and the two configurations agree on ∂V_{x_s}, both receive one common sample, and otherwise they resample independently. The marginals are the chains $\mathcal{K}$ and $\mathcal{K}_Y$. A link where the configurations differ at time t lies in $V_{x_{s_m}}$ for some times $s_1 < \cdots < s_m \leq t$ with $x_{s_1} \notin Y$ and $x_{s_l} \rightsquigarrow x_{s_{l+1}}$, by induction on t. If the set of such links meets S, then $x_{s_m} \in Y_1(S)$ and $x_{s_{m-l}} \in Y_{l+1}(S)$, so $m > m_0$. For fixed m there are at most $t^m/m!$ choices of times, $n^4|S|$ of x_{s_m} and $q_n n^4$ of each earlier cube, each sequence having probability L^{-4m}; the difference of f is at most $2\|f\|_\infty$. □

For finite $S, S' \subset E$ let $d(S,S')$ be the least $|b-b'|_\infty$ over base points b of links of S and b' of links of S'.

Theorem 6.3 (decay from one cube)**.** *Suppose $\mathrm{BP}(n,\varpi)$ holds for μ. Then $\varpi \leq 4$, and for bounded real $f, g \in \mathcal{A}$ depending on the links of finite sets S_f, S_g with $d = d(S_f,S_g) \geq 3n-2$,*

$$|\operatorname{Cov}_\mu(f,g)| \leq \Big(e + \frac{k|S_f||S_g|}{64e^2}\Big)\|f\|_\infty\|g\|_\infty\, e^{-\varpi k/(e^2 q_n)}, \qquad k = \Big\lfloor \frac{d+2}{n}\Big\rfloor - 2.$$

In particular $|\operatorname{Cov}_\mu(f,g)| \leq C(1+d)e^{-\varpi d/(e^2 q_n n)}$ for every d, with C depending only on $|S_f|, |S_g|, \|f\|_\infty, \|g\|_\infty$.

Proof. The bound $\varpi \leq 4$. Take $f = \operatorname{Re}\operatorname{Tr} U_p$ for one plaquette. Each $I - \mathcal{P}_x$ is an orthogonal projection annihilating constants, so $\mu((f-\mathcal{P}_x f)^2) \leq \operatorname{Var}_\mu(f)$, and it vanishes unless V_x contains a link of p, which happens for at most $4n^4$ points x.

Spectral part. On the functions of $\mathcal{A}$ with mean zero, $\mathcal{K}$ is self-adjoint with $0 \leq \mathcal{K} \leq I$, and $\mathrm{BP}(n,\varpi)$ reads $\langle h, (I-\mathcal{K})h\rangle \geq \epsilon\|h\|^2$ with $\epsilon = \varpi n^4/L^4$. So $\|\mathcal{K}^T h\|_2 \leq e^{-\epsilon T}\|h\|_2$, and from $f - \mu(f) = \mathcal{K}^T(f-\mu(f)) + \sum_{t<T}\mathcal{K}^t(I-\mathcal{K})f$,

$$\operatorname{Cov}_\mu(f,g) = \langle \mathcal{K}^T(f-\mu f), g-\mu g\rangle + \sum_{t=0}^{T-1}\langle \mathcal{K}^t f, (I-\mathcal{K})g\rangle,$$

whose first term is at most $e^{-\epsilon T}\|f\|_\infty\|g\|_\infty$.

Local part. Let $Z_g = Y_1(S_g)$, so $|Z_g| \le n^4|S_g|$ and $(I-\mathcal{K})g = L^{-4}\sum_{x\in Z_g}(g-\mathcal{P}_x g)$; if h does not depend on the links of V_x then $\langle h, g-\mathcal{P}_x g\rangle = 0$. Put $m_0 = k-1$ and $Y = Y_{m_0}(S_f)$. A link of V_x, $x \in Z_g$, lying in S_f or in ∂V_y with $y \in Y$ would join base points of S_f and S_g at distance at most $n(m_0+3)-3 \le n\lfloor (d+2)/n\rfloor - 3 < d$, which is impossible. So $\mathcal{K}_Y^t f$ does not depend on the links of V_x, $x \in Z_g$, and by Lemma 6.2

$$\big|\langle \mathcal{K}^t f, (I-\mathcal{K})g\rangle\big| = \big|\langle \mathcal{K}^t f - \mathcal{K}_Y^t f, (I-\mathcal{K})g\rangle\big| \le \frac{4|S_f||S_g|n^4}{q_n L^4}\|f\|_\infty\|g\|_\infty \sum_{m\ge k}\frac{(q_n\tau_t)^m}{m!}.$$

Summing over $t < T$ with $\tau = Tn^4/L^4$, the second term is at most

$$\frac{4|S_f||S_g|\tau}{q_n}\,\|f\|_\infty\|g\|_\infty \sum_{m\ge k}\frac{(q_n\tau)^m}{m!}.$$

Choice of T. Take $T = \lfloor kL^4/(e^2 q_n n^4)\rfloor$, so that $z = q_n\tau \le k/e^2$ and $\epsilon T \ge \varpi k/(e^2 q_n) - 1$, using $\varpi \le 4$ and $L^4 \ge 4n^4$. Since $k! \ge (k/e)^k$, $\sum_{m\ge k} z^m/m! \le (ez/k)^k e^z \le e^{-k(1-e^{-2})}$, and $\tau/q_n \le k/(256e^2)$. As $\varpi/(e^2 q_n) < 1 - e^{-2}$, both terms are bounded as stated. For the second form use $k > d/n - 3$, and for $d < 3n-2$ the trivial bound $\|f\|_\infty\|g\|_\infty$. □

A *torus limit* at β is a state ω on the gauge-invariant local functions of $\mathbb{Z}^4$ with $\omega(O) = \lim_j \mu_{L_j}(O)$ along some $L_j \to \infty$; by compactness every sequence has a subsequence along which one exists.

Corollary 6.4 (lattice gap)**.** *Let $\beta \ge 0$ and suppose* $\mathrm{BP}(n,\varpi)$ *holds for μ_L for all L in an infinite set of even integers $L \ge 2n+4$. Then every torus limit along this set is invariant under translations, reflection positive, and its transfer matrix has spectral gap at least*

$$m_n = \frac{\varpi}{e^2 q_n n} \ge \frac{16\varpi}{625 e^2 n}.$$

Proof. Translation invariance and reflection positivity are identities and inequalities among finitely many local functions of μ_L [43], and pass to ω. For a continuous gauge-invariant F at positive times the supports of $\vartheta(\bar F)$ and $\tau_t F$ are at distance d with $|d-t| \le c_F$, and Theorem 6.3 on Λ_L gives $|\operatorname{Cov}_{\mu_L}(\vartheta(\bar F), \tau_t F)| \le C_F(1+t)e^{-m_n t}$ uniformly in L. In the limit $|\omega(\vartheta(\bar F)\tau_t F) - |\omega(F)|^2| \le C_F(1+t)e^{-m_n t}$, and Proposition 2.1, applied with every $m < m_n$, gives the gap. □

6.1. **The continuum limit.** Let $a_k \to 0$ be lattice spacings with $a_k = 2^{-k}a_0$, and β_k couplings. Let ω_k be lattice states with transfer matrices $\mathbb{T}_k$ on spaces $\mathcal{H}_k$ with vacua Ω_k. Let $\mathcal{T}$ be a complex vector space of test functions on $\mathbb{R}^4$ supported in $\{t > 0\}$, and for each k let $f \mapsto O_k(f)$ be a linear map to gauge-invariant local observables at positive times; the example is $O_k(f) = Z_k \sum_x a_k^4 f(a_k x)\Phi(x)$ with a local field Φ and a renormalisation constant Z_k.

Definition 6.5. A *continuum limit* of (ω_k, O_k) is a Hilbert space $\mathcal{H}$ with a unit vector Ω, a self-adjoint $H \ge 0$ with $H\Omega = 0$, and a linear map $\psi : \mathcal{T} \to \mathcal{H}$ with dense range such that for all $f, g \in \mathcal{T}$ and dyadic $\tau \ge 0$

$$\langle \psi(f), e^{-\tau H}\psi(g)\rangle = \lim_k \omega_k\big(\vartheta(\overline{O_k(f)})\,\tau_{\tau/a_k} O_k(g)\big), \qquad \langle \Omega, \psi(f)\rangle = \lim_k \omega_k(O_k(f)),$$

over the k with $\tau/a_k \in \mathbb{Z}$. It is *nontrivial* if $\mathcal{H} \ne \mathbb{C}\Omega$.

The Osterwalder–Schrader reconstruction produces such a limit from convergent renormalised Schwinger functions satisfying the axioms [41, 42]; the definition records only what is used below.

Proof of Theorem B. Let ω_k be torus limits at β_k along the tori on which $\mathrm{BP}(n_k,\varpi)$ holds. By Corollary 6.4, $\mathbb{T}_k$ has gap at least $16\varpi/(625e^2n_k)$ on $\Omega_k^\perp$. For $f\in\mathcal{T}$ put $O=O_k(f)$ and $\varphi_k=\Psi_O-\omega_k(O)\Omega_k$. For dyadic τ and k large, $t=\tau/a_k$ is an integer and $16\varpi t/(625e^2n_k)\ge\Delta\tau$ since $n_ka_k\le\ell$, so

$$\omega_k(\vartheta(\bar O)\tau_tO)-|\omega_k(O)|^2=\langle\varphi_k,\mathbb{T}_k^t\varphi_k\rangle\le e^{-\Delta\tau}\|\varphi_k\|^2.$$

Both sides converge by Definition 6.5: with $\varphi=\psi(f)-\langle\Omega,\psi(f)\rangle\Omega$, $\langle\varphi,e^{-\tau H}\varphi\rangle\le e^{-\Delta\tau}\|\varphi\|^2$. These vectors form a subspace dense in $\Omega^\perp$, and the inequality extends to $\Omega^\perp$ and to all $\tau\ge0$ by continuity. For $\varphi\in\Omega^\perp$ with spectral measure μ_φ of H this says $\int e^{-\tau E}d\mu_\varphi(E)\le e^{-\Delta\tau}\mu_\varphi([0,\infty))$ for all τ, which forces $\mu_\varphi([0,\Delta-\eta])=0$ for every $\eta>0$. So 0 is a simple eigenvalue and the spectrum on $\Omega^\perp$ lies in $[\Delta,\infty)$, and is not empty if $\Omega^\perp\ne\{0\}$. □

The proof uses the cube condition only through the decay of two-point functions of gauge-invariant observables, and the constant Δ depends on ϖ and ℓ alone.

7. The cube condition at strong coupling

Keep the setting of Sec. 6, with a bi-invariant metric δ of diameter D on G, invariant under inversion, and $d_E(U,U')=\sum_e\delta(U_e,U'_e)$. Two distinct links are *partners* if they share a plaquette; two links share at most one plaquette, and every link has eighteen partners, three in each of its six plaquettes. The direction field σ of Sec. 3 has period two, so it is defined on Λ_L. By Lemma 3.2(i) the links of F with an end point v are exactly $(v,\sigma(v))$ and $(v-e_{\sigma(v)},\sigma(v))$, and $\sigma(v-e_{\sigma(v)})=\sigma(v)$.

Proposition 7.1. *At $\beta=0$, $\mathrm{BP}(n,1)$ holds for every n.*

Proof. For $\varphi\equiv1$, μ is a product of Haar measures. Write $f=\sum_S f_S$ with f_S a function of the links of S and $\int f_S\,dU_e=0$ for $e\in S$; the f_S are orthogonal and $\mathcal{P}_xf=\sum_{S\cap V_x=\emptyset}f_S$, so $\sum_x\mu((f-\mathcal{P}_xf)^2)=\sum_{S\ne\emptyset}\|f_S\|^2\#\{x:S\cap V_x\ne\emptyset\}\ge n^4\,\mathrm{Var}_\mu(f)$. □

Gauge fixing inside a cube. For a cube $Q_x=x+[0,n)^4$ write $\hat v=v-x$ and call v *interior* if $\hat v\in[1,n-1]^4$; let I_x be the set of interior vertices. For $z\in I_x$ put $\tau(z)=(z-e_{\sigma(z)},\sigma(z))$, a link of F with head z, and $T_x=\{\tau(z):z\in I_x\}$. Every link with an end point in I_x has its base point in Q_x, so $T_x\subset V_x$, the map τ is injective, and no link outside V_x touches I_x. Call the links of $V_x\setminus T_x$ *free*. For $\psi\in G^{V_x\setminus T_x}$ let $[\psi,U]$ be the configuration equal to ψ on the free links, to 1 on T_x and to U outside V_x, and let $\gamma^T_{x,U}$ be the probability measure on $G^{V_x\setminus T_x}$ with density proportional to $\prod_{p\cap V_x\ne\emptyset}\varphi([\psi,U]_p)$.

Lemma 7.2. *For every bounded measurable $f\in\mathcal{A}$, $\mathcal{P}_xf(U)=\int f([\psi,U])\,\gamma^T_{x,U}(d\psi)$.*

Proof. Let $\sigma'\in G^{V_x}$ and write (σ',U) for the configuration equal to σ' on V_x and to U elsewhere. Order I_x by the sum of the coordinates of $\hat z$; then $z-e_{\sigma(z)}$ precedes z or is not interior. Put $g_v=1$ for $v\notin I_x$ and recursively $g_z=g_{z-e_{\sigma(z)}}\sigma'_{\tau(z)}$. The gauge transformation g sets every link of T_x to 1 and fixes every link outside V_x, so $g\cdot(\sigma',U)=[\psi,U]$ with $\psi=(g\cdot\sigma')|_{V_x\setminus T_x}$. Since g depends only on σ'_{T_x}, for fixed σ'_{T_x} the map $\sigma'_{V_x\setminus T_x}\mapsto\psi$ multiplies each coordinate on both sides by fixed elements, so $\sigma'\mapsto(\psi,\sigma'_{T_x})$ preserves Haar measure. Plaquette variables change by conjugation and φ is a class function, so the density of $\gamma^0_{V_x}(\cdot\mid U)$ at σ' is proportional to $\prod_{p\cap V_x\ne\emptyset}\varphi([\psi,U]_p)$, while $f(\sigma',U)=f([\psi,U])$. Integrating out σ'_{T_x} gives the claim. □

A link (b,μ) of V_x is a *face link* if $\hat b_i=0$ for some $i\ne\mu$, that is, if both its end points lie in one lower face $\{\hat v_i=0\}$ of the cube.

Lemma 7.3. (i) *If all four links of a plaquette are free, its link of F is a face link.*
(ii) *Every free link has at most fifteen free partners.*

Proof. (i) Let $\phi = (y, s)$ be the link of F of the plaquette p and $h = y + e_s$. If h were interior, then $\sigma(h) = s$ and $\phi = \tau(h) \in T_x$. Since $y \in Q_x$, $\hat{h}_i = \hat{y}_i \in [0, n-1]$ for $i \neq s$ and $\hat{h}_s \in [1, n]$, so $\hat{h}_s = n$ or $\hat{y}_i = 0$ for some $i \neq s$. If $\hat{h}_s = n$, the link of p parallel to its second direction ν through h has a base point with s-coordinate n and is not in V_x. Hence ϕ is a face link.

(ii) Let $k = (b, \mu)$ be free and $m(p)$ the number of free links of a plaquette $p \ni k$; k has $\sum_{p \ni k}(m(p) - 1)$ free partners. If k is a face link, $\hat{b}_i = 0$ with $i \neq \mu$, the plaquette of k with base point $b - e_i$ in the plane $\{\mu, i\}$ has its other three links based at i-coordinate -1, outside V_x; the other five plaquettes give at most 3 each, 15 in all. Otherwise k has at most $12 + M$ free partners, where M counts the plaquettes $p \ni k$ with $m(p) = 4$, whose link ϕ of F is a face link by (i), so $\phi \neq k$. If $k \in F$ then $M = 0$; let k be active and ϕ lie in the face $\{\hat{v}_i = 0\}$. If ϕ shares an end point v with k, then $v = b$ (for $v = b + e_\mu$ either k is a face link or $\hat{b}_i = -1$), $\hat{b}_i = 0$ and $i = \mu$; there are two links of F at b, each in at most one plaquette of k. If ϕ is opposite to k, it is $(b \pm e_\nu, \mu)$ in a plaquette of the plane $\{\mu, \nu\}$; as k is not a face link, $\nu = i$, the sign is $-$ and $\hat{b}_i = 1$, and by Lemma 3.2(ii) the plane $\{\mu, \nu\}$ is the one plane in which the link of F is opposite to k, so this happens for at most one plaquette. Hence $M \leq 3$. □

For a free link e the conditional law $\gamma_e^T(\cdot \mid \psi, U)$ of ψ_e under $\gamma_{x,U}^T$ has density proportional to $\prod_{p \ni e} \varphi([\psi, U]_p)$. Let C_{ek}^T be the supremum of $W_1(\gamma_e^T(\cdot \mid \psi, U), \gamma_e^T(\cdot \mid \psi', U'))/\delta([\psi, U]_k, [\psi', U']_k)$ over pairs for which $[\psi, U]$ and $[\psi', U']$ differ only at a link k, free or outside V_x; it vanishes unless e and k are partners. For $j \notin V_x$ let $b(V_x, j)$ be the number of links of V_x that are partners of j, and

$$R_x(j) = \sup \frac{W_1^{d_x}(\gamma_{x,U}^T, \gamma_{x,U'}^T)}{\delta(U_j, U_j')}, \qquad d_x(\psi, \psi') = \sum_{e \in V_x \setminus T_x} \delta(\psi_e, \psi_e'),$$

the supremum over configurations that differ only at j.

Lemma 7.4. *If $C_{ek}^T \leq c$ for all free e and all k, with $15c < 1$, then $R_x(j) \leq c\, b(V_x, j)/(1 - 15c)$.*

Proof. Let U, U' differ only at j, let π minimise $\int d_x \, d\pi$ over couplings of $\gamma_{x,U}^T$ and $\gamma_{x,U'}^T$, and let $\varrho(e) = \int \delta(\psi_e, \psi_e') \, d\pi$. Replacing (ψ_e, ψ_e') by a measurably chosen optimal coupling of the two conditional laws of ψ_e [51, Cor. 5.22] gives another coupling, so by minimality and the triangle inequality $\varrho(e) \leq c \sum_k \varrho(k) + c[e \text{ a partner of } j]\delta(U_j, U_j')$, the sum over the free partners k of e. Summing over e and using Lemma 7.3(ii), $(1 - 15c) \sum_e \varrho(e) \leq c\, b(V_x, j)\delta(U_j, U_j')$. □

Lemma 7.5. *For $n \geq 2$, $L \geq 2n + 4$ and a link j, $\sum_{x:\, j \notin V_x} b(V_x, j) = 18n^3 - 3n^2$.*

Proof. Let b be the base point of j and ν its direction. For each $\mu \neq \nu$ the partners in the two plaquettes of the plane $\{\mu, \nu\}$ have base points $b + \Delta$, $\Delta \in \{e_\nu, e_\mu, 0, -e_\mu, -e_\mu, e_\nu - e_\mu\}$. With $z = b - x$ and $Q = [0, n)^4$, a link based at $b + \Delta$ lies in V_x exactly when $z + \Delta \in Q$, and $j \in V_x$ when $z \in Q$. The sum is $\sum_\Delta (n^4 - |Q \cap (Q - \Delta)|)$, which is 0 for $\Delta = 0$, n^3 for $\Delta = \pm e_\mu, e_\nu$ and $2n^3 - n^2$ for $\Delta = e_\nu - e_\mu$, that is $6n^3 - n^2$ for each of the three μ. □

Lemma 7.6. (i) *For $G = \mathrm{SU}(2)$, the chordal metric and $\varphi = e^{2\beta \operatorname{Re} \operatorname{Tr}(\cdot)}$, $C_{ek}^T \leq |\beta| = |\beta_W|/4$.*

(ii) *Let G have semisimple Lie algebra, δ the geodesic distance of $\langle X, Y \rangle = -\operatorname{Tr}(XY)$, and $6N|\beta| < \kappa_G$. Then $C_{ek}^T \leq N|\beta|/(\kappa_G - 6N|\beta|)$.*

Proof. (i) The conditional law of ψ_e is the von Mises–Fisher law ν_b with $b = 4\beta \sum_{p \ni e} \bar{s}_p$, where s_p is the product of the other three variables of p, as in Sec. 4. A partner k enters one s_p, and moving it by δ_k moves $\bar{s}_p$ by δ_k, since left and right multiplication and inversion are isometries; the sharp quarter, Theorem C.4, gives the bound. Links of T_x do not move.

(ii) The conditional law is $\nu \propto e^{-V} d\psi_e$ with $V = -N\beta \sum_{p \ni e} \operatorname{Re}\operatorname{Tr}[\psi, U]_p$; by (5.1) each term has second derivative $\operatorname{Re}\operatorname{Tr}(Y^2 W)$ with $|Y| = |X|$, so $\operatorname{Ric}_G + \operatorname{Hess} V \geq K_1 = \kappa_G - 6N|\beta|$. Lemma 5.5 with one factor and the argument of Lemma 5.6 give $|\operatorname{Cov}_\nu(g,h)| \leq \|\nabla g\|_\infty \|\nabla h\|_\infty / K_1$. Move the partner k along a unit-speed geodesic; the derivative of $\nu_s(g)$ is $-\operatorname{Cov}_{\nu_s}(g, \partial_s V_s)$, and $\partial_s V_s$ comes from the one plaquette containing e and k, with gradient in ψ_e of norm at most $N|\beta|$ by (5.3). For smooth 1-Lipschitz g this gives $|\nu_0(g) - \nu_\delta(g)| \leq N|\beta|\delta/K_1$; convolution on G preserves Lipschitz constants for a bi-invariant metric, and Kantorovich–Rubinstein duality concludes. □

Proposition 7.7. *If $\sum_{x : i \notin V_x} R_x(i) \leq \alpha n^4$ for every link i, with $\alpha < 1$, then $\mathrm{BP}(n, 1-\alpha)$ holds.*

Proof. For a function h let $\operatorname{Lip}(h)$ be the least constant with $|h(U) - h(U')| \leq \operatorname{Lip}(h)\delta(U_i, U_i')$ whenever U, U' differ only at one link i; then $|h(U) - h(U')| \leq \operatorname{Lip}(h) d_E(U, U')$. Let $h \in \mathcal{A}$ be Lipschitz and U, U' differ only at i. If $i \in V_x$, $\mathcal{P}_x h(U) = \mathcal{P}_x h(U')$. If $i \notin V_x$, Lemma 7.2 and an optimal coupling π of $\gamma^T_{x,U}$ and $\gamma^T_{x,U'}$ give

$$|\mathcal{P}_x h(U) - \mathcal{P}_x h(U')| \leq \int |h([\psi, U]) - h([\psi', U'])| \, d\pi \leq \operatorname{Lip}(h)(R_x(i) + 1)\delta(U_i, U_i').$$

Since i lies in n^4 of the cubes, $\operatorname{Lip}(\mathcal{K}h) \leq (1 - \kappa_0)\operatorname{Lip}(h)$ with $\kappa_0 = (1-\alpha)n^4/L^4$, and as $\mathcal{K}$ preserves $\mathcal{A}$, $\operatorname{Var}_\mu(\mathcal{K}^t h) \leq \frac{1}{2}(1-\kappa_0)^{2t}\operatorname{Lip}(h)^2 D^2 |E|^2$. The restriction of $\mathcal{K}$ to the mean-zero part of $\mathcal{A}$ is self-adjoint with $0 \leq \mathcal{K} \leq I$, and $\operatorname{Var}_\mu(\mathcal{K}^t h) = \int \lambda^{2t} d\nu_h$ for its spectral measure ν_h at $h - \mu(h)$, so ν_h lives on $[0, 1-\kappa_0]$. Lipschitz functions are dense in $L^2(\mu)$, and averaging over the compact gauge group, the orthogonal projection onto $\mathcal{A}$, does not increase Lip, because gauge transformations act link by link by isometries; so Lipschitz functions are dense in $\mathcal{A}$, and $\langle f, (I - \mathcal{K}) f \rangle \geq \kappa_0 \operatorname{Var}_\mu(f)$ for real $f \in \mathcal{A}$, which is $\mathrm{BP}(n, 1-\alpha)$. □

Theorem 7.8. *Let $n \geq 2$ and $L \geq 2n + 4$. If $C^T_{ek} \leq c$ for all cubes, free links e and links k, with $15c < 1$, then $\mathrm{BP}(n, 1 - \alpha_n)$ holds whenever*

$$\alpha_n = \frac{c\,(18n - 3)}{(1 - 15c)\, n^2} < 1.$$

In particular:

(i) *for $G = \mathrm{SU}(2)$ and $|\beta_W| < \frac{4}{15}$, with $\alpha_n = |\beta_W|(18n-3)/((4 - 15|\beta_W|)n^2)$;*
(ii) *for $G \subset \mathrm{U}(N)$ with semisimple Lie algebra and $21N|\beta| < \kappa_G$, with $\alpha_n = N|\beta|(18n - 3)/((\kappa_G - 21N|\beta|)n^2)$.*

For fixed β in these ranges α_n decreases to 0, so $\mathrm{BP}(n, \varpi)$ holds with $\varpi \to 1$ as $n \to \infty$, uniformly in the torus.

Proof. Lemmas 7.4 and 7.5 give $\sum_{x : i \notin V_x} R_x(i) \leq c(18n^3 - 3n^2)/(1 - 15c) = \alpha_n n^4$, and Proposition 7.7 applies. Lemma 7.6 gives $c = |\beta_W|/4$ for SU(2), and $c = N|\beta|/(\kappa_G - 6N|\beta|)$ in general, for which $15c < 1$ reads $21N|\beta| < \kappa_G$ and $c/(1-15c) = N|\beta|/(\kappa_G - 21N|\beta|)$. □

For SU(2) the condition $\alpha_n < 1$ reads $|\beta_W|(15n^2 + 18n - 3) < 4n^2$; at $|\beta_W| = \frac{1}{4}$ it holds for $n \geq 18$, and its threshold increases to $\frac{4}{15}$ as $n \to \infty$. At strong coupling the condition is therefore *easier* for large cubes, because the response of a cube to one boundary link is spread over n^4 links while it touches only $O(n^3)$ of them.

A weighted count. For a free link e let $N(e)$ be its set of free partners. The proof of Lemma 7.4 gives, for each free e,

$$\varrho(e) \leq \sum_{k \in N(e)} C^T_{ek} \varrho(k) + C^T_{ej} [e \text{ a partner of } j] \, \delta(U_j, U_j'). \tag{7.1}$$

Lemma 7.9. *Let w be a function on the free links with values in $[w_-, w_+]$, $w_- > 0$, with $\sum_{k \in N(e)} w(k) \leq \lambda w(e)$ for every free e. If $C^T_{ek} \leq c$ for all free e and all k, and $c\lambda < 1$, then $R_x(j) \leq (w_+/w_-)\, c\, b(V_x, j)/(1 - c\lambda)$.*

Proof. Multiply (7.1) by $w(e)$ and sum over e. Since $k \in N(e)$ exactly when $e \in N(k)$, $\sum_e w(e)\varrho(e) \le c\lambda \sum_k w(k)\varrho(k) + c\, w_+ b(V_x, j)\delta(U_j, U_j')$, and $\sum_e \varrho(e) \le w_-^{-1} \sum_e w(e)\varrho(e)$. □

Write a link of V_x as $(x+b, \mu)$ with $b \in [0, n-1]^4$, let $z(e) = \#\{i \ne \mu : b_i = 0\}$ be the number of lower faces containing it, and call it a *foot* if $b_\mu = 0$. On free links put $w = 45$ if $z = 1$ and e is active and not a foot, $w = 47$ if $z = 1$ and $e \in F$ or e is a foot, $w = 48$ if $z = 0$ and e is an active foot, and $w = 40$ otherwise.

Proposition 7.10. *For every $n \ge 2$, every cube and every free link e, $\sum_{k \in N(e)} w(k) \le \frac{72}{5} w(e)$.*

Proof. Work in the coordinates b, so that $(y, \nu) \in F$ when $\sigma(x+y) = \nu$. Let $e = (b, \mu)$. For $\nu \ne \mu$ the other links of the plaquette p_ν^+ of base point b in the plane $\{\mu, \nu\}$ are $P_1 = (b + e_\mu, \nu)$, $P_2 = (b + e_\nu, \mu)$, $P_3 = (b, \nu)$, and those of the plaquette p_ν^- of base point $b - e_\nu$ are $P_4 = (b - e_\nu, \mu)$, $P_5 = (b - e_\nu + e_\mu, \nu)$, $P_6 = (b - e_\nu, \nu)$. If e is active, with $s = \sigma(x+b)$, $t = \sigma(x + b + e_\mu)$ and ρ the fourth direction, the links of F among them are P_3, P_6 for $\nu = s$, P_1, P_5 for $\nu = t$ and P_2, P_4 for $\nu = \rho$: the first four by Lemma 3.2(i) at b and at $b + e_\mu$, and in the plane $\{\mu, \rho\}$ the link of F is opposite to e by Lemma 3.2(ii). If $e \in F$, none of them is in F.

Reduction to the orthant. Call a link with base point in $O = \mathbb{Z}^4_{\ge 0}$ *O-free* if it is active or has $z \ge 1$, extend w to such links by the same rule, and count as partners of e the O-free links of the plaquettes of e with base point in O. If $b_\nu = 0$, the links of p_ν^- other than e lie outside V_x. A link $(y, \mu') \in V_x$ lies in T_x exactly when it is in F and its head is interior, that is, $z = 0$ and $y_{\mu'} \le n - 2$; so free links are O-free except the *exceptional* ones, links of F with $z = 0$ and $y_{\mu'} = n - 1$, of weight 40. An exceptional link ϕ in a plaquette p is its link of F, and p has a link ℓ based at μ'-coordinate n, which lies outside V_x, is active, has its base point in O (as $z(\phi) = 0$), and has $w(\ell) \ge 40$; exchanging ϕ for ℓ shows that the orthant sum dominates the sum over $N(e)$ for non-exceptional e. An exceptional e has at most two partners in V_x in each plaquette, each of weight at most 48, and $12 \cdot 48 = 576 = \frac{72}{5} \cdot 40$.

The orthant count. Fix $\nu \ne \mu$, put $f = [b_\mu = 0]$, $\beta = b_\nu$, and let z' be the number of zeros among b_i, $i \notin \{\mu, \nu\}$. From the base points, $z(P_1) = z(P_2) = z(P_5) = z'$, $z(P_3) = z(P_6) = z' + f$, $z(P_4) = z' + [\beta = 1]$; P_2, P_4 are feet when $f = 1$, P_1, P_3 when $\beta = 0$, P_5, P_6 when $\beta = 1$; and p_ν^- counts only when $\beta \ge 1$. An O-free link has weight 40 if $z \ge 2$, 47 or 45 if $z = 1$ according as it is in F or a foot or neither, and 48 or 40 if $z = 0$ (then it is active) according as it is a foot or not; a link of F with $z = 0$ contributes nothing. Let $C(\text{role}, f, \beta, z')$ be the resulting weight of the O-free links among $P_1, \dots, P_6$, the role of ν being s, t, ρ, or F when $e \in F$. For $z' = 2$, $C = 120$ if $\beta = 0$ and $C = 240$ otherwise; for $z' \le 1$ the values are

		$z'=0$, $\beta=$			$z'=1$, $\beta=$		
role	f	0	1	≥ 2	0	1	≥ 2
s	0	88	173	160	139	271	274
s	1	143	277	270	134	259	264
t	0	88	173	160	139	271	274
t	1	95	187	186	134	261	268
ρ	0	96	223	160	141	271	274
ρ	1	95	227	170	134	259	264
F	0	136	261	240	139	269	270
F	1	143	275	266	134	259	264

For instance, for the role s, $f = 1$, $\beta = 1$, $z' = 0$ the weights of $P_1, \dots, P_6$ are $40, 48, 47, 47, 48, 47$. The orthant sum is $\sum_{\nu \ne \mu} C(\text{role}(\nu), f, b_\nu, z'_\nu)$, where $z'_\nu = z(e) - 1$ if $b_\nu = 0$ and $z'_\nu = z(e)$ otherwise, and $\frac{72}{5} w(e)$ is 576, 648, 676.8 or 691.2 for $w(e) = 40, 45, 47, 48$. For active e with $z = 0$ the sum is at most $173 + 173 + 223 = 569$ ($f = 0$, $w = 40$) or $277 + 187 + 227 = 691$ ($f = 1$, $w = 48$). For active e with $z = 1$ it is at most

$96+274+274=644$ ($f=0$, $w=45$), and for $f=1$, $w=47$, at most $143+268+264$, $95+264+264$ or $95+264+268$ as the zero direction has role s, t or ρ, all at most 675. For active e with $z=2$ it is at most $141+139+240=520$, for $z=3$ it is 360, with $w=40$. For $e\in F$ (then $z\ge 1$) it is at most $136+270+270=676$ or $143+264+264=671$ when $z=1$, $w=47$; at most 518 when $z=2$; and 360 when $z=3$, with $w=40$. □

Theorem 7.11. *Let $n\ge 2$ and $L\ge 2n+4$. If $C^T_{ek}\le c$ for all cubes, free links e and links k, with $\frac{72}{5}c<1$, then $\mathrm{BP}(n,1-\alpha_n)$ holds whenever $\alpha_n=6c(18n-3)/((5-72c)n^2)<1$. In particular:*

(i) *for $G=\mathrm{SU}(2)$ and $|\beta_W|<\frac{5}{18}$, with $\alpha_n=3|\beta_W|(18n-3)/(2(5-18|\beta_W|)n^2)$;*
(ii) *for $G=\mathrm{SU}(3)$, the Hilbert–Schmidt distance and $0\le\beta_W\le 0.45$.*

Proof. Lemma 7.9 with Proposition 7.10, $\lambda=\frac{72}{5}$ and $w_+/w_-=\frac{6}{5}$, and Lemma 7.5 give $\sum_{x:\,i\notin V_x}R_x(i)\le\frac{6}{5}c(18n^3-3n^2)/(1-\frac{72}{5}c)=\alpha_n n^4$, and Proposition 7.7 applies; its proof uses only that δ is bi-invariant and invariant under inversion. For (i), $c=|\beta_W|/4$ by Lemma 7.6. For (ii) let $r=2\beta_W$, $Z_*(r)=\int_{\mathrm{SU}(3)}e^{r\,\mathrm{Re}\,\mathrm{Tr}\,U}dU$, $\epsilon(r)=1-5e^{-r/5}/(9Z_*(r))$ and $\gamma(r)=\frac{283}{128}-\frac{r}{2}$. The conditional law of a free link is $\nu_b\propto e^{\mathrm{Re}\,\mathrm{Tr}(Ub)}dU$ with b a sum of $N\beta=r/6$ times six products of elements of SU(3), the links of T_x being 1, so the one-link estimate of the SU(3) theorem of [8] applies and gives $c=r\sqrt{\epsilon(r)}/(6\gamma(r))$, increasing in r. With the moments $m_k=\int(\mathrm{Re}\,\mathrm{Tr}\,U)^k dU$, $(m_0,\dots,m_8)=(1,0,\frac12,\frac14,\frac34,\frac{15}{16},\frac{65}{32},\frac{231}{64},\frac{973}{128})$, and $|m_k|\le 3^{k-8}m_8$ for $k\ge 8$,

$$Z_*(r)\le\sum_{k=0}^{8}\frac{m_k r^k}{k!}+\frac{m_8}{3^8}\frac{(3r)^9}{9!}\Big(1-\frac{3r}{10}\Big)^{-1}.$$

At $r=0.9$ the sum is at most 1.2599146 and the tail at most $3.4\cdot10^{-5}$, so $Z_*(0.9)\le 1.2599486$; with $e^{-0.18}\ge 1-0.18+0.0162-0.000972$ this gives $\epsilon(0.9)\le 0.63172<0.7949^2$, and with $\gamma(0.9)=\frac{1127}{640}$, $\frac{72}{5}c\le 2.16\cdot 0.7949\cdot 640/1127\le 0.9751$. □

For fixed n the threshold of (i) exceeds that of Theorem 7.8(i) exactly when $n^2-6n+1>0$, that is for $n\ge 6$; both theorems hold. The same arithmetic at $r=0.88$, where $Z_*\le 1.246559$, $\epsilon\le 0.62627$ and $\gamma=\frac{5667}{3200}$, gives $15c\le 0.9832$, so Theorem 7.8 holds for SU(3) with $0\le\beta_W\le 0.44$.

Without the gauge fixing every link has eighteen partners, and the same argument needs $18c<1$, that is $|\beta_W|<\frac29$ for SU(2). No choice of gauge-fixed links and weights goes much further:

Proposition 7.12. *Let $T\subset V_x$ have at most $(n-1)^4$ links and A be the partner matrix of $V_x\setminus T$. Then $\lambda_{\max}(A)\ge 6r(4-r)(3-r)/(4-r^2)$ with $r=(1-1/n)^2$, which tends to 12. If $c\lambda_{\max}(A)\ge 1$, the inequalities $0\le\varrho\le cA\varrho+s$ used in Lemma 7.4 admit solutions with $\sum_e\varrho(e)$ arbitrarily large, so for SU(2) the argument cannot pass $|\beta_W|=\frac13$.*

Proof. The $P=6n^2(n-1)^2=6rn^4$ plaquettes with all links in V_x contain at most $6|T|=aP$ links of T in all, $a\le r$, so their numbers $m(p)$ of links outside T have mean at least $4-a\ge 3$; by convexity $\mathbf{1}^{\top}A\mathbf{1}\ge\sum_p m(p)(m(p)-1)\ge P(4-a)(3-a)$, and the Rayleigh quotient of $\mathbf{1}$ is at least $6r(4-a)(3-a)/(4-ar)$, which decreases in a because $r/(4-ar)\le 1/(4-a)$. For a Perron vector $v\ge 0$ of A, $\varrho=tv$ satisfies $cA\varrho\ge\varrho$ for every $t>0$. □

8. Stars: the cube condition for SU(2) past $\beta_W=\frac13$

Proposition 7.12 shows that no argument built on single-link updates and gauge-fixed links passes $\beta_W=\frac13$ for SU(2). This section replaces single links by *stars*, the eight links at a vertex of one parity, and passes that value: the cube condition holds uniformly in the cube size for $|\beta_W|\le\frac{7}{20}$ (Theorem 8.14), and a block condition giving a lattice gap holds for $|\beta_W|\le\frac{9}{25}$ (Theorem 8.5). The gain is visible at first order: the perturbation of one plaquette at a vertex enters the law of the star only through the relative variable of two of

its links, so a star responds with the coefficient c where two single links respond with $2c$, and no plaquette is internal to a star.[3]

Setting. Throughout the section $G = \mathrm{SU}(2)$, identified with the unit quaternions $S^3 \subset \mathbb{R}^4$, with the Euclidean inner product $\langle \cdot, \cdot \rangle$ of $\mathbb{R}^4$ and the chordal metric $\delta(U, V) = |U - V|$, which is bi-invariant, invariant under inversion, and of diameter 2. For unit quaternions $\operatorname{Re}\operatorname{Tr} U = 2\langle U, 1\rangle$, and for unit p, q and $x, y \in \mathbb{R}^4$, $\langle px, y\rangle = \langle x, \bar{p}y\rangle$, $\langle xq, y\rangle = \langle x, y\bar{q}\rangle$ and $|pxq| = |x|$. The Wilson measure is $\mu(dU) = Z^{-1} \prod_p e^{2\beta \operatorname{Re}\operatorname{Tr} U_p} \prod_e dU_e$ on Λ_L, L even, with $\beta_W = 4\beta$, and $\mathcal{A}$, V_x, $\mathcal{P}_x$ and $\mathrm{BP}(n, \varpi)$ are as in Sec. 6. For a set V of links, $\mathcal{P}_V$ is the conditional expectation given the links outside V. For $b \in \mathbb{R}^4$, $\nu_b(dx) \propto e^{\langle x, b\rangle} dx$ is the von Mises–Fisher law on S^3, and the sharp quarter (Theorem C.4 in Appendix C) states

$$W_1(\nu_b, \nu_{b'}) \le \theta\, |b - b'| \quad \text{for all } b, b' \in \mathbb{R}^4, \qquad \theta = \tfrac{1}{4}, \tag{8.1}$$

with W_1 the Kantorovich distance for δ. This is the only property of ν_b used in the section. We put

$$c = 4\theta|\beta| = |\beta| = \tfrac{1}{4}|\beta_W|.$$

Every argument below uses only (8.1), and with a constant θ in place of $\frac{1}{4}$ the ranges become $4\theta|\beta| \le \frac{7}{80}$ and $4\theta|\beta| \le \frac{9}{100}$.

8.1. **Even stars and two-paths.** Call a vertex *even* if the sum of its coordinates is even. For an even vertex v the *star* B_v is the set of the eight links with an end point at v. Write the eight directions as $\pm 1, \dots, \pm 4$, and for the link of B_v in direction σ let X_σ be its parallel transporter from v outwards: $X_{+\mu} = U_{(v,\mu)}$ and $X_{-\mu} = U^{-1}_{(v-e_\mu,\mu)}$. Two directions are *adjacent* if they are distinct and not opposite; each direction is adjacent to six others, and there are 24 adjacent pairs.

Lemma 8.1. *Let $L \ge 4$ be even.*

(i) *Every link has exactly one even end point, so the stars partition E.*

(ii) *Every plaquette has exactly two even corners v, v', which are opposite corners, and two of its links lie in B_v, two in $B_{v'}$. The plaquettes with corner v correspond bijectively to the adjacent pairs $\{\sigma, \tau\}$: the plaquette $p_{\sigma\tau}$ has corners v, $v + \sigma$, $v + \sigma + \tau$, $v + \tau$.*

(iii) *For such a plaquette, $\operatorname{Re}\operatorname{Tr} U_p = 2\langle h_p, X_\sigma^{-1} X_\tau\rangle$, where the* two-path *$h_p$ is the transporter from $v + \sigma$ to $v + \tau$ through the opposite even corner $v' = v + \sigma + \tau$, the product of the two links of p in $B_{v'}$, suitably inverted. h_p is unchanged by a gauge transformation at v', and changing one of its two links by η in δ changes h_p by exactly η.*

(iv) *The conditional law of $(X_\sigma)_\sigma$ given the links outside B_v is*

$$\gamma_{v,h}(dX) = \frac{1}{Z_v} \prod_{\{\sigma,\tau\}} e^{4\beta\langle h_{\sigma\tau}, X_\sigma^{-1} X_\tau\rangle} \prod_\sigma dX_\sigma,$$

with one ordering fixed in each adjacent pair. It depends on the configuration only through the 24 two-paths $h = (h_{\sigma\tau})$, and it is invariant under $X_\sigma \mapsto gX_\sigma$, the gauge transformation at v.

(v) *A link i with even end point v_i lies in exactly six plaquettes; their even corners other than v_i are six distinct even vertices, and for each of them, v, the link i is a factor of exactly one two-path h_p with p at v.*

Proof. (i) The end points of a link differ by a unit vector, so their parities differ. (ii) The corners of a plaquette are y, $y + e_\mu$, $y + e_\mu + e_\nu$, $y + e_\nu$, of parities $\epsilon, \epsilon + 1, \epsilon, \epsilon + 1$, and each link joins consecutive corners. A plaquette with even corner v is spanned by two adjacent directions at v, and conversely; since $L \ge 4$ these 24 squares are distinct. (iii) Going around p

[3] A Dobrushin certificate bounds the response of a block to a change of its boundary by a matrix of local influences, and contraction means that this matrix has spectral radius below one. Grouping links into blocks changes the matrix: interactions inside a block are treated exactly instead of being bounded.

from v, U_p is conjugate to $X_\sigma h_p X_\tau^{-1}$ or to its inverse, and $\operatorname{Re}\operatorname{Tr}$ is a class function, invariant under inversion on $\mathrm{SU}(2)$; so $\operatorname{Re}\operatorname{Tr} U_p = \operatorname{Re}\operatorname{Tr}(X_\sigma h_p X_\tau^{-1}) = 2\langle X_\sigma h_p, X_\tau\rangle = 2\langle h_p, X_\sigma^{-1}X_\tau\rangle$. A gauge transformation g at v' multiplies the first factor of h_p on the right by g^{-1} and the second on the left by g, and multiplication by unit quaternions is an isometry. (iv) The plaquettes containing a link of B_v are those with corner v, by (ii), and their weights are given by (iii); the gauge transformation at v changes no two-path. (v) If i joins v_i to $w = v_i + \sigma$, its plaquettes are spanned by σ and the six directions $\tau \perp \sigma$, with opposite even corners $v_i + \sigma + \tau$, which are distinct since $L \geq 4$; in the plaquette with corner v, the link i is one of the two links at v_i, hence a factor of the two-path. □

For the star at v define the quotient pseudometric

$$\bar{d}_v(X, X') = \min_{g\in G} \sum_\sigma \delta(gX_\sigma, X'_\sigma).$$

8.2. **The seven-spin system.** Fix an adjacent pair $\{a, b\}$, let S_a be the set of the seven directions other than a, A_7 its adjacency matrix and e_b the indicator of b. In S_a, the direction b is adjacent to $\bar{a} = -a$ and to the four directions C of the other two axes; $\bar{a}$ is adjacent to b, $\bar{b} = -b$ and C; $\bar{b}$ to $\bar{a}$ and C; and each element of C to b, $\bar{a}$, $\bar{b}$ and the two elements of C on the other axis. The largest degree is 6. For $0 \leq c < \frac{1}{6}$ put

$$s(c) = \mathbf{1}^{\mathsf{T}}(I - cA_7)^{-1}e_b = \sum_{k\geq 0} c^k\, \mathbf{1}^{\mathsf{T}} A_7^k e_b,$$

a nondecreasing function of c with $s(c) \geq 1$. The symmetries fixing a and b permute C transitively, so $x = (I - cA_7)^{-1}e_b$ is constant, equal to z, on C, and $(x_b, x_{\bar{a}}, x_{\bar{b}}, z)$ is the unique solution of

$$\begin{aligned} x_b &= 1 + c(x_{\bar{a}} + 4z), \qquad & x_{\bar{a}} &= c(x_b + x_{\bar{b}} + 4z),\\ x_{\bar{b}} &= c(x_{\bar{a}} + 4z), & z &= c(x_b + x_{\bar{a}} + x_{\bar{b}} + 2z), \end{aligned} \tag{8.2}$$

with $s(c) = x_b + x_{\bar{a}} + x_{\bar{b}} + 4z$.

Lemma 8.2. (i) $s(\frac{1}{12}) = \frac{26}{15}$, *with* $(x_b, x_{\bar{a}}, x_{\bar{b}}, z) = \frac{1}{210}(221, 28, 11, 26)$.
(ii) $s(\frac{7}{80}) = \frac{3480}{1933}$, *with* $(x_b, x_{\bar{a}}, x_{\bar{b}}, z) = \frac{1}{181702}(192531, 26320, 10829, 24360)$.
(iii) $s(\frac{9}{100}) = \frac{5450}{2957}$, *with* $(x_b, x_{\bar{a}}, x_{\bar{b}}, z) = \frac{1}{348926}(371363, 53100, 22437, 49050)$.
Hence $6c\,s(c)$ *equals* $\frac{13}{15}$ *at* $c = \frac{1}{12}$, $\frac{1827}{1933}$ *at* $c = \frac{7}{80}$ *and* $\frac{2943}{2957}$ *at* $c = \frac{9}{100}$, *and* $6c\,s(c) \leq \frac{2943}{2957}$ *for* $0 \leq c \leq \frac{9}{100}$.

Proof. Since $c\lambda_{\max}(A_7) \leq 6c < 1$, the matrix $I - cA_7$ is invertible and it suffices to check (8.2). (i) Multiplied by $12 \cdot 210$ the four equations read $12 \cdot 221 - (28 + 104) = 2520$, $12 \cdot 28 - (221 + 11 + 104) = 0$, $12 \cdot 11 - 132 = 0$ and $12 \cdot 26 - (221 + 28 + 11 + 52) = 0$; the sum is $364/210 = 26/15$. (ii) Multiplied by $80 \cdot 181702 = 14536160$:

$$\begin{aligned} 80 \cdot 192531 - 7(26320 + 4 \cdot 24360) &= 15402480 - 866320 = 14536160,\\ 80 \cdot 26320 - 7(192531 + 10829 + 4 \cdot 24360) &= 2105600 - 7 \cdot 300800 = 0,\\ 80 \cdot 10829 - 7(26320 + 4 \cdot 24360) &= 866320 - 7 \cdot 123760 = 0,\\ 80 \cdot 24360 - 7(192531 + 26320 + 10829 + 2 \cdot 24360) &= 1948800 - 7 \cdot 278400 = 0. \end{aligned}$$

The sum is $327120/181702 = 3480/1933$, as $327120 = 94 \cdot 3480$ and $181702 = 94 \cdot 1933$. (iii) Multiplied by $100 \cdot 348926$: $100 \cdot 371363 - 9(53100 + 196200) = 34892600$, $100 \cdot 53100 - 9(371363 + 22437 + 196200) = 5310000 - 9 \cdot 590000 = 0$, $100 \cdot 22437 - 9(53100 + 196200) = 0$ and $100 \cdot 49050 - 9(371363 + 53100 + 22437 + 98100) = 4905000 - 9 \cdot 545000 = 0$. The sum is $643100/348926 = 5450/2957$, as $348926 = 118 \cdot 2957$ and $643100 = 118 \cdot 5450$. The last claim follows from the monotonicity of s. □

Lemma 8.3 (one star)**.** *Assume* (8.1) *and* $c < \frac{1}{6}$*. Let* h, h' *be two families of two-paths at* v *that differ only at one adjacent pair* $\{a, b\}$*. Then*

$$W_1^{\bar{d}_v}(\gamma_{v,h}, \gamma_{v,h'}) \le c\, s(c)\, |h_{ab} - h'_{ab}|.$$

Proof. Under $\gamma_{v,h}$ put $Z_\sigma = X_a^{-1} X_\sigma$. The map $X \mapsto (X_a, Z)$ preserves Haar measure, and by Lemma 8.1(iv) the density depends only on Z, with $Z_a = 1$; so X_a is Haar distributed and independent of $Z_S = (Z_\sigma)_{\sigma \in S_a}$, whose law $\mu_h^{(a)}$ has density proportional to $\prod e^{4\beta \langle h_{\sigma\tau}, Z_\sigma^{-1} Z_\tau \rangle}$. Given a coupling of $\mu_h^{(a)}$ and $\mu_{h'}^{(a)}$, take $X_a = X'_a$ Haar and independent, $X_\sigma = X_a Z_\sigma$ and $X'_\sigma = X_a Z'_\sigma$; this couples $\gamma_{v,h}$ and $\gamma_{v,h'}$, and with $g = 1$ in the definition, $\bar{d}_v(X, X') \le \sum_{\sigma \in S_a} \delta(Z_\sigma, Z'_\sigma)$. It suffices to bound the Kantorovich distance of $\mu_h^{(a)}$ and $\mu_{h'}^{(a)}$ for the cost $\sum_{\sigma \in S_a} \delta$.

For $\sigma \in S_a$ the conditional law of Z_σ given the others is ν_{B_σ} with $B_\sigma = 4\beta \sum_{\tau \text{ adj. } \sigma} w_{\sigma\tau}$, where $w_{\sigma\tau} = Z_\tau \bar{h}_{\sigma\tau}$ or $Z_\tau h_{\tau\sigma}$ according to the ordering of the pair, since $\langle h_{\sigma\tau}, Z_\sigma^{-1} Z_\tau \rangle = \langle Z_\sigma, Z_\tau \bar{h}_{\sigma\tau} \rangle$; changing Z_τ by η changes $w_{\sigma\tau}$ by η, and changing h_{ab} changes only B_b, by at most $4|\beta||h_{ab} - h'_{ab}|$. Let π minimise $\int \sum_\sigma \delta(Z_\sigma, Z'_\sigma)\, d\pi$ over couplings of $\mu_h^{(a)}$ and $\mu_{h'}^{(a)}$ (one exists by compactness) and put $\varrho_\sigma = \int \delta(Z_\sigma, Z'_\sigma)\, d\pi$. Replacing (Z_σ, Z'_σ) by a measurably chosen optimal coupling of the two conditional laws [51, Cor. 5.22] gives another coupling that changes only the σ-term of the cost, so by minimality and (8.1)

$$\varrho_\sigma \le \theta \int |B_\sigma - B'_\sigma|\, d\pi \le c \sum_{\tau \in S_a,\ \tau \text{ adj. } \sigma} \varrho_\tau + c\, |h_{ab} - h'_{ab}|\, [\sigma = b],$$

that is, $\varrho \le cA_7 \varrho + c|h_{ab} - h'_{ab}| e_b$ componentwise. Iterating, $\varrho \le \sum_{k<K} (cA_7)^k c|h_{ab} - h'_{ab}| e_b + (cA_7)^K \varrho$, and $(cA_7)^K \to 0$ since the spectral radius of cA_7 is at most $6c < 1$. Summing the components gives $\sum_\sigma \varrho_\sigma \le c\, s(c) |h_{ab} - h'_{ab}|$. □

8.3. **The star condition and a lattice gap up to** $\beta_W = \frac{9}{25}$**.** Let $\mathcal{P}_v = \mathcal{P}_{B_v}$ and let $N_s = L^4/2$ be the number of even vertices. For a function h on G^E let $\mathrm{Lip}(h)$ be the least constant with $|h(U) - h(U')| \le \mathrm{Lip}(h) \delta(U_i, U'_i)$ whenever U, U' differ only at one link i; changing links one at a time, $|h(U) - h(U')| \le \mathrm{Lip}(h) \sum_e \delta(U_e, U'_e)$.

Proposition 8.4 (star condition)**.** *Assume* (8.1) *and put* $\gamma = c\, s(c)$*, with* $c < \frac{1}{6}$ *and* $6\gamma < 1$*. Then for every real* $f \in \mathcal{A}$

$$\sum_{v \text{ even}} \mu\big((f - \mathcal{P}_v f)^2\big) \ge (1 - 6\gamma)\, \mathrm{Var}_\mu(f).$$

Proof. Let $h \in \mathcal{A}$ be Lipschitz, gauge invariant at every configuration, and let U, U' differ only at i. If $i \in B_v$, then $\mathcal{P}_v h(U) = \mathcal{P}_v h(U')$. Otherwise let π be a coupling of $\gamma_v(\cdot \mid U)$ and $\gamma_v(\cdot \mid U')$ minimising the integral of $\bar{d}_v$, a continuous pseudometric on a compact space. For $\sigma, \sigma' \in G^{B_v}$ and $g \in G$ the gauge transformation g at v changes only the links of B_v, so by gauge invariance $h(\sigma', U') = h(g\sigma', U')$ and

$$|h(\sigma, U) - h(\sigma', U')| \le \mathrm{Lip}(h) \Big(\sum_{e \in B_v} \delta(\sigma_e, (g\sigma')_e) + \delta(U_i, U'_i) \Big);$$

the infimum over g gives $\mathrm{Lip}(h)(\bar{d}_v(\sigma, \sigma') + \delta(U_i, U'_i))$. By Lemma 8.1(iv), (v), the laws $\gamma_v(\cdot \mid U)$ and $\gamma_v(\cdot \mid U')$ differ only through the two-path of one plaquette, which moves by $\delta(U_i, U'_i)$, if v is one of the six vertices of Lemma 8.1(v), and coincide otherwise. Lemma 8.3 gives

$$|\mathcal{P}_v h(U) - \mathcal{P}_v h(U')| \le \mathrm{Lip}(h)\, \big(1 + \gamma\, [v \text{ one of the six}]\big)\, \delta(U_i, U'_i).$$

For $\mathcal{K}_s = N_s^{-1} \sum_v \mathcal{P}_v$, summing over v (exactly one star contains i) gives $\mathrm{Lip}(\mathcal{K}_s h) \le (1 - \kappa_0)\, \mathrm{Lip}(h)$ with $\kappa_0 = (1 - 6\gamma)/N_s$. Each $\mathcal{P}_v$ maps $\mathcal{A}$ to itself, by gauge covariance of γ_v, and for a Lipschitz $g \in \mathcal{A}$, $\mathrm{Var}_\mu(g) \le \frac{1}{2} \iint (g(U) - g(U'))^2\, \mu(dU) \mu(dU') \le 2\, \mathrm{Lip}(g)^2 |E|^2$. So $\mathrm{Var}_\mu(\mathcal{K}_s^t h) \le 2(1 - \kappa_0)^{2t}\, \mathrm{Lip}(h)^2 |E|^2$. On the mean-zero part of $\mathcal{A}$ the operator $\mathcal{K}_s$ is

self-adjoint with $0 \le \mathcal{K}_s \le I$, and $\mathrm{Var}_\mu(\mathcal{K}_s^t h) = \int \lambda^{2t}\, d\nu_h(\lambda)$ for its spectral measure ν_h at $h - \mu(h)$; hence ν_h lives on $[0, 1-\kappa_0]$. Lipschitz functions are dense in $L^2(\mu)$, and the orthogonal projection onto $\mathcal{A}$, averaging over the compact gauge group, which acts link by link by isometries, does not increase Lip; so such h are dense in $\mathcal{A}$, and the spectral projection of $\mathcal{K}_s$ on $(1-\kappa_0, 1]$ vanishes on the mean-zero part of $\mathcal{A}$. Thus $\langle f, (I-\mathcal{K}_s) f\rangle \ge \kappa_0 \,\mathrm{Var}_\mu(f)$, and $\langle f, (I-\mathcal{P}_v) f\rangle = \mu((f-\mathcal{P}_v f)^2)$. □

Theorem 8.5. *Let $G = \mathrm{SU}(2)$ and $|\beta_W| \le \frac{9}{25}$. Then for every even $L \ge 4$ and every real $f \in \mathcal{A}$,*

$$\sum_{v \text{ even}} \mu((f-\mathcal{P}_v f)^2) \ge \frac{14}{2957}\,\mathrm{Var}_\mu(f),$$

and for every $n \ge 2$ and even $L \ge 2n+4$, $\mathrm{BP}(n, \varpi_n)$ holds with

$$\varpi_n = \frac{14}{2957}\cdot\frac{(n-1)^4}{n^4\lceil (n-1)^4/2\rceil}, \qquad \varpi_2 = \frac{14}{16\cdot 2957}.$$

For $0 \le \beta_W \le \frac{9}{25}$ every torus limit is translation invariant and reflection positive, and its transfer matrix has spectral gap at least $\varpi_2/(2e^2 q_2) = 7/(625\cdot 2957\, e^2)$.

Proof. Here $c = |\beta| \le \frac{9}{100}$, and by Lemma 8.2 $1 - 6c\, s(c) \ge \frac{14}{2957}$; the first claim is Proposition 8.4. For a cube V_x, an even vertex v with $v - x \in [1, n-1]^4$ has its eight links based in $x + [0,n)^4$, so $B_v \subset V_x$. The σ-algebra of the links outside V_x is contained in that of the links outside B_v, so $\mu((f-\mathcal{P}_x f)^2) = \mu(f^2) - \mu((\mathcal{P}_x f)^2) \ge \mu(f^2) - \mu((\mathcal{P}_v f)^2) = \mu((f-\mathcal{P}_v f)^2)$. Each cube contains at most $N = \lceil (n-1)^4/2\rceil$ such vertices, and each even v is in this position in exactly $(n-1)^4$ cubes. Averaging over those v in each cube,

$$\sum_x \mu((f-\mathcal{P}_x f)^2) \ge \frac{(n-1)^4}{N}\sum_{v \text{ even}} \mu((f-\mathcal{P}_v f)^2) \ge \frac{(n-1)^4}{N}\cdot\frac{14}{2957}\,\mathrm{Var}_\mu(f),$$

which is $\mathrm{BP}(n, \varpi_n)$. The last claim is Corollary 6.4 with $n = 2$ and $q_2 = \frac{625}{16}$. □

The same proof works whenever $6c\, s(c) < 1$. The exact threshold is the root c_* of $6c\, s(c) = 1$, and Lemma 8.2(iii) shows $c_* > \frac{9}{100}$.[4] At $\beta_W = \frac{1}{3}$ the star condition holds with $6c\, s(c) = \frac{13}{15}$, while the single-link argument of Sec. 7 is exactly marginal there. The constant ϖ_n of Theorem 8.5 decays like $\frac{28}{2957} n^{-4}$, so that theorem is a cube condition for each fixed n, enough for the lattice gap but not for Theorem B. The lower faces of a cube are the reason, and the next proposition shows that they stop every certificate that treats their links one at a time.

Proposition 8.6 (the face barrier)**.** *For a lower face $\{\hat v_i = 0\}$ of the cube V_x, $\hat v = v - x$, let F_i be the set of links $(b,\mu) \in V_x$ with $\hat b_i = 0$ and $\mu \ne i$, and let A_F be the matrix on F_i with $(A_F)_{ek} = 1$ when $e \ne k$ lie in a common plaquette all of whose links are in F_i. Then $\lambda_{\max}(A_F) \ge 12(1-\frac{1}{n})^2$. Consequently, if a certificate for the response of a cube treats the links of F_i as single sites with influence matrix $C \ge cA_F$ on $F_i \times F_i$, the inequalities $0 \le \varrho \le C\varrho + \mathfrak{s}$ admit solutions with $\sum \varrho$ arbitrarily large as soon as $12c(1-\frac{1}{n})^2 \ge 1$; as $n \to \infty$ such a certificate cannot pass $c = \frac{1}{12}$, that is $|\beta_W| = \frac{1}{3}$.*

Proof. $|F_i| = 3n^3$. A plaquette in a plane $\{\mu,\nu\} \not\ni i$ with base point y, $\hat y_i = 0$, has all four links in F_i exactly when its three base points lie in $x + [0,n)^4$, that is $\hat y_\mu, \hat y_\nu \le n-2$; there are $3n(n-1)^2$ of them. Two links share at most one plaquette, so $\mathbf{1}^{\mathsf T} A_F \mathbf{1} = 12\cdot 3n(n-1)^2$, and the Rayleigh quotient of $\mathbf{1}$ is $12(1-\frac{1}{n})^2$. If $c\lambda_{\max}(A_F) \ge 1$ and $w \ge 0$ is a Perron vector of A_F, extended by 0, then $\varrho = tw$ satisfies $C\varrho \ge tcA_F w \ge \varrho$ on F_i for every $t > 0$. □

[4] Solving (8.2) numerically gives $c_* \approx 0.09023$, that is $\beta_W \approx 0.3609$. No statement of the paper uses this value.

For single-link updates the hypothesis $C \geq cA_F$ holds with $c = |\beta|$: at $b = 0$ the derivative of $b \mapsto \nu_b$ has Kantorovich norm exactly $\frac{1}{4}$ (Lemma C.3(i)), and the staples of a face link can cancel while one of them moves. The forest of Sec. 7 does not help, since its links T_x have interior heads and the face links are free. What does help is to let the gauge transformation act at the vertices of the faces as well, and to pay for it.

8.4. **The cube as a block system.** Fix x and $n \geq 2$, and write $\hat{v} = v - x$, computed in a window $x + [-2, n+2]^4$ that embeds in $\mathbb{Z}^4$ because $L \geq 2n + 4$. Let $V = V_x$ and $Q = \{v : \hat{v} \in [0, n-1]^4\}$. Call v *lower* if $v \in Q$, with zero set $Z(v) = \{a : \hat{v}_a = 0\}$ and $z(v) = |Z(v)|$, and *upper* if $\hat{v} \in [0, n]^4$ has exactly one coordinate equal to n. If k is a link with odd end point o and direction μ, its six plaquettes are indexed by (ν, ϵ), $\nu \neq \mu$, $\epsilon = \pm 1$; the even corner of the plaquette (ν, ϵ) other than the even end point of k is $o + \epsilon e_\nu$, and there k is a factor of the two-path of this plaquette only (Lemma 8.1(v)).

Lemma 8.7. *Let $n \geq 2$.*

(i) *If v is lower and even, $B_v \cap V$ consists of B_v minus the $z(v)$* outside links *$(v - e_a, a)$, $a \in Z(v)$; we call the directions $-a$, $a \in Z(v)$,* fixed *and the others* free. *If v is upper and even with $\hat{v}_a = n$, then $B_v \cap V = \{(v - e_a, a)\}$. Other even vertices carry no link of V. Hence V is partitioned into the* blocks *$B_v \cap V$, v even and lower or upper, and the* owner *$w(k)$ of $k \in V$ is its even end point.*

(ii) *Let $k \in V$ have odd end point o, direction μ and owner $o + \varsigma e_\mu$, $\varsigma = \pm 1$, and let p be its plaquette (ν, ϵ), $v = o + \epsilon e_\nu$. The links of p at v are in the directions $-\epsilon\nu$ and $\varsigma\mu$. If v is lower, both lie in V. If v is upper with $\hat{v}_a = n$, the links of p at v other than $(v - e_a, a)$ lie outside V.*

(iii) *Let v be lower and even and $a \in Z(v)$. The outside link $(v - e_a, a)$ is a factor of no two-path h_p at a lower or upper vertex. If a plaquette p at v has one of its links at v outside V, then both links of p at the opposite even corner lie outside V.*

Proof. (i) The outgoing links (v, b) of a lower v have base point in Q, and an incoming link $(v - e_b, b)$ lies in V if and only if $\hat{v}_b \geq 1$. If $\hat{v}_a = n$ for exactly one a, no outgoing link and no incoming link $(v - e_b, b)$, $b \neq a$, has its base point in Q, while $(v - e_a, a)$ has. If $\hat{v} \notin [0, n]^4$ or two coordinates equal n, no link at v has its base point in Q.

(ii) The corners of p are o, $o + \varsigma e_\mu$, $o + \varsigma e_\mu + \epsilon e_\nu$ and v, so the links at v go to o, in direction $-\epsilon\nu$, and to $v + \varsigma e_\mu$, in direction $\varsigma\mu$. Let v be lower. If $\epsilon = +1$ the first link is (o, ν), and $\hat{o} = \hat{v} - e_\nu$ lies in Q, its ν-coordinate being $\hat{o}_\nu \geq 0$; if $\epsilon = -1$ it is (v, ν), with base point $v \in Q$. The second link is (v, μ) if $\varsigma = +1$; if $\varsigma = -1$ it is $(v - e_\mu, \mu)$, which lies in V because $\hat{v}_\mu = \hat{o}_\mu \geq 1$, as $k = (o - e_\mu, \mu) \in V$. If v is upper, (i) gives the claim.

(iii) The plaquettes of $(v - e_a, a)$ have the even corners v and $v - e_a \pm e_\nu$, whose a-coordinate is -1, so these are neither lower nor upper; and a link is a factor of a two-path only at the even corner opposite to its own even end point. If p at v is spanned by $-a$ and τ, the opposite corner $v' = v - e_a + \tau$ and both its neighbours in p, $v - e_a$ and $v + \tau$, are joined to it by links based at a-coordinate -1. □

For a block vertex v the conditional law $\gamma_v(\cdot \mid \sigma, U)$ of the links of $B_v \cap V$, given the other links of V (values σ) and the links off V (values U), has density proportional to the product of the weights $e^{4\beta\langle h_p, X_\sigma^{-1} X_\tau\rangle}$ of the plaquettes p at v with a link at v in V. It depends on (σ, U) only through these two-paths and the values of the links at v off V.

8.5. **The soft quotient.** At a vertex of a lower face the gauge transformation moves the outside links at that vertex, and the conditional law of the cube does not see it. The following pseudometric allows it anyway, at the price of the displacement it causes on those links. The price is legitimate in the Lipschitz contraction argument of Proposition 7.7, because the test function depends on all links.

Definition 8.8. For a lower even v and configurations X, X' of $B_v \cap V$, written as outward transporters, put

$$\tilde d_v(X,X') = \min_{g\in G}\Big[\sum_{\sigma\ \text{free}} \delta(gX_\sigma, X'_\sigma) + z(v)\,\delta(g,1)\Big],$$

and for an upper even v let $\tilde d_v$ be δ on its single link. For $\sigma,\sigma' \in G^V$ let $D(\sigma,\sigma') = \sum_v \tilde d_v(\sigma_{B_v}, \sigma'_{B_v})$, the sum over the block vertices, and for $j \notin V$

$$\tilde R_x(j) = \sup\Big\{ \frac{W^D(\gamma^0_{V,U}, \gamma^0_{V,U'})}{\delta(U_j, U'_j)} : U, U' \text{ differ only at } j \Big\},$$

where $\gamma^0_{V,U} = \gamma^0_V(\cdot \mid U)$ and W^D is the Kantorovich distance for the cost D.

The minimum exists by compactness, and $\tilde d_v$ and D are continuous, symmetric, since $\delta(gX,X') = \delta(X, g^{-1}X')$ and $\delta(g,1) = \delta(g^{-1},1)$, and satisfy the triangle inequality, by composing the minimising elements. For $z(v) = 0$, $\tilde d_v = \bar d_v$.

Lemma 8.9. *Let h be a function on G^E, gauge invariant at every configuration, with $|h(W) - h(W')| \le \mathrm{Lip}(h) \sum_e \delta(W_e, W'_e)$. If U, U' differ only at $j \notin V$, then for all $\sigma, \sigma' \in G^V$*

$$|h(\sigma,U) - h(\sigma',U')| \le \mathrm{Lip}(h)\big(D(\sigma,\sigma') + \delta(U_j, U'_j)\big).$$

Proof. For each lower even v choose g_v attaining $\tilde d_v(\sigma_{B_v}, \sigma'_{B_v})$, and let g be the gauge transformation equal to g_v at these vertices and to 1 elsewhere; then $h(\sigma,U) = h(g\cdot(\sigma,U))$. A link of V has exactly one even end point v, and g acts on it by $X \mapsto g_v X$ on its outward transporter at v, or trivially if v is upper. A link $e \notin V$ has at most one end point where $g \ne 1$, namely an even lower v, and then e is one of the $z(v)$ outside links at v, by Lemma 8.7(i); there $\delta((g\cdot U)_e, U'_e) \le \delta(g_v,1) + \delta(U_e, U'_e)$ by bi-invariance. Summing over all links, $|h(g\cdot(\sigma,U)) - h(\sigma',U')| \le \mathrm{Lip}(h)\big(\sum_v \tilde d_v + \delta(U_j, U'_j)\big)$. □

8.6. **Block responses.**

Lemma 8.10 (Dobrushin on S^3)**.** *Assume (8.1). Let S be finite and ν, ν' probability measures on G^S with continuous positive densities with respect to Haar measure, such that for every $s \in S$ the conditional law of x_s given $x_{S\setminus s}$ is $\nu_{B_s(x)}$ under ν and $\nu_{B'_s(x)}$ under ν', where $|B_s(x) - B_s(y)| \le 4|\beta| \sum_t a_{st}\delta(x_t, y_t)$ and $|B_s(x) - B'_s(x)| \le 4|\beta|\eta_s$ for all x, y, with $a_{st} \ge 0$, $a_{ss} = 0$ and $\eta_s \ge 0$. Then there is a coupling π of ν, ν' whose $\varrho_s = \int \delta(x_s, x'_s)\, d\pi$ satisfy $\varrho \le cA\varrho + c\eta$ componentwise. Consequently:*

(a) *if $\sum_s a_{st} \le d$ for every t and $cd < 1$, then $\sum_s \varrho_s \le c\sum_s \eta_s/(1-cd)$;*
(b) *if the spectral radius of cA is less than 1 and $y \in [0,\infty)^S$ satisfies $y \ge cAy + c\eta$, then $\varrho \le y$;*
(c) *if $y \ge 0$ satisfies $y \ge c_0 Ay + \eta$ and $c \le c_0$, then $cy \ge cA(cy) + c\eta$, so $\varrho \le cy$ when the spectral radius of cA is less than 1.*

Proof. Let π minimise $\int \sum_s \delta(x_s, x'_s)\, d\pi$ over couplings. Fix s and replace under π the pair (x_s, x'_s) by a sample of an optimal coupling of $\nu_{B_s(x)}$ and $\nu_{B'_s(x')}$, chosen measurably in $(x_{S\setminus s}, x'_{S\setminus s})$ [51, Cor. 5.22]. This is again a coupling of ν, ν', and only the s-term of the cost changes, so by minimality, (8.1) and the triangle inequality, $\varrho_s \le \theta \int |B_s(x) - B'_s(x')|\, d\pi \le c\sum_t a_{st}\varrho_t + c\eta_s$. (a) Summing over s, $\sum \varrho \le cd \sum \varrho + c\sum \eta$. (b) The vector $y - \varrho$ satisfies $y - \varrho \ge cA(y-\varrho)$, hence $y - \varrho \ge (cA)^K (y-\varrho) \to 0$. (c) $cy - c^2Ay \ge c(y - c_0 Ay) \ge c\eta$; apply (b). □

From now on $c \le c_0 = \frac{7}{80}$, that is $|\beta_W| \le \frac{7}{20}$, and $\kappa_0 = s(c_0) = \frac{3480}{1933} < 1.8004$, so that $6c_0\kappa_0 = \frac{1827}{1933}$. For a direction a and a family h of two-paths let $\mu_h^{(a)}$ be the law on G^{S_a} with density proportional to $\prod_{\{\sigma,\tau\}} e^{4\beta\langle h_{\sigma\tau}, Z_\sigma^{-1} Z_\tau\rangle}$, $Z_a = 1$, as in the proof of Lemma 8.3. By

that proof, Lemma 8.2(ii) and Lemma 8.10(c), if h, h' differ only at the pair $\{a, b\}$ there is a coupling of $\mu_h^{(a)}$ and $\mu_{h'}^{(a)}$ with

$$\int \sum_{\sigma \in S_a} \delta(Z_\sigma, Z'_\sigma) \le \kappa_0 c \, |h_{ab} - h'_{ab}| \qquad (c \le c_0). \tag{8.3}$$

Lemma 8.11. *Let $c \le c_0$ and v a block vertex, and write $\gamma^{h,Y}$ for its block law as a function of the two-paths h at v and of the values Y of the links at v off V. Put $\kappa_1 = \kappa_0$, $\kappa_2 = \frac{9}{4}$, $\kappa_3 = \frac{40}{13}$ and $\kappa_4 = \frac{160}{59}$.*

(a) *Let v be lower, and let (h, Y) and (h', Y) differ only at the two-path h_p of a plaquette p whose two links at v are free. Then $W^{\tilde{d}_v}(\gamma^{h,Y}, \gamma^{h',Y}) \le \kappa_{z(v)} c \, |h_p - h'_p|$, with κ_0 for $z(v) = 0$.*

(b) *In particular, for $z(v) = 1$ the constant is the full-star constant $\kappa_0 c$.*

(c) *Let v be lower and p a plaquette at v with one link at v free and one fixed. Then the bound of* (a) *holds with $\kappa_0 c$ if $z(v) = 1$ and with $\frac{16}{9} c$ if $z(v) \ge 2$.*

(d) *Let v be lower and let (h, Y) and (h, Y') differ only at one outside link, by η in δ. Then $W^{\tilde{d}_v}(\gamma^{h,Y}, \gamma^{h,Y'}) \le \eta$.*

(e) *Let v be upper. If (h, Y) and (h', Y') differ only at one two-path h_p with $p \ni (v - e_a, a)$, or only at one outside link at v, the distance is at most $c|h_p - h'_p|$, respectively $c\eta$.*

Proof. Representation. Let v be lower and F its set of fixed directions, let $\Phi_h(X) = \prod e^{4\beta \langle h_{\sigma\tau}, X_\sigma^{-1} X_\tau \rangle}$ over all 24 adjacent pairs, so that $\Phi_h(gX) = \Phi_h(X)$, and note that $\gamma^{h,Y}$ has density proportional to Φ_h evaluated at $X_F = Y_F$. Fix a free direction σ_0. If $F = \emptyset$, the map $X \mapsto (X_{\sigma_0}, (X_{\sigma_0}^{-1} X_\rho)_{\rho \ne \sigma_0})$ preserves Haar measure and carries γ^h to Haar $\otimes \, \mu_h^{(\sigma_0)}$, as in the proof of Lemma 8.3. If $F \ne \emptyset$, fix $f_1 \in F$ and let $\Psi(X)$ consist of $W_\rho = X_{\sigma_0}^{-1} X_\rho$ for the free $\rho \ne \sigma_0$ and of $W_m = X_{\sigma_0}^{-1} Y_{f_1}$. Then Ψ is a bijection of the free coordinates, with inverse $X_{\sigma_0} = Y_{f_1} W_m^{-1}$, $X_\rho = Y_{f_1} W_m^{-1} W_\rho$, and it preserves Haar measure: $X_{\sigma_0} \mapsto X_{\sigma_0}^{-1} Y_{f_1}$ does, and for fixed X_{σ_0} so does each $X_\rho \mapsto X_{\sigma_0}^{-1} X_\rho$. Since $\Phi_h(X) = \Phi_h(X_{\sigma_0}^{-1} X)$ and $X_{\sigma_0}^{-1} X_f = W_m K_f$ with $K_f = Y_{f_1}^{-1} Y_f$, the image of $\gamma^{h,Y}$ has density proportional to Φ_h at $X_{\sigma_0} = 1$, $X_\rho = W_\rho$, $X_f = W_m K_f$. If (W, W') couples the images for (h, Y) and (h', Y) and $X = \Psi^{-1} W$, $X' = \Psi^{-1} W'$, then $g = Y_{f_1} W_m'^{-1} W_m Y_{f_1}^{-1}$ satisfies $g X_{\sigma_0} = X'_{\sigma_0}$, $\delta(g X_\rho, X'_\rho) = \delta(W_\rho, W'_\rho)$ and $\delta(g, 1) = \delta(W_m, W'_m)$, so

$$\tilde{d}_v(X, X') \le \sum_\rho \delta(W_\rho, W'_\rho) + z(v) \, \delta(W_m, W'_m). \tag{8.4}$$

(a), (b) for $z(v) \le 1$. Take σ_0 and τ to be the two directions of p at v. For $z(v) = 1$, $F = \{f_1\}$, the image is exactly $\mu_h^{(\sigma_0)}$, the coordinate W_m being Z_{f_1}, and the change of $h_p = h_{\sigma_0 \tau}$ is a change of the field on Z_τ; (8.3) and (8.4) give $\kappa_0 c |h_p - h'_p|$. For $z(v) = 0$ couple with the same X_{σ_0} and use $g = 1$; this is Lemma 8.3 with (8.3).

(a) for $z(v) = 2$. Let $F = \{-i, -j\}$; the free directions are $P = +i$, $Q = +j$, $K_\pm = \pm k$ and $L_\pm = \pm l$. Up to the symmetries $i \leftrightarrow j$, $k \leftrightarrow l$, $k \mapsto -k$ and $l \mapsto -l$ of the star, which fix F, the pair of p is one of $\{P, Q\}$, $\{P, K_+\}$ and $\{K_+, L_+\}$. In the image the terms of the density are: for adjacent free $\rho, \rho' \ne \sigma_0$, $\langle h, W_\rho^{-1} W_{\rho'} \rangle$; for $\{\sigma_0, \rho\}$, a field on W_ρ; for $\{f, \rho\}$, $\langle h, (W_m K_f)^{-1} W_\rho \rangle = \langle K_f h, W_m^{-1} W_\rho \rangle$, linear in W_m and in W_ρ; for $\{\sigma_0, f\}$, a field on W_m; and for $\{-i, -j\}$, a constant. So Lemma 8.10 applies with $a_{\rho\rho'} = 1$ for adjacent free $\rho, \rho' \ne \sigma_0$, $a_{\rho m} = a_{m\rho}$ equal to the number of $f \in F$ adjacent to ρ (one for P, Q, two for $K_\pm, L_\pm$), and $\eta = |h_p - h'_p| e_\tau$. Every row sum of A is at most 9, so $c_0 A$ has spectral radius at most $\frac{63}{80}$. By (8.4) the cost is at most $\sum_\rho \varrho_\rho + 2\varrho_m$. We give $\hat{y} \ge c_0 A \hat{y} + e_\tau$ and use Lemma 8.10(c); each item lists $\hat{y}$ and the inequalities multiplied by 80 and by the denominator of $\hat{y}$.

The pair $\{P, Q\}$, $\sigma_0 = P$, $\tau = Q$. The spins are Q, $K_\pm$, $L_\pm$, each of the last four adjacent to Q and to the two of the other axis, and m; $100\hat{y} = (Q, K_\pm, L_\pm, m) = (108, 16, 21)$. The

inequalities are $8640 \ge 7(4\cdot 16+21)+8000 = 8595$, $1280 \ge 7(108+2\cdot 16+2\cdot 21) = 1274$ and $1680 \ge 7(108+8\cdot 16) = 1652$, and the cost is at most $(108+4\cdot 16+2\cdot 21)/100 = 2.14$.

The pair $\{P, K_+\}$, $\sigma_0 = K_+$, $\tau = P$. The spins are P and Q, adjacent to each other and to K_-, $L_\pm$; K_-, adjacent to $P, Q, L_\pm$; $L_\pm$, adjacent to P, Q, K_-; and m; $200\hat{y} = (P, Q, K_-, L_\pm, m) = (216, 32, 35, 32, 40)$. The inequalities are $17280 \ge 7(32+35+64+40)+16000 = 17197$, $2560 \ge 7(216+35+64+40) = 2485$, $2800 \ge 7(216+32+64+80) = 2744$, $2560 \ge 7(216+32+35+80) = 2541$ and $3200 \ge 7(216+32+70+128) = 3122$, and the cost is at most $(216+32+35+64+80)/200 = 2.135$.

The pair $\{K_+, L_+\}$, $\sigma_0 = K_+$, $\tau = L_+$, with the same spins and adjacencies; $200\hat{y} = (P, Q, K_-, L_+, L_-, m) = (32, 32, 36, 219, 19, 54)$. The inequalities are $2560 \ge 7(32+36+219+19+54) = 2520$ for P and for Q, $2880 \ge 7(64+219+19+108) = 2870$, $17520 \ge 7(64+36+108)+16000 = 17456$, $1520 \ge 7(64+36+108) = 1456$ and $4320 \ge 7(64+72+2\cdot 238) = 4284$, and the cost is at most $(64+36+219+19+108)/200 = 2.23$.

In all cases the response is at most $2.23\,c|h_p - h'_p| \le \kappa_2 c|h_p - h'_p|$.

(a) for $z(v) \ge 3$, and (c), (d) for $z(v) \ge 2$. Use the coordinates X_σ, σ free, directly, and $g = 1$ in $\tilde{d}_v$. Each conditional law is ν_B with $B = 4\beta$ times the sum of the unit vectors of the plaquettes at v through σ, those through free neighbours depending isometrically on them; so Lemma 8.10(a) applies with A the adjacency matrix of the free directions, whose largest degree is $d = 5, 4, 3$ for $z(v) = 2, 3, 4$ (for $z = 2$, $+i$ is adjacent to $+j$, $\pm k$, $\pm l$; for $z = 3$, $+i$ to $+j$, $+k$, $\pm l$; for $z = 4$ the free directions are the four positive ones). A change of h_p with both links free gives $\sum \eta = 2|h_p - h'_p|$, with one link free $\sum \eta = |h_p - h'_p|$, and a change of the outside link in direction f by η gives η on each free neighbour of f, at most d of them. With $c \le c_0$: $\frac{2c}{1-4c} \le \frac{40}{13}c$, $\frac{2c}{1-3c} \le \frac{160}{59}c$, $\frac{c}{1-cd} \le \frac{c}{1-5c_0} = \frac{16}{9}c$ and $\frac{dc}{1-dc} \le \frac{5c_0}{1-5c_0} = \frac{7}{9}$.

(c), (d) for $z(v) = 1$. Let $F = \{f\}$ and $X_f = Y$. The map $X \mapsto Z = (Y^{-1}X_\rho)_\rho$ carries $\gamma^{h,Y}$ to $\mu_h^{(f)}$, which does not depend on Y. A change of h_p, p spanned by f and τ, is a change of the field on Z_τ, so (8.3) gives a coupling with $\int \sum \delta(YZ_\rho, YZ'_\rho) \le \kappa_0 c|h_p - h'_p|$, with $g = 1$. For a change $Y \to Y'$ couple $X = YZ$ and $X' = Y'Z$ and take $g = Y'Y^{-1}$; then $\tilde{d}_v(X, X') \le \delta(Y'Y^{-1}, 1) = \delta(Y, Y')$.

(e) The block law is ν_b with $b = 4\beta$ times the sum of the six vectors of the plaquettes through $(v - e_a, a)$; such a change moves b by at most $4|\beta||h_p - h'_p|$, respectively $4|\beta|\eta$, since an outside link lies in at most one plaquette with $(v - e_a, a)$. Apply (8.1). □

Part (b) is the point of the soft quotient: a star with one fixed link is exactly as good as a full star, and the price $z(v)\delta(g, 1)$ never feeds back into the system, because an outside link of a lower vertex is a factor of no two-path at a block (Lemma 8.7(iii)). The margin is lost at the edges of the cube, $z(v) = 2$, and the next lemma pays for them with weights.

8.7. **The column inequality.** For a block vertex v and a plaquette p at v let $\Gamma_v(p)$ be the constant of Lemma 8.11(a), (c) or (e), and $\Gamma_v(p) = 0$ if no link of p at v lies in V. Put $u_v = 1$ for upper v and $u_v = u_{z(v)}$ for lower v, with

$$u_0 = u_1 = 1, \quad u_2 = \tfrac{5}{6}, \quad u_3 = \tfrac{7}{10}, \quad u_4 = \tfrac{3}{5}; \qquad \lambda = \tfrac{97}{100}.$$

Lemma 8.12. *Let $n \ge 4$ and $c \le c_0$. For every $k \in V$, with plaquettes $p_{(\nu,\epsilon)}$ and even corners $v_{(\nu,\epsilon)} = o + \epsilon e_\nu$ as above,*

$$\sum_{(\nu,\epsilon)} u_{v_{(\nu,\epsilon)}} \Gamma_{v_{(\nu,\epsilon)}}(p_{(\nu,\epsilon)}) \le \lambda\, u_{w(k)}.$$

Proof. Put $f(z) = u_z \kappa_z c$: $f(0) = f(1) = \kappa_0 c < 1.8004\,c$, $f(2) = \frac{15}{8}c$, $f(3) = \frac{28}{13}c < 2.1539\,c$ and $f(4) = \frac{96}{59}c < 1.6272\,c$, all at least c. By Lemma 8.7(ii) a lower $v_{(\nu,\epsilon)}$ has both links of p free, so it contributes $f(z(v_{(\nu,\epsilon)}))$; an upper one contributes at most c, and other vertices contribute 0. Let o, μ and ς be as in Lemma 8.7(ii).

Case A: $\hat{o}$ has a coordinate equal to n. Since $k \in V$ is (o, μ), which needs $o \in Q$, or $(o - e_\mu, \mu)$, which needs $\hat{o} - e_\mu \in [0, n-1]^4$, this coordinate is $\hat{o}_\mu$, and $w(k) = o - e_\mu$ with

$\hat{w}_\mu = n-1 \geq 1$; so $z(w) \leq 3$ and $u_w \geq \frac{7}{10}$. Every $v_{(\nu,\epsilon)}$ has μ-coordinate n, so the sum is at most $6c \leq 6c_0 = 0.525 < \frac{7}{10} \cdot 0.97$.

Case B: $o \in Q$. Let $Z = z(o)$ be the number of zeros of $\hat{o}$, $a = \#\{\nu \neq \mu : \hat{o}_\nu = 0\}$ and $b = \#\{\nu \neq \mu : \hat{o}_\nu = 1\}$. As $n \geq 4$, the classes $\{0\}$, $\{1\}$, $[2, n-2]$ and $\{n-1\}$ are disjoint. For $\nu \neq \mu$: if $\hat{o}_\nu = 0$, $\epsilon = -1$ gives no block and $\epsilon = +1$ a lower vertex with $z = Z - 1$; if $\hat{o}_\nu = 1$, $\epsilon = +1$ gives $z = Z$ and $\epsilon = -1$ gives $z = Z + 1$; if $2 \leq \hat{o}_\nu \leq n-2$ both give $z = Z$; and if $\hat{o}_\nu = n-1$, $\epsilon = -1$ gives $z = Z$ and $\epsilon = +1$ an upper vertex, which contributes at most $c \leq f(Z)$. Hence the sum is at most

$$a\, f(Z-1) + b\big(f(Z) + f(Z+1)\big) + 2(3-a-b) f(Z).$$

If $\hat{o}_\mu = 0$, then $\varsigma = +1$, $Z = a+1$ and $z(w) = a$, and the sum is at most $h(a,b) = af(a) + b(f(a+1) + f(a+2)) + 2(3-a-b)f(a+1)$, to be compared with λu_a. If $\hat{o}_\mu \geq 1$, then $Z = a$ and w is upper, with $u_w = 1$, or lower with $z(w) \in \{a, a+1\}$, so $u_w \geq u_{a+1}$; the sum is at most $g(a,b) = af(a-1) + b(f(a) + f(a+1)) + 2(3-a-b)f(a)$, to be compared with λu_{a+1}. Here $a + b \leq 3$. Since $f(0) = f(1) \leq f(2) \leq f(3)$, the quantities $h(0,b)$, $g(1,b)$, $h(1,b)$ and $g(2,b)$ increase with b, and $g(0,b) = 6f(0)$. Every term is c times a constant, so it suffices to take $c = c_0$; with $\hat{\lambda} = \lambda/c_0 = \frac{388}{35} > 11.0857$ and f in units of c, the cases are

$$\begin{aligned}
&h(0,3) = 3f(0) + 3f(2) < 5.4012 + 5.625 = 11.0262 < \hat{\lambda},\\
&g(0,b) = 6f(0) < 10.8024 < \hat{\lambda},\\
&g(1,2) = 3f(0) + 2f(2) < 9.1512 < \tfrac{5}{6}\hat{\lambda}, \quad \tfrac{5}{6}\hat{\lambda} > 9.2380,\\
&h(1,2) = f(0) + 2f(2) + 2f(3) < 9.8582 < \hat{\lambda},\\
&g(2,1) = 2f(0) + f(2) + f(3) < 7.6297 < \tfrac{7}{10}\hat{\lambda}, \quad \tfrac{7}{10}\hat{\lambda} > 7.7599,\\
&h(2,0) = 2f(2) + 2f(3) < 8.0578 < \tfrac{5}{6}\hat{\lambda},\\
&h(2,1) = 2f(2) + f(3) + f(4) < 7.5311 < \tfrac{5}{6}\hat{\lambda},\\
&g(3,0) = 3f(2) = 5.625 < \tfrac{3}{5}\hat{\lambda}, \quad \tfrac{3}{5}\hat{\lambda} > 6.6514,\\
&h(3,0) = 3f(3) < 6.4617 < \tfrac{7}{10}\hat{\lambda}.
\end{aligned}$$

All inequalities hold. □

8.8. **The cube condition uniformly in the cube size.**

Proposition 8.13. *Let $c \leq c_0$, $n \geq 4$ and $L \geq 2n+4$ even. For every link j,*

$$\sum_{x:\, j \notin V_x} \tilde{R}_x(j) \leq 259\, n^3.$$

Proof. Fix x, $j \notin V = V_x$ and U, U' differing only at j, and write $\eta = \delta(U_j, U'_j)$. Let π minimise $\int D\, d\pi$ over couplings of $\gamma^0_{V,U}$ and $\gamma^0_{V,U'}$, and put $\varrho_v = \int \tilde{d}_v\, d\pi$. As in Lemma 8.10, replacing the block of v by an optimal coupling for $\tilde{d}_v$ of $\gamma_v(\cdot \mid \sigma, U)$ and $\gamma_v(\cdot \mid \sigma', U')$ changes only the v-term, so $\varrho_v \leq \int \Omega_v\, d\pi$ with $\Omega_v = W^{\tilde{d}_v}(\gamma_v(\cdot \mid \sigma, U), \gamma_v(\cdot \mid \sigma', U'))$.

Pointwise bound. Changing the two-paths at v one at a time, and then the outside link j if it is a link at v, the triangle inequality for $W^{\tilde{d}_v}$ and Lemma 8.11 give $\Omega_v \leq \sum_p \Gamma_v(p) |h_p(\sigma, U) - h_p(\sigma', U')| + r_v \eta$, where $r_v \leq 1$ if v is lower and j is an outside link at v, $r_v \leq c$ if v is upper and j is a link at v, and $r_v = 0$ otherwise. For each block vertex w fix g^*_w attaining $\tilde{d}_w(\sigma_{B_w}, \sigma'_{B_w})$, with $g^*_w = 1$ for upper w, and put $\delta_k = \delta(g^*_{w(k)} X_k(\sigma), X_k(\sigma'))$ for $k \in V$, so that $\sum_{k \in B_w \cap V} \delta_k \leq \tilde{d}_w$. If both links of p at the opposite corner v' lie in V, then v' is lower, an upper vertex carrying one link of V, and by invariance of h_p under $X \mapsto g^*_{v'} X$ at v', $|h_p(\sigma, U) - h_p(\sigma', U')|$ is at most the sum of their δ_k. If exactly one lies in V, the other is off V and is not an outside link of a lower vertex (Lemma 8.7(iii)), so v' is upper and the bound is $\delta_k + [j \in p]\eta$. If none lies in V, the bound is $[j \in p]\eta$. Each $k \in V$

is a factor of a two-path at v for at most one p at v, and these v are the vertices $v_{(\nu,\epsilon)}$ of k, so Lemma 8.12 gives, pointwise,

$$\sum_v u_v\Omega_v \le \lambda\sum_{k\in V} u_{w(k)}\delta_k + B\eta \le \lambda\sum_w u_w\tilde d_w + B\eta,$$

with $B=\sum_{p\ni j} u_{v(p)}\Gamma_{v(p)}(p)+\max_v u_v r_v$, where $v(p)$ is the even corner of p opposite to the even end point of j.

Constants. By Lemma 8.11, $u_v\Gamma_v(p)\le\frac{28}{13}c$ in all cases ($\max_z f(z)=f(3)$; with one fixed link $\kappa_0 c$ or $\frac{16}{9}c$; upper vertices c), and only the plaquettes $p\ni j$ containing a link of V contribute. Integrating, $\sum_v u_v\varrho_v\le\lambda\sum_v u_v\varrho_v + B_x(j)\eta$, with $B_x(j)=\frac{28}{13}c\,P_x(j)+R_x(j)$, where $P_x(j)$ is the number of plaquettes $p\ni j$ with a link in V_x, and $R_x(j)=1$ if j is an outside link at a lower vertex of the cube, c if j is a link off V_x at an upper even vertex of the cube, and 0 otherwise. All terms are finite, so $\sum_v u_v\varrho_v\le B_x(j)\eta/(1-\lambda)$ and

$$W^D(\gamma^0_{V,U},\gamma^0_{V,U'})\le\sum_v\varrho_v\le\frac{B_x(j)\,\eta}{u_4(1-\lambda)}=\frac{500}{9}B_x(j)\,\eta.$$

Summation over x. A plaquette counted in $P_x(j)$ contains a partner k of j with $k\in V_x$, and distinct plaquettes give distinct partners. For $j=(b,\nu)$ the eighteen partners have base points $b+\Delta$ with $\sum\|\Delta\|_1=18$, $\Delta\in\{0,e_\mu,e_\nu,-e_\mu,-e_\mu,e_\nu-e_\mu\}$ for each $\mu\ne\nu$, and $\#\{x:k\in V_x,\ j\notin V_x\}=|Q_0\setminus(Q_0-\Delta)|\le n^3\|\Delta\|_1$ with $Q_0=[0,n)^4$, by moving one coordinate at a time; so $\sum_x P_x(j)\le 18n^3$. For R_x, let v be the even end point of j. If v is the head of $j=(v-e_a,a)$, it is a lower vertex with $a\in Z(v)$ for exactly n^3 cubes, and an upper vertex of the cube with j off V_x for at most $3n^3$ cubes, the coordinate equal to n not being a; if v is the base point of j, it is never lower with $j\notin V_x$ and is upper for at most $4n^3$ cubes. So $\sum_x R_x(j)\le(1+3c)n^3$. With $c\le c_0$,

$$\sum_x\tilde R_x(j)\le\frac{500}{9}\Big(\frac{28}{13}\cdot 18c_0+1+3c_0\Big)n^3=\frac{500}{9}\cdot\frac{4841}{1040}\,n^3<258.61\,n^3.\qquad\square$$

Theorem 8.14. *Let $G=\mathrm{SU}(2)$ and $|\beta_W|\le\frac{7}{20}$.*

(i) *For every $n\ge 260$ and even $L\ge 2n+4$, $\mathrm{BP}(n,1-259/n)$ holds; in particular $\mathrm{BP}(n,\frac12)$ holds for $n\ge 518$.*

(ii) *For every $n\ge 2$ and even $L\ge 2n+4$, $\mathrm{BP}(n,\varpi)$ holds with $\varpi=\frac{106}{1933}\cdot 517^{-4}$, independent of n and L.*

Proof. (i) Let $\mathcal K=L^{-4}\sum_x\mathcal P_x$ and let h be gauge invariant at every configuration with $\mathrm{Lip}(h)<\infty$, and let U,U' differ only at i. If $i\in V_x$, then $\mathcal P_x h(U)=\mathcal P_x h(U')$. If $i\notin V_x$, Lemma 8.9 and any coupling π of $\gamma^0_{V_x,U}$ and $\gamma^0_{V_x,U'}$ give $|\mathcal P_x h(U)-\mathcal P_x h(U')|\le\mathrm{Lip}(h)(\int D\,d\pi+\delta(U_i,U_i'))$, and the infimum over π gives $\mathrm{Lip}(h)(\tilde R_x(i)+1)\delta(U_i,U_i')$. As i lies in n^4 cubes, Proposition 8.13 gives $\mathrm{Lip}(\mathcal Kh)\le(1-\kappa)\,\mathrm{Lip}(h)$ with $\kappa=(1-259/n)n^4/L^4>0$, and $\mathcal Kh$ is again gauge invariant at every configuration, by gauge covariance of $\gamma^0_{V_x,U}$. The spectral argument of Proposition 8.4 gives $\langle h,(I-\mathcal K)h\rangle\ge\kappa\,\mathrm{Var}_\mu(h)$, first for such h and then, by density, for every real $f\in\mathcal A$; and $L^{-4}\sum_x\mu((f-\mathcal P_x f)^2)=\langle f,(I-\mathcal K)f\rangle$, which is $\mathrm{BP}(n,1-259/n)$.

(ii) For $n\ge 518$ use (i). For $2\le n\le 517$ use the proof of Theorem 8.5, which gives $\mathrm{BP}(n,(1-6c\,s(c))(n-1)^4/(n^4\lceil(n-1)^4/2\rceil))$ whenever $6c\,s(c)<1$; here $1-6c\,s(c)\ge 1-\frac{1827}{1933}=\frac{106}{1933}$ by Lemma 8.2(ii), and $(n-1)^4\ge\lceil(n-1)^4/2\rceil$, so $\mathrm{BP}(n,\frac{106}{1933}n^{-4})$ holds, and $\frac{106}{1933}n^{-4}\ge\varpi$. $\square$

Theorem 8.14 extends the uniform range $\frac{5}{18}$ of Theorem 7.11 past $\beta_W=\frac13$, where Proposition 7.12 stops every gauge-fixed single-link scheme and Proposition 8.6 every scheme that treats face links as single sites. The faces cost nothing here; the margin is lost

at the cube edges, through the column $h(0,3)$ of Lemma 8.12.[5] The value $\beta_W \le \frac{7}{20}$ is far below the scaling region of four-dimensional SU(2), where β_W is of order 2 and above, and the next proposition explains why no one-step contraction of this kind reaches it.

8.9. **Single links near flat configurations.** Let $A(\kappa) = \int x_1 \, \nu_{\kappa e_1}(dx)$ be the mean resultant length of ν_b at $|b| = \kappa$. By symmetry $\int x \, \nu_b(dx) = A(|b|)b/|b|$; A is nondecreasing, its derivative being the variance of x_1, and $A(\kappa) \to 1$ as $\kappa \to \infty$.

Proposition 8.15. *Let $\beta > 0$, let e be a link whose six plaquettes contain no link fixed at a value other than 1, and let k be a partner of e. The single-link influence satisfies*

$$C_{ek} = \sup \frac{W_1(\nu_b, \nu_{b'})}{\delta(U_k, U_k')} \ge \frac{A(24\beta)}{6}.$$

Hence in any single-link scheme in which every link has at least twelve free partners, as for the forest of Theorem A, every row sum of the influence matrix is at least $2A(24\beta)$, which tends to 2 as $\beta \to \infty$.

Proof. Take all links equal to 1, so every staple is 1 and $b = 24\beta \cdot 1$. Change U_k so that the staple of the plaquette containing e and k becomes $\cos\vartheta + t \sin\vartheta$ with a unit imaginary quaternion t; then $\delta(U_k, U_k') = 2\sin\frac{\vartheta}{2}$ and $b' = 4\beta(5 + \cos\vartheta + t\sin\vartheta)$. The function $x \mapsto \langle x, t\rangle$ is 1-Lipschitz, so $W_1(\nu_b, \nu_{b'}) \ge \langle \int x \, d\nu_{b'} - \int x \, d\nu_b, t\rangle = A(|b'|)\, 4\beta \sin\vartheta/|b'|$. Divide by $2\sin\frac{\vartheta}{2}$ and let $\vartheta \to 0$. For the limit of A: the law of x_1 under $\nu_{\kappa e_1}$ has density proportional to $e^{\kappa t}\sqrt{1-t^2}$ on $[-1,1]$, and for $0 < \epsilon < \frac{1}{2}$ the mass of $[-1, 1-2\epsilon]$ divided by that of $[1-\epsilon, 1]$ is at most $e^{-\kappa\epsilon}\pi / \int_{1-\epsilon}^{1} \sqrt{1-t^2}\, dt \to 0$. □

At weak coupling the Gibbs measure concentrates near flat configurations, so even an influence matrix computed only on typical configurations does not contract there: per link it approaches $\frac{1}{6}$ per partner, and $12 \cdot \frac{1}{6} = 2$.[6] Larger blocks with better internal estimates push the threshold to some finite β_W, but the mass at weak coupling is non-perturbative, $m \sim \Lambda e^{-c\beta}$, and a contraction can appear only at block sizes of order $e^{c\beta}$ lattice units. Section 11 and Appendix G make this precise.

9. Free massive fields

Here $d \ge 1$, $n \ge 2$, $L \ge n+1$, $\mathbb{T} = (\mathbb{Z}/L\mathbb{Z})^d$, $\mu > 0$ is a lattice mass, $-\Delta$ is the lattice Laplacian and $Q = -\Delta + \mu^2$. For a positive definite Q on $\mathbb{R}^N$ let γ_Q be the centred Gaussian law with covariance $C = Q^{-1}$, and for $V \subset \{1, \dots, N\}$ let $\mathcal{P}_V$ be the conditional expectation given the coordinates outside V. For the cubes $V_x = x + \{0, \dots, n-1\}^d$, $x \in \mathbb{T}$, put

$$\varpi(Q) = \inf_f \frac{n^{-d} \sum_x \|f - \mathcal{P}_{V_x} f\|^2}{\mathrm{Var}_{\gamma_Q}(f)}, \tag{9.1}$$

the infimum over nonconstant $f \in L^2(\gamma_Q)$, the best constant in $\mathrm{BP}(n, \cdot)$. With $s = n\mu$ and $\mu = ma$, $n = \ell/a$, the parameter $s = m\ell$ is the mass times the physical size of the cube.

Lemma 9.1 (linear functions suffice)**.** *For a positive definite Q and any family of coordinate sets (V_x), the infimum in (9.1) equals its infimum over the linear functions $\ell_h(\phi) = \langle h, \phi\rangle$, and*

$$\|\ell_h - \mathcal{P}_V \ell_h\|^2 = \langle h_V, Q_{VV}^{-1} h_V\rangle, \qquad \mathrm{Var}(\ell_h) = \langle h, Q^{-1} h\rangle,$$

with h_V the restriction of h to V and Q_{VV} the block of Q on V.

[5] A linear programme over weights depending on the block, with the exact constants for $z(v) = 2$, suggests on cubes of side up to 8 that the method reaches the star threshold $\beta_W \approx 0.3609$ of Theorem 8.5; with weights depending only on $z(v)$ it stops near $\beta_W \approx 0.345$. These are numerical observations and no statement rests on them.

[6] Numerically $A(\kappa) = \frac{1}{2}$ at $\kappa \approx 2.45$, that is $\beta_W \approx 0.41$.

Proof. The conditional law of ϕ_V given ϕ_{V^c} is Gaussian with covariance Q_{VV}^{-1}, which gives the formula for ℓ_h. Write $\phi = Q^{-1/2}\xi$ with ξ standard Gaussian. Conditioning on ϕ_{V^c} is conditioning on the Gaussian subspace $K_V = \{\langle u, \xi\rangle : u \in Q^{-1/2}\mathbb{R}^{V^c}\}$, and on the m-th Wiener chaos, identified with the symmetric tensors $\mathrm{Sym}^m(\mathbb{R}^N)$, it acts as $\pi_V^{\otimes m}$, with π_V the orthogonal projection onto $Q^{-1/2}\mathbb{R}^{V^c}$ [30, Ch. 4]. The chaoses are orthogonal and invariant under every $\mathcal{P}_V$, so it suffices to treat one chaos $m \geq 1$. The projection $I - \pi_V^{\otimes m}$ dominates $(I - \pi_V) \otimes I^{\otimes(m-1)}$, as both are projections that commute and the range of the second lies in that of the first. For a symmetric tensor F, therefore,

$$\sum_x \|F - \pi_{V_x}^{\otimes m} F\|^2 \geq \Big\langle F, \Big(\sum_x (I - \pi_{V_x})\Big) \otimes I^{\otimes(m-1)} F \Big\rangle \geq \lambda \|F\|^2,$$

where λ is the least eigenvalue of $\sum_x (I - \pi_{V_x})$ on $\mathbb{R}^N$, which is the infimum of the quotient over linear functions ($m = 1$). □

Theorem 9.2 (the exact constant)**.** *Let $V = \{0, \dots, n-1\}^d$, $-\Delta_D$ its Dirichlet Laplacian and $\mathbf{1}$ its constant function. Then*

$$\varpi(-\Delta + \mu^2) = \frac{\mu^2}{n^d} \langle \mathbf{1}, (-\Delta_D + \mu^2)^{-1} \mathbf{1} \rangle,$$

attained only at multiples of $\ell_{\mathbf{1}}$ among linear functions, and

$$\Big(\frac{5}{6}\Big)^d \frac{s^2}{s^2 + 10d} \leq \varpi(-\Delta + \mu^2) \leq \frac{s^2}{s^2 + 4dn^2 \sin^2 \frac{\pi}{2n+2}} \leq \frac{s^2}{s^2 + 16d/9}.$$

Proof. Since $L \geq n+1$, the block $Q_{V_x V_x}$ is the translate of $Q_{VV} = -\Delta_D + \mu^2$; put $G_V = Q_{VV}^{-1}$. By Lemma 9.1, $n^d \varpi$ is the least value of $\langle h, Sh\rangle / \langle h, Q^{-1} h\rangle$ with $S = \sum_x R_{V_x}^{\mathsf{T}} Q_{V_x V_x}^{-1} R_{V_x}$, that is, the least eigenvalue of $Q^{1/2} S Q^{1/2}$. Translations commute with Q and S, so the plane waves $e_k(y) = e^{ik\cdot y}$ diagonalise it with eigenvalues $\lambda_k = \hat{q}(k) \langle e_k^V, G_V e_k^V \rangle$, where $\hat{q}(k) = \mu^2 + \sum_i (2 - 2\cos k_i)$ and e_k^V is the restriction to V.

Let B map f to the function $u \mapsto \sum_{w \sim u,\, w \notin V} f(w)$ on V. Then $Q_{VV} R_V f = R_V Q f + B f$, so $Q_{VV} e_k^V = \hat{q}(k) e_k^V + B e_k$, and pairing G_V of this with e_k^V gives

$$\lambda_k = n^d - \sum_{u,u' \in V} \sum_{w \sim u',\, w \notin V} G_V(u, u') \cos\big(k \cdot (w - u)\big).$$

Write $Q_{VV} = (2d + \mu^2)(I - K)$ with K entrywise nonnegative and $\|K\| \leq 2d/(2d + \mu^2) < 1$; the Neumann series shows that G_V has positive entries. Hence $\lambda_k - \lambda_0 = \sum G_V(u, u')(1 - \cos(k \cdot (w - u))) \geq 0$, with equality only at $k = 0$ (take $u = u'$ on a face and $w = u + e_i$). Since $Q\mathbf{1} = \mu^2 \mathbf{1}$, $\lambda_0 = \mu^2 \langle \mathbf{1}, G_V \mathbf{1}\rangle$, which is the formula.

Upper bound. $G_V \leq (\lambda_1 + \mu^2)^{-1}$ with $\lambda_1 = 4d \sin^2 \frac{\pi}{2n+2}$ the least eigenvalue of $-\Delta_D$, and $\|\mathbf{1}\|^2 = n^d$; then $\sin t \geq 2t/\pi$ gives $n^2 \lambda_1 \geq 4dn^2/(n+1)^2 \geq 16d/9$.

Lower bound. By the variational formula $\langle \mathbf{1}, G_V \mathbf{1}\rangle \geq \langle \mathbf{1}, w\rangle^2 / \langle w, Q_{VV} w\rangle$, take $w(u) = \prod_i \tau(u_i)$ with the parabolic profile $\tau(s) = 6(s+1)(n-s)/(n(n+1)(n+2))$ on $\{0, \dots, n-1\}$, zero outside. Then $\sum \tau = 1$, $S_2 = \sum \tau^2 = \alpha_n / n$ and $D_2 = \sum_{s \in \mathbb{Z}} (\tau(s+1) - \tau(s))^2 = \beta_n / n^3$ with

$$\alpha_n = \frac{6(n^2 + 2n + 2)}{5(n+1)(n+2)} \leq \frac{6}{5}, \qquad \beta_n = \frac{12n^2}{(n+1)(n+2)} = 10 \rho_n \alpha_n, \quad \rho_n = \frac{n^2}{n^2 + 2n + 2} < 1,$$

by the power sums $\sum_{u=1}^n u(n+1-u) = n(n+1)(n+2)/6$, $\sum_{u=1}^n u^2 (n+1-u)^2 = n(n+1)(n+2)(n^2+2n+2)/30$ and $\sum_{u=0}^n (n - 2u)^2 = n(n+1)(n+2)/3$. The Dirichlet form of the product is $\langle w, -\Delta_D w\rangle = d S_2^{d-1} D_2$ and $\langle \mathbf{1}, w\rangle = 1$, so

$$\varpi \geq \frac{\mu^2}{n^d (\mu^2 S_2^d + d S_2^{d-1} D_2)} = \frac{1}{\alpha_n^d + d \alpha_n^{d-1} \beta_n s^{-2}} = \alpha_n^{-d} \frac{s^2}{s^2 + 10 d \rho_n} \geq \Big(\frac{5}{6}\Big)^d \frac{s^2}{s^2 + 10d}. \quad \square$$

At $n = 2$ the two bounds coincide with $s^2/(s^2 + 4d)$. For $d = 4$ the lower bound is 0.0118 at $s = 1$ and 0.0438 at $s = 2$; the upper bound vanishes as $s \to 0$, so a massless free field satisfies no cube condition uniform in the lattice spacing. The same proof, with the Hodge identity of the torus, gives the cube condition for the free massive vector (Proca) field on gauge-invariant functions, with the same lower bound and the upper bound $s^2/(1 + s^2)$ [8].

10. Mass generation at one scale implies the cube condition

Theorem B reduces the continuum gap to statement (D), a cube condition uniform along the tuning. This section reduces (D) in turn to a hypothesis of a different kind, closer to the physics: a uniform decay of correlations between the gauge-invariant observables of two slabs of a box separated by a slab of physical width about $\ell/2$, for every boundary condition. If the maximal correlation is at most $c < \frac{1}{2}$, then the cube condition holds with $\varpi = (1-2c)^4/32$ (Theorem D). The value $\frac{1}{2}$ is the exact limit of the method (Proposition G.4 in Appendix G); the hypothesis implies exponential decay of gauge-invariant correlations under every boundary condition (Proposition 10.10); and it holds in two dimensions along the weak-coupling tuning (Theorem G.8). In four dimensions it is a form of mass generation at the physical scale and it is not proved here.

10.1. **Boxes, boundary conditions and gauge-invariant observables.** In this section $G \subset \mathrm{U}(N)$ is compact and nontrivial, φ is a positive continuous class function on G, λ denotes normalised Haar measure on G or on a product of copies of G, and $\omega_\varphi = \log(\max\varphi/\min\varphi)$. Links of $\mathbb{Z}^4$ are written (y,ν) with base point y, from y to $y + e_\nu$. A *box* is a set $Q = \prod_{i=1}^4 [\alpha_i, \alpha_i + s_i) \cap \mathbb{Z}$ with integers $s_i \geq 1$, its *widths*, and $V(Q)$ is the set of links with base point in Q. For an exterior configuration $\omega \in G^{E(\mathbb{Z}^4)\setminus V(Q)}$ let ν_Q^ω be the probability measure on $G^{V(Q)}$ with density proportional to $\prod_{p\cap V(Q)\neq\emptyset} \varphi((\sigma,\omega)_p)$ with respect to λ, a finite product. It is the conditional law of $U_{V(Q)}$ given the other links under any Gibbs measure with weight φ, and on a torus Λ_L with all $s_i \leq L-3$ it is the conditional law of the torus measure.[7] For $A \subset V(Q)$ write E_A for the conditional expectation under ν_Q^ω given $\sigma_{V(Q)\setminus A}$; its regular version is integration against the measure on G^A with density proportional to $\prod_{p\cap A\neq\emptyset} \varphi(U_p)$, which is jointly continuous in all variables. In particular, if $Q' \subset Q$ is a box, the conditional law of $\sigma_{V(Q')}$ under ν_Q^ω given $\sigma_{V(Q)\setminus V(Q')} = \eta$ is $\nu_{Q'}^{(\eta,\omega)}$.

A vertex v is *interior* to Q if the eight links incident to v lie in $V(Q)$. Let $\mathcal{G}_Q$ be the group of $h \in G^{\mathbb{Z}^4}$ with $h_v = 1$ at every vertex not interior to Q, acting by $(h\sigma)_{(y,\nu)} = h_y \sigma_{(y,\nu)} h_{y+e_\nu}^{-1}$; it fixes every link outside $V(Q)$, whose end points are not interior. Let $\mathcal{A}_Q^\omega$ be the set of real $F \in L^2(\nu_Q^\omega)$ with $F \circ h = F$ almost surely for every $h \in \mathcal{G}_Q$: the gauge-invariant observables of the box, the boundary condition having fixed the gauge on the boundary.

Lemma 10.1. *Let Q be a box, ω an exterior configuration and $\nu = \nu_Q^\omega$.*

(i) *ν is invariant under $\mathcal{G}_Q$.*

(ii) *For every $S \subset V(Q)$, each $h \in \mathcal{G}_Q$ maps $\sigma(\sigma_S)$ onto itself, and $E_A(F \circ h) = (E_A F) \circ h$ for $A \subset V(Q)$ and $F \in L^2(\nu)$.*

(iii) *$\mathcal{A}_Q^\omega$ is a closed subspace that contains the constants and is preserved by every E_A. Every $F \in \mathcal{A}_Q^\omega$ has a version $\tilde F$ with $\tilde F(h\sigma) = \tilde F(\sigma)$ for all h and σ, and if F is $\sigma(\sigma_S)$-measurable, $\tilde F$ can be taken $\sigma(\sigma_S)$-measurable.*

Proof. (i) h multiplies each link on the left and on the right by fixed elements, so it preserves λ. Since h fixes the links outside $V(Q)$, $(h\sigma,\omega) = h\cdot(\sigma,\omega)$, and the holonomy of a plaquette with base point y is conjugated by h_y; φ is a class function, so the density is invariant.

[7] The plaquettes meeting $V(Q)$ have their links based in $\prod_i [\alpha_i - 1, \alpha_i + s_i]$, which embeds in Λ_L when $s_i \leq L - 3$.

(ii) $(h\sigma)_e$ is a function of σ_e alone and $h^{-1} \in \mathcal{G}_Q$, so h maps $\sigma(\sigma_S)$ onto itself. For a bounded $\sigma(\sigma_{V(Q)\setminus A})$-measurable K, by (i), $\nu((F \circ h)K) = \nu(F\,(K \circ h^{-1})) = \nu((E_A F)(K \circ h^{-1})) = \nu(((E_A F) \circ h)K)$, and $(E_A F) \circ h$ is $\sigma(\sigma_{V(Q)\setminus A})$-measurable.

(iii) By (i) each map $F \mapsto F \circ h$ is unitary, so its fixed space is closed, and so is the intersection of these spaces; (ii) gives the invariance under E_A. The group $\mathcal{G}_Q \cong G^I$, I the finite set of interior vertices, is compact and acts continuously. For a measurable version F_0 put $\tilde{F}(\sigma) = \int F_0(h\sigma)\,dh$ where $\int |F_0(h\sigma)|\,dh < \infty$, and $\tilde{F}(\sigma) = 0$ elsewhere. By the invariance of Haar measure under right translations, the set where the integral converges and $\tilde{F}$ are invariant under every $k \in \mathcal{G}_Q$. For each h, $F_0(h\sigma) = F_0(\sigma)$ for almost every σ, so by Fubini's theorem, for almost every σ the equality holds for almost every h, and $\tilde{F} = F_0$ almost surely. If F_0 is a function of σ_S, so is $\sigma \mapsto F_0(h\sigma)$ for each h, and so is $\tilde{F}$. □

Definition 10.2. Let Q be a box, i a coordinate, and $Q = Q_- \cup Q_0 \cup Q_+$ a splitting into three nonempty boxes, consecutive in the coordinate i in this order: the *outer parts* $Q_\pm$ and the *middle* Q_0. For an exterior configuration ω put $\nu = \nu_Q^\omega$ and

$$\rho^\omega(Q_-, Q_+) = \sup |\nu(XY)|,$$

the supremum over $X \in \mathcal{A}_Q^\omega$ measurable with respect to $\sigma_{V(Q_-)}$ and $Y \in \mathcal{A}_Q^\omega$ measurable with respect to $\sigma_{V(Q_+)}$ with $\nu(X) = \nu(Y) = 0$ and $\nu(X^2) = \nu(Y^2) = 1$ (the supremum of the empty set being 0).

This is the maximal correlation[8] between the gauge-invariant observables of the two outer parts, the middle being integrated out, under the boundary condition ω. By the Cauchy–Schwarz inequality $\rho^\omega \le 1$.

Definition 10.3. Let $d \ge 2$, $m \ge 4$ and $w \ge 1$ be integers and $0 \le c \le 1$.

(i) $(\mathrm{M}_{d,m}(c))$: $\rho^\omega(Q_-, Q_+) \le c$ for every box Q all of whose widths lie in $[2d-1, md-1]$, every coordinate i in which Q has width $3d-1$, the splitting in which Q_-, Q_0, Q_+ have widths d, $d-1$, d in the coordinate i, and every exterior configuration ω.

(ii) $(\mathrm{M}_w^*(c))$: $\rho^\omega(Q_-, Q_+) \le c$ for every box Q, every coordinate i, every splitting whose middle has width w in the coordinate i, the outer parts being of any widths, and every ω.

The boxes of $(\mathrm{M}_{d,m}(c))$ are fat: all their widths are at least $2d-1$. The restriction matters. In two dimensions the same condition for boxes of width one in a transverse direction fails along the weak-coupling tuning, through a broken global symmetry of a pinned strip, while the fat form holds there (Theorem G.8 and Proposition G.14 in Appendix G). At $\varphi \equiv 1$ the links are independent under ν_Q^ω, so $\rho^\omega = 0$ and both conditions hold with $c = 0$. Along the tuning, with $d \approx \ell/(2a)$, $(\mathrm{M}_{d,m}(c))$ is a statement at the physical separation $(d-1)a \approx \ell/2$ in boxes of physical size at most about $m\ell/2$, uniform in the boundary condition, and stated for all square-integrable gauge-invariant observables of the two regions rather than for pairs of local ones.

10.2. **Two projections and a chain of projections.**

Lemma 10.4. *Let P_1, P_2 be orthogonal projections on a real Hilbert space $\mathcal{K}$, $\mathcal{N}$ a closed subspace of* $\operatorname{ran} P_1 \cap \operatorname{ran} P_2$*, and suppose that $|\langle x, y\rangle| \le c\|x\|\|y\|$ for all $x \in \operatorname{ran} P_1 \ominus \mathcal{N}$ and $y \in \operatorname{ran} P_2 \ominus \mathcal{N}$. Put $Q_k = I - P_k$ and $X = Q_1 + Q_2$. Then $X^2 \ge (1-c)X$, that is, $Q_1Q_2 + Q_2Q_1 \ge -c(Q_1 + Q_2)$.*

Proof. For $n \in \mathcal{N}$, $P_k n = n$, so $Xn = 0$; as X is self-adjoint it also preserves $\mathcal{N}^\perp$. Let $g \in \mathcal{N}^\perp$. Then $P_k g \perp \mathcal{N}$, since $\langle P_k g, n\rangle = \langle g, P_k n\rangle = \langle g, n\rangle = 0$, so with $p = \|P_1 g\|$ and $q = \|P_2 g\|$, $|\langle P_1 g, P_2 g\rangle| \le cpq$. Put $s = \langle g, (P_1 + P_2)g\rangle = p^2 + q^2$. Then $\|(P_1 + P_2)g\|^2 =$

[8]The maximal correlation of two random variables, the supremum of the correlations of square-integrable functions of each, was introduced by Hirschfeld and by Gebelein [24]; for jointly Gaussian variables it equals the absolute value of the ordinary correlation.

$p^2+q^2+2\langle P_1g,P_2g\rangle \le (1+c)(p^2+q^2)$ and $s \le \|g\|\|(P_1+P_2)g\| \le \|g\|((1+c)s)^{1/2}$, so $s \le (1+c)\|g\|^2$ and $\langle g, Xg\rangle = 2\|g\|^2 - s \ge (1-c)\|g\|^2$. So $X=0$ on $\mathcal{N}$, and on $\mathcal{N}^\perp$ the operator $X-(1-c)$ is nonnegative and commutes with $X \ge 0$; hence $X^2-(1-c)X = X^{1/2}(X-(1-c))X^{1/2} \ge 0$ there. Finally $X^2 - X = Q_1Q_2 + Q_2Q_1$. □

The angle between $\operatorname{ran} P_1 \ominus \mathcal{N}$ and $\operatorname{ran} P_2 \ominus \mathcal{N}$ is the Friedrichs angle of the two ranges, and the lemma is a form of the classical relation between that angle and the rate of the alternating projections [17].

Lemma 10.5. *Let $Q_0,\dots,Q_M$ be orthogonal projections on a real Hilbert space $\mathcal{K}$ with $Q_tQ_{t'} = Q_{t'}Q_t$ whenever $|t-t'|\ge 2$ and $Q_tQ_{t+1}+Q_{t+1}Q_t \ge -c(Q_t+Q_{t+1})$ for $0 \le t < M$, with $0 \le c < \frac12$. Then $H = \sum_t Q_t$ satisfies $H^2 \ge (1-2c)H$, and $\langle g, Hg\rangle \ge (1-2c)\|g\|^2$ for every $g \perp \ker H$.*

Proof. $H^2 = \sum_t Q_t + \sum_{t<M}(Q_tQ_{t+1}+Q_{t+1}Q_t) + \sum Q_tQ_{t'}$, the last sum over the ordered pairs with $|t-t'|\ge 2$. Each such $Q_tQ_{t'}$ is a product of commuting projections, hence a projection and nonnegative. The middle sum is at least $-c\sum_{t<M}(Q_t+Q_{t+1}) \ge -2cH$. So $H^2-(1-2c)H \ge 0$, and by the spectral theorem the spectrum of the nonnegative operator H lies in $\{0\}\cup[1-2c,\infty)$, which gives the second claim. □

Lemma 10.6. *Let Q be a box, ω an exterior configuration, and $R_0,\dots,R_M$ subsets of $V(Q)$ with union $V(Q)$. If $F \in L^2(\nu_Q^\omega)$ is, for every t, almost surely equal to a $\sigma(\sigma_{V(Q)\setminus R_t})$-measurable function, then F is almost surely constant.*

Proof. The density of $\nu = \nu_Q^\omega$ with respect to λ is bounded above and below by positive constants, so $L^2(\nu) = L^2(\lambda)$ with the same null sets. Let Λ_t be the conditional expectation under λ given $\sigma_{V(Q)\setminus R_t}$, which integrates out σ_{R_t}. By hypothesis $F = F_t$ almost surely with F_t a function of $\sigma_{V(Q)\setminus R_t}$, so $\Lambda_t F = F_t = F$. By Fubini's theorem $\Lambda_0\cdots\Lambda_M$ integrates out all of $\bigcup_t R_t = V(Q)$, so $F = \Lambda_0\cdots\Lambda_M F$ is constant. □

10.3. **One direction.** Fix a box $R = \prod_i I_i$ and a coordinate i with $I_i = [\alpha, \alpha+(M+2)d-1)$ for an integer $M \ge 0$. For $0 \le t \le M$ let R_t be the box obtained from R by replacing I_i with $\alpha + dt + [0, 2d-1)$, and for $t < M$ let Q_t be the box obtained by replacing I_i with $\alpha + dt + [0, 3d-1)$, split into Q_t^-, Q_t^0, Q_t^+ by $\alpha+dt+[0,d)$, $\alpha+dt+[d,2d-1)$ and $\alpha+dt+[2d-1,3d-1)$. Then

(10.1)
$$V(Q_t) = V(R_t)\cup V(R_{t+1}), \quad V(Q_t^-) = V(R_t)\setminus V(R_{t+1}), \quad V(Q_t^+) = V(R_{t+1})\setminus V(R_t),$$

and $\bigcup_t V(R_t) = V(R)$, since the intervals $\alpha+dt+[0,2d-1)$ cover I_i.

Proposition 10.7 (one direction)**.** *In this setting let ω be an exterior configuration of R, $\nu = \nu_R^\omega$, $P_t = E_{V(R_t)}$, and suppose that $\rho^{\omega'}(Q_t^-, Q_t^+) \le c < \frac12$ for every $t < M$ and every exterior configuration ω' of Q_t. Then every $F \in \mathcal{A}_R^\omega$ satisfies*

$$(1-2c)\operatorname{Var}_\nu(F) \le \sum_{t=0}^{M} \nu\big((F-P_tF)^2\big).$$

Proof. For $M=0$, $P_0F = \nu(F)$ and the claim is trivial; let $M \ge 1$. Put $\mathcal{K} = \mathcal{A}_R^\omega$. By Lemma 10.1(iii) each P_t preserves $\mathcal{K}$ and restricts to an orthogonal projection of $\mathcal{K}$; let $Q_t = I - P_t$ on $\mathcal{K}$.

Non-adjacent moves commute. Let $t' \ge t+2$. The base points of the links of $V(R_t)$ have i-th coordinate at most $\alpha+dt+2d-2$ and those of $V(R_{t'})$ at least $\alpha+dt+2d$, so no plaquette meets both sets. Given σ off $V(R_t)\cup V(R_{t'})$, the conditional density of $(\sigma_{V(R_t)}, \sigma_{V(R_{t'})})$ is a product of factors each of which involves the links of only one of the two sets. So they are conditionally independent, and $P_tP_{t'}F$ and $P_{t'}P_tF$ are both the integral of F against the product of the two conditional laws. Hence $P_tP_{t'} = P_{t'}P_t$, and the Q_t commute.

Adjacent moves. Fix $t < M$ and write $a = V(Q_t^-)$, $b = V(Q_t^0)$, $c' = V(Q_t^+)$, $e = V(R) \setminus V(Q_t)$. By (10.1), $V(R) \setminus V(R_t) = c' \cup e$ and $V(R) \setminus V(R_{t+1}) = a \cup e$, so the ranges of P_t and of P_{t+1} in $\mathcal{K}$ are the elements of $\mathcal{K}$ measurable with respect to $(\sigma_{c'}, \sigma_e)$ and to (σ_a, σ_e). Let $\mathcal{N}$ be the elements of $\mathcal{K}$ measurable with respect to σ_e, a closed subspace of both ranges. We check the hypothesis of Lemma 10.4. Let x and y be in the ranges of P_t and P_{t+1} and orthogonal to $\mathcal{N}$. The conditional expectation E_e given σ_e is $E_{V(Q_t)}$, which preserves $\mathcal{K}$, so $E_e x \in \mathcal{N}$ and $\|E_e x\|^2 = \langle x, E_e x\rangle = 0$; likewise $E_e y = 0$. By Lemma 10.1(iii) for the box R take versions $x = \tilde{x}(\sigma_{c'}, \sigma_e)$ and $y = \tilde{y}(\sigma_a, \sigma_e)$ invariant at every point under $\mathcal{G}_R$. Every vertex interior to Q_t is interior to R, so $\mathcal{G}_{Q_t} \subset \mathcal{G}_R$, and $\mathcal{G}_{Q_t}$ fixes σ_e. Hence for each $\eta \in G^e$ the functions $X_\eta = \tilde{x}(\cdot, \eta)$ of $\sigma_{V(Q_t^+)}$ and $Y_\eta = \tilde{y}(\cdot, \eta)$ of $\sigma_{V(Q_t^-)}$ are invariant under $\mathcal{G}_{Q_t}$ at every point. The conditional law of $\sigma_{V(Q_t)}$ given $\sigma_e = \eta$ is $\nu_\eta = \nu_{Q_t}^{(\eta,\omega)}$, so for almost every η, $X_\eta, Y_\eta \in L^2(\nu_\eta)$, $\nu_\eta(X_\eta) = (E_e x)(\eta) = 0$, $\nu_\eta(Y_\eta) = 0$, and by the hypothesis with $\omega' = (\eta, \omega)$,

$$|\nu_\eta(X_\eta Y_\eta)| \le c\, \nu_\eta(X_\eta^2)^{1/2} \nu_\eta(Y_\eta^2)^{1/2}.$$

Integrating over η and using the Cauchy–Schwarz inequality, $|\langle x, y\rangle| \le c\|x\|\|y\|$, and Lemma 10.4 gives $Q_t Q_{t+1} + Q_{t+1} Q_t \ge -c(Q_t + Q_{t+1})$ on $\mathcal{K}$.

Conclusion. Lemma 10.5 applies. If $Hg = 0$ for $g \in \mathcal{K}$, then $\sum_t \|Q_t g\|^2 = \langle g, Hg\rangle = 0$, so $g = P_t g$ for all t and g is constant by Lemma 10.6. For $F \in \mathcal{K}$ the function $g = F - \nu(F)$ is orthogonal to the constants, hence to $\ker H$, and $\langle g, Hg\rangle = \sum_t \nu((F - P_t F)^2)$. □

10.4. **The patch gap and the cube condition.** For $d \ge 2$, $m \ge 4$ and $r \in \mathbb{Z}^4$ let $\Pi = r + [0, md-1)^4$, the box of base points of a patch, and for $j \in [0, m-2]^4$ let $B_j = V(r + dj + [0, 2d-1)^4)$, as in Appendix F. For an exterior configuration ω of Π write $\nu = \nu_\Pi^\omega$ and $P_j = E_{B_j}$.

Theorem 10.8 (patch gap from slab decorrelation)**.** *Let $d \ge 2$, $m \ge 4$, $0 \le c < \frac{1}{2}$, and assume* ($\mathrm{M}_{d,m}(c)$)*. Then for every r, every ω and every $F \in \mathcal{A}_\Pi^\omega$,*

$$(1-2c)^4 \operatorname{Var}_\nu(F) \le \sum_{j \in [0,m-2]^4} \nu\big((F - P_j F)^2\big).$$

Proof. For $0 \le k \le 4$ and $\tau \in [0, m-2]^k$ let $R_\tau^{(k)} = r + \prod_{i \le k}(d\tau_i + [0, 2d-1)) \times [0, md-1)^{4-k}$ and $P_\tau^{(k)} = E_{V(R_\tau^{(k)})}$, under ν. We prove by induction on k that

$$(1-2c)^k \operatorname{Var}_\nu(F) \le \sum_{\tau \in [0,m-2]^k} \nu\big((F - P_\tau^{(k)} F)^2\big) \qquad (F \in \mathcal{A}_\Pi^\omega).$$

For $k = 0$, $R^{(0)} = \Pi$ and $P^{(0)} F = \nu(F)$. Let $k < 4$, fix τ, write $R = R_\tau^{(k)}$, and split $\sigma = (\sigma', \omega')$ with $\sigma' = \sigma_{V(R)}$ and $\omega' = \sigma_{V(\Pi) \setminus V(R)}$. Take F invariant at every point under $\mathcal{G}_\Pi$ (Lemma 10.1(iii)). Since $\mathcal{G}_R \subset \mathcal{G}_\Pi$ fixes ω', each $F(\cdot, \omega')$ is invariant under $\mathcal{G}_R$. The conditional law of σ' given ω' is $\nu' = \nu_R^{(\omega', \omega)}$, so

$$\nu\big((F - P_\tau^{(k)} F)^2\big) = \int \operatorname{Var}_{\nu'}\big(F(\cdot, \omega')\big)\, d\nu(\omega').$$

In the coordinate $k+1$ the box R has width $md - 1 = ((m-2)+2)d - 1$. The boxes R_u and Q_u of Proposition 10.7 built from R with $i = k+1$ and $M = m-2$ are $R_u = R_{(\tau,u)}^{(k+1)}$ and boxes whose widths are $2d-1$ in the coordinates $\le k$, $3d-1$ in the coordinate $k+1$ and $md-1$ in the coordinates $> k+1$, split as in ($\mathrm{M}_{d,m}(c)$); all these widths lie in $[2d-1, md-1]$. By Proposition 10.7, for almost every ω',

$$(1-2c) \operatorname{Var}_{\nu'}(F(\cdot, \omega')) \le \sum_{u=0}^{m-2} \nu'\big((F(\cdot, \omega') - P_u' F(\cdot, \omega'))^2\big),$$

with P'_u the conditional expectation under ν' given σ' off $V(R^{(k+1)}_{(\tau,u)})$. The conditional law of the links of that set given the other links of Π is the same under ν and under ν', both having density proportional to $\prod_p \varphi(U_p)$ over the plaquettes meeting it; so $P'_u F(\cdot,\omega') = (P^{(k+1)}_{(\tau,u)}F)(\cdot,\omega')$, and integrating over ω' gives $\nu((F-P^{(k+1)}_{(\tau,u)}F)^2)$. Multiplying the induction hypothesis by $1-2c$ and inserting these bounds proves the case $k+1$. For $k=4$, $V(R^{(4)}_\tau)=B_\tau$. □

Theorem D (mass generation implies the cube condition)**.** *Let $d\ge 2$, $0\le c<\frac12$ and $m\ge 4$ with $m-1\ge 148/(1-2c)^4$, and assume $(\mathrm{M}_{d,m}(c))$. Then for every even L with $d\mid L$ and $L\ge 2md$ the measure μ_L with weight φ satisfies*

$$\mathrm{BP}\Big(2d-1,\frac{(1-2c)^4}{32}\Big).$$

Consequently, let (β_k,a_k) be a tuning with $a_k\to 0$ and Wilson weights, and $d_k\ge 2$ integers with $(2d_k-1)a_k\le \ell$. If $(\mathrm{M}_{d_k,m}(c))$ holds for every k with $c<\frac12$ and m as above independent of k, then statement (D) holds with $n_k=2d_k-1$ and $\varpi=(1-2c)^4/32$, and every continuum limit as in Theorem B along the tori with $d_k\mid L$ satisfies

$$\sigma(H)\subset\{0\}\cup\Big[\frac{(1-2c)^4}{1250\,e^2\ell},\infty\Big),$$

with 0 a simple eigenvalue; a nontrivial limit has a mass gap at least this large.

Proof. Theorem 10.8 says that the gauge-invariant patch gap of Definition F.3 satisfies $\gamma_{d,m}\ge(1-2c)^4$, and Theorem F.4 with $\gamma=(1-2c)^4$ gives $\mathrm{BP}(2d-1,(1-2c)^4/32)$. Along the tuning this holds on an infinite set of even tori with $L\ge 2(2d_k-1)+4$ for each k, with $(2d_k-1)a_k\le\ell$, and Theorem B gives the gap $16\varpi/(625e^2\ell)$ with $\varpi=(1-2c)^4/32$. □

The patch gap obtained, $(1-2c)^4$, does not decay with m, whereas the finite-size criterion needs only $148/(m-1)$; this slack is what allows a large but fixed number m of blocks per patch. The threshold $c<\frac12$ cannot be raised for this argument: for the massless Gaussian chain every adjacent pair of moves has maximal correlation exactly $\frac12$, the non-adjacent moves commute, and the gap of the slab dynamics tends to zero (Proposition G.4).

10.5. **Decay of correlations under every boundary condition.**

Lemma 10.9. *Let Q be a box, ω an exterior configuration, i a coordinate, and $Y_0,\dots,Y_K$ the sets $V(Q\cap\{x_i\in J_k\})$ for consecutive nonempty integer intervals $J_0<J_1<\cdots<J_K$ whose union is the i-th side of Q. Under ν^ω_Q, for each k, the families $(\sigma_{Y_l})_{l<k}$ and $(\sigma_{Y_l})_{l>k}$ are conditionally independent given σ_{Y_k}.*

Proof. The links in $V(Q)$ of a plaquette are based at points whose i-th coordinates take at most two consecutive values, so they lie in one Y_l or in two consecutive ones. Grouping the factors of the density by the least l met, the density of ν^ω_Q with respect to λ is $\prod_l \psi_l(\sigma_{Y_l},\sigma_{Y_{l+1}})$, with ψ_K a function of σ_{Y_K}, which gives the conditional independence. □

Proposition 10.10. *Assume $(\mathrm{M}^*_w(c))$ with $w\ge 1$ and $0\le c\le 1$. Let Q be a box, ω an exterior configuration, $\nu=\nu^\omega_Q$, i a coordinate, $s\in\mathbb{Z}$, $r\ge 1$, and let $f,g\in\mathcal{A}^\omega_Q$ be measurable with respect to $\sigma_{V(Q\cap\{x_i\le s\})}$ and $\sigma_{V(Q\cap\{x_i\ge s+r\})}$. Then*

$$|\operatorname{Cov}_\nu(f,g)|\le c^{\lfloor r/(w+1)\rfloor}\|f-\nu(f)\|\,\|g-\nu(g)\|.$$

Proof. If one of the two parts of Q is empty, f or g is constant and the covariance vanishes. Otherwise the i-th side of Q contains $[s,s+r]$. Let $J=\lfloor r/(w+1)\rfloor$; for $J=0$ the claim is the Cauchy–Schwarz inequality. Let $J\ge 1$ and split the i-th side into consecutive intervals: T_0, the values $\le s$; for $1\le k\le J$, $M_k=[s+(k-1)(w+1)+1,\ s+(k-1)(w+1)+w]$; for $1\le k\le J-1$, $T_k=\{s+k(w+1)\}$; and T_J, the values $\ge s+J(w+1)$, which contain the values $\ge s+r$. Use the same letters for the corresponding sets of links. By Lemma 10.9 the

sequence $T_0, M_1, T_1, \dots, M_J, T_J$ is Markov, and by Lemma 10.1 the conditional expectations given σ_{T_k} preserve $\mathcal{A}_Q^\omega$.

For $1 \le k \le J$ let $u \in \mathcal{A}_Q^\omega$ be a centred function of $\sigma_{T_{k-1}}$ and $v = E[u \mid \sigma_{T_k}]$, which is centred and lies in $\mathcal{A}_Q^\omega$. The box Q splits in the coordinate i into the values $\le s+(k-1)(w+1)$, the middle M_k of width w, and the values $\ge s + k(w+1)$; u is a function of the first part and v of the last, so by $(\mathrm{M}_w^*(c))$, $\|v\|^2 = \nu(uv) \le c\|u\|\|v\|$, that is, $\|v\| \le c\|u\|$. Let $f_0 = f - \nu(f)$ and $u_k = E[f_0 \mid \sigma_{T_k}]$. By the Markov property $u_k = E[u_{k-1} \mid \sigma_{T_k}]$, so $\|u_k\| \le c\|u_{k-1}\|$ and $\|u_J\| \le c^J\|f_0\|$. As g is σ_{T_J}-measurable, $\mathrm{Cov}_\nu(f,g) = \nu(u_J(g-\nu(g)))$, and the Cauchy–Schwarz inequality gives the claim. □

So $(\mathrm{M}_{d-1}^*(c))$ with $c<1$, which contains $(\mathrm{M}_{d,m}(c))$ for every m, gives exponential decay of the covariances of gauge-invariant observables of a box, uniformly in the box and in the boundary condition, at rate $\log(1/c)/d$ per lattice spacing, that is, of order $\log(1/c)/\ell$ in physical units along the tuning. This is close to the decay hypothesis of Chatterjee [12, Def. 2.3], under which an area law for Wilson loops follows; there the observables are local functions of a cube with prescribed boundary, and the geometries differ, so neither condition is claimed to imply the other. The converse direction is the weak point of any pairwise form: a bound $K_1 e^{-K_2 r}$ for pairs of local functions, summed over the pairs of two slabs of transverse size $(md)^3$, bounds ρ^ω only up to a factor of order $(md)^6$, which is not uniform in the lattice spacing.

10.6. **What is known about the hypothesis.** *Strong coupling at fixed d and m.* The hypothesis is not vacuous: $(\mathrm{M}_{d,m}(c))$ holds with $c = 1 - e^{-24\omega_\varphi(md)^4}$. Let Q, the splitting and ω be as in Definition 10.3, $A_1 = V(Q_- \cup Q_0)$, $A_2 = V(Q_0 \cup Q_+)$, $\nu = \nu_Q^\omega$, and let γ_2 be the best constant in

$$\gamma_2 \operatorname{Var}_\nu(g) \le \sum_{k=1,2} \nu((g - E_{A_k} g)^2), \qquad g \in L^2(\nu) \text{ real}.$$

Under λ the links are independent and $A_1 \cup A_2 = V(Q)$, so the Efron–Stein inequality [22] gives the constant 1 for λ. The logarithm of the density of ν with respect to λ has oscillation Ω at most ω_φ times the number of plaquettes meeting $V(Q)$, which is at most $24(md)^4$, and a change of the reference measure by a density whose logarithm has oscillation Ω changes the constant by at most the factor $e^{-\Omega}$ [29].[9] So $\gamma_2 \ge e^{-24\omega_\varphi(md)^4}$. Now let X, Y be as in Definition 10.2 with $r = \nu(XY) \ge 0$, and $g = X+Y$, so $\nu(g) = 0$ and $\|g\|^2 = 2+2r$. Since X is a function of links off A_2, $E_{A_2}X = X$, and $\|X - E_{A_1}X\|^2 = 1 - \|E_{A_1}X\|^2 \le 1 - r^2$ because Y is a unit vector in the range of E_{A_1} and $|\langle E_{A_1}X, Y\rangle| = |\langle X, Y\rangle| = r$; symmetrically for Y. Hence

$$\gamma_2(2+2r) \le \sum_k \|g - E_{A_k} g\|^2 \le 2(1 - r^2),$$

so $r \le 1 - \gamma_2$. The range of couplings obtained in this way shrinks with md, and it is far smaller than that of Theorem C.

Two dimensions. In two dimensions the seam between the outer parts of a box is one-dimensional, and after an axial gauge the middle slab integrates out exactly to a convolution power of the weight. This proves the two-dimensional form of the hypothesis along the weak-coupling tuning, with $c \to 0$ as the physical size of the box grows, uniformly in the boundary condition (Theorem G.8 in Appendix G). In four dimensions the seam carries a three-dimensional layer of links, and $(\mathrm{M}_{d,m}(c))$ is a bound, uniform in the boundary condition, on the second singular value of the $(d-1)$-step transfer operator of the box on gauge-invariant functions. Boundary conditions with values in the centre, 't Hooft's twisted ones [50], leave a global conjugation symmetry of the box, and in the two-dimensional strip such a symmetry is broken at weak coupling (Proposition G.14); in four dimensions the

[9] Write $d\nu/d\lambda = e^V/Z$. Then $\operatorname{Var}_\nu(g) \le Z^{-1} e^{\sup V} \operatorname{Var}_\lambda(g)$, and $\lambda((g - E_A^\lambda g)^2) \le Z e^{-\inf V} \nu((g - E_A g)^2)$, as both conditional expectations project onto the functions of the links outside A.

hypothesis asks that the twisted sectors be disordered at the scale $\ell/2$, which is the form that confinement takes here. We know no boundary condition that refutes $(\mathrm{M}_{d,m}(c))$ along the four-dimensional tuning, and no proof of it.

An area law does not suffice. The next proposition shows that the Wilson loops of the defining representation carry no information about the observables that (D) constrains.

Proposition 10.11 (central extensions)**.** *Let $\tilde G \subset \mathrm{U}(\tilde N)$ and $G \subset \mathrm{U}(N)$ be compact, $\pi : \tilde G \to G$ a continuous surjective homomorphism with finite central kernel Z, φ a positive continuous class function on G and $\tilde\varphi = \varphi \circ \pi$. On Λ_L let $\tilde\mu$, μ be the measures with weights $\tilde\varphi$, φ, and $\Pi : \tilde G^E \to G^E$ the map π applied to every link.*

(i) *$\tilde\varphi$ is a positive continuous class function and $\Pi_*\tilde\mu = \mu$.*

(ii) *For every x and $F \in L^2(\mu)$, $\tilde{\mathcal{P}}_x(F \circ \Pi) = (\mathcal{P}_x F) \circ \Pi$ and $\mathrm{Var}_{\tilde\mu}(F \circ \Pi) = \mathrm{Var}_\mu(F)$; if F is gauge invariant, so is $F \circ \Pi$. Hence $\mathrm{BP}(n, \varpi)$ for $\tilde\mu$ implies $\mathrm{BP}(n, \varpi)$ for μ.*

(iii) *If some $z \in Z$ acts on $\mathbb{C}^{\tilde N}$ as a scalar $\zeta \neq 1$, then $\tilde\mu(\operatorname{Tr} \tilde U_C) = 0$ for every closed lattice path C that traverses some link exactly once.*

Proof. (i) $\pi(hgh^{-1}) = \pi(h)\pi(g)\pi(h)^{-1}$, so $\tilde\varphi$ is a class function. The image under π of Haar measure on $\tilde G$ is a left-invariant probability measure on G, hence Haar measure, so $\Pi_*\tilde\lambda = \lambda$. As π is a homomorphism, $\pi(\tilde U_p) = (\Pi\tilde U)_p$, so $\tilde\mu$ has density $\Psi \circ \Pi$ with respect to $\tilde\lambda$, with $\Psi(U) = \tilde Z^{-1} \prod_p \varphi(U_p)$, and $\Pi_*((\Psi \circ \Pi)\tilde\lambda) = \Psi\lambda$; both being probability measures, $\Psi\lambda = \mu$.

(ii) $\tilde{\mathcal{P}}_x(F \circ \Pi)(\tilde U)$ is the integral of $F(\Pi\tilde\sigma, \Pi\tilde U_{\mathrm{out}})$ against the density proportional to $\prod_{p \cap V_x \neq \emptyset} \varphi((\Pi\tilde\sigma, \Pi\tilde U_{\mathrm{out}})_p)$ on $\tilde G^{V_x}$; by the change of variables $\sigma = \Pi\tilde\sigma$ and (i) on V_x it equals $(\mathcal{P}_x F)(\Pi\tilde U)$. The variance identity follows from (i), and $\Pi(\tilde g\tilde U) = \pi(\tilde g)\Pi(\tilde U)$ for a gauge transformation $\tilde g$ gives the invariance. Then for gauge-invariant F, $\varpi \operatorname{Var}_\mu(F) = \varpi \operatorname{Var}_{\tilde\mu}(F \circ \Pi) \le n^{-4} \sum_x \tilde\mu((F \circ \Pi - \tilde{\mathcal{P}}_x(F \circ \Pi))^2) = n^{-4} \sum_x \mu((F - \mathcal{P}_x F)^2)$.

(iii) Let e be a link traversed once by C and T the map $\tilde U_e \mapsto z\tilde U_e$, the other links being fixed. It preserves $\tilde\lambda$, and as z is central with $\pi(z) = 1$, each plaquette variable through e becomes $z^{\pm1}\tilde U_p$ with $\tilde\varphi(z^{\pm1}\tilde U_p) = \tilde\varphi(\tilde U_p)$; so $\tilde\mu$ is T-invariant. Moving the central factor out of the product, $\operatorname{Tr} \tilde U_C \circ T = \zeta^{\pm1} \operatorname{Tr} \tilde U_C$, so $\tilde\mu(\operatorname{Tr} \tilde U_C) = \zeta^{\pm1}\tilde\mu(\operatorname{Tr} \tilde U_C) = 0$. □

For instance, with $\tilde G = G = \mathrm{U}(1)$, $\pi(z) = z^2$ and the Wilson weight φ of U(1), every Wilson loop of charge one under $\tilde\varphi(z) = e^{\beta \operatorname{Re} z^2}$ that traverses some link once vanishes for every β, so Wilson's area law holds with every string tension, while by part (ii) the cube condition for $\tilde\varphi$ is at least as hard as for the U(1) theory, in which it fails at weak coupling in the Coulomb phase (Section 11). The vanishing is kinematic, a consequence of a local symmetry and not of confinement; what the example shows is that an argument deriving (D) from an area law of the defining representation cannot be valid for all compact groups and weights. For SU(N) with the Wilson action no such example is known, and nothing is claimed about whether an area law and (D) hold together there.

11. Weak coupling: what a proof of the mass gap still needs

The continuum limit of four-dimensional Yang–Mills theory is taken along the tuning dictated by asymptotic freedom: for SU(N) the bare coupling $g^2 = 2N/\beta_W$ tends to zero like $1/(2b_0 \log(1/(a\Lambda)))$ with $b_0 = 11N/(48\pi^2)$, the first coefficient of the β-function [28, 44] and a fixed physical scale Λ, so $\beta_W \to \infty$ logarithmically in $1/a$. Theorems A and 7.8 hold for $\beta_W < N^2/29$ and $\beta_W < N^2/42$ respectively; at fixed β the lattice gap in physical units, $m_G(\beta)/a$, diverges as $a \to 0$, and no continuum limit exists at fixed β. The results of this paper therefore say nothing about the continuum theory without statement (D). Three facts show what (D) asks for.

(a) A cube condition is a mass. By Theorem 9.2 the best constant for a free field is at most $s^2/(s^2 + 16d/9)$ with $s = m\ell$. At weak coupling the leading approximation of the Wilson measure, in any gauge, is a massless Gaussian field, for which the constant vanishes as $a \to 0$ at fixed ℓ. Hence (D) does not follow from perturbation theory to any finite order:

it holds only if the interaction generates a mass of order $1/\ell$, and it is a quantitative form of that mass generation.

(b) The cube must grow. For SU(N), $\beta \geq 1$ and the torus of side $2n+4$, BP(n, ϖ) implies $\varpi \leq C_N n^4(1+\log\beta)/\beta$ [8]: on a small torus at weak coupling the Polyakov loops order, and the cube dynamics can change their centre charge only through the curvature, which is small. Along the tuning $n_k = \ell/a_k$ grows like a power of $1/a_k$ while β_k grows like $\log(1/a_k)$, so this obstruction is consistent with (D) but excludes every argument at fixed cube size.

(c) The group must be non-abelian. For $G = \mathrm{U}(1)$ with the Villain action at large β the theory has a massless photon [23], and the cube condition fails at every cube size, since by Theorem 6.3 it would make the field-strength correlations summable [8]. Every proof of (D) must therefore use the non-abelian structure of G, exactly as a proof of the mass gap must.

The two ends. Bałaban's renormalisation group [2, 3, 4] controls the effective actions of the lattice theory from the scale a up to scales at which the running coupling is still small, uniformly in the lattice spacing, in finite volume. The theorems of Secs. 5–7 control the theory when the coupling is large: they need $N\beta$ small, that is, g large. The running coupling grows towards the infrared, so a proof of (D) would follow the renormalisation group from the weak-coupling regime, where Bałaban's estimates hold, through the regime of coupling of order one, to a scale ℓ at which the effective theory satisfies a strong-coupling criterion of the kind proved here, and would then transport the resulting cube estimate back to the fine lattice. Neither the middle step nor the final transport is available: the effective actions are not of Wilson form, Bałaban's analysis stops before the coupling is of order one, and no known criterion accepts the effective theory at that scale. This gap between the two ends is the Yang–Mills mass gap problem in the form isolated here.

Statement (E) is the construction problem. Bałaban's ultraviolet stability bounds give compactness of suitably smeared observables in finite volume, but the infinite-volume limit, Euclidean invariance and nontriviality of the limit are open [31]. Given (E), Theorem B reduces the mass gap to (D) with the explicit constant $\Delta \geq 16\varpi/(625e^2\ell)$.

12. Conclusion

Lattice Yang–Mills theory in four dimensions has a mass gap with the explicit lower bound $m_G(\beta)$ for every compact simple gauge group in the range $N\beta < \frac{2}{29}\kappa_G$, and for SU(2) up to $\beta_W = \frac{1}{3}$ with the bound $-\log(3\beta_W)$ and up to $\beta_W = \frac{9}{25}$ by resampling stars. A single spectral gap ϖ of the dynamics that resamples cubes of a fixed physical size ℓ, uniform along the continuum tuning, gives every nontrivial continuum limit the mass gap $16\varpi/(625e^2\ell)$. That spectral gap holds at strong coupling, for SU(2) up to $\beta_W = \frac{7}{20}$ uniformly in the cube size and for SU(3) up to $\beta_W = 0.45$, and for massive free fields; it fails for massless fields, at fixed cube size at weak coupling, and in the abelian Coulomb phase.

The cube gap follows from a decay of correlations, below the sharp threshold $\frac{1}{2}$, between the gauge-invariant observables on the two sides of a slab of physical width $\ell/2$, uniformly in the boundary condition. This hypothesis is a form of mass generation at one scale. It holds in two dimensions along the weak-coupling tuning and fails there for strips one link wide; in four dimensions it is not proved. Curvature arguments, with or without gauge fixing, stop at explicit couplings of order one; spectral gaps for blocks of fixed lattice size force a trivial continuum limit; and a lattice gap tending to zero at the right rate produces a gap in a continuum limit that is not identified with Yang–Mills theory. Statements (D) and (E) remain open, and with them the Clay problem; what this paper contributes is an explicit strong-coupling theory for every gauge group, a reduction of the continuum gap to one estimate on one scale and of that estimate to a decay of correlations across one slab, and a map of what the present methods can reach.

Appendix A. The line field

Table 2 lists, for each residue class of x modulo two, the direction $\sigma(x)$ and the values $\pi_x(\mu) = \sigma(x + e_\mu)$. In every row π_x fixes $\sigma(x)$ and is a three-cycle on the other directions.

Table 2. The line field: x modulo two (digits $x_0x_1x_2x_3$), $\sigma(x)$, and $\sigma(x+e_\mu)$ for $\mu = 0, 1, 2, 3$.

x	σ	0	1	2	3	x	σ	0	1	2	3
0000	0	0	2	3	1	0001	1	2	1	3	0
1000	0	0	3	1	2	1001	2	1	3	2	0
0100	2	3	0	2	1	0101	1	3	1	0	2
1100	3	2	0	1	3	1101	3	1	2	0	3
0010	3	1	2	0	3	0011	3	2	0	1	3
1010	1	3	1	0	2	1011	2	3	0	2	1
0110	2	1	3	2	0	0111	0	0	3	1	2
1110	1	2	1	3	0	1111	0	0	2	3	1

Appendix B. The blocks of Lemma 5.2

Diagonal blocks. Let $w_a = e^{i\phi}$. On the diagonal part $f_p = -\cos\phi\,(y_1 + y_2 - y_3 - y_4)^2$ and $c_p = 2(y_1y_4 - y_1y_2 + y_2y_3 - y_3y_4)$, so

$$\mathcal{F} = \sum_i \alpha_i y_i^2 + \cos\phi\,(y_1 + y_2 - y_3 - y_4)^2 - \tfrac{9}{5}(y_1y_4 - y_1y_2 + y_2y_3 - y_3y_4),$$

with $y_k = 0$, $\alpha_{k\pm1} = \frac{21}{10}$, $\alpha_{k+2} = \frac{5}{4}$. It is affine in $\cos\phi$, so positivity at $\cos\phi = \pm1$ suffices. For $k = 1$ and $\cos\phi = 1$ the matrix in (y_2, y_3, y_4) is

$$\begin{pmatrix} \frac{31}{10} & -\frac{19}{10} & -1 \\ -\frac{19}{10} & \frac{9}{4} & \frac{19}{10} \\ -1 & \frac{19}{10} & \frac{31}{10} \end{pmatrix},$$

with leading principal minors $\frac{31}{10}$, $\frac{673}{200}$, $\frac{8421}{2000}$. In general the leading principal minors are $\frac{31}{10}, \frac{673}{200}, \frac{8421}{2000}$ (fixed link e_1 or e_4), $\frac{31}{10}, \frac{861}{100}, \frac{8421}{2000}$ (e_2) and $\frac{9}{4}, \frac{673}{200}, \frac{8421}{2000}$ (e_3) at $\cos\phi = 1$, and $\frac{11}{10}, \frac{53}{200}, \frac{21}{2000}$ (e_1, e_4), $\frac{11}{10}, \frac{21}{100}, \frac{21}{2000}$ (e_2) and $\frac{1}{4}, \frac{53}{200}, \frac{21}{2000}$ (e_3) at $\cos\phi = -1$; all are positive.

Pair blocks. Let $w_a = e^{i(\chi+\vartheta)}$ and $w_b = e^{i(\chi-\vartheta)}$. The terms of f_p carry $w_a + \bar{w}_b = 2\cos\chi\, e^{i\vartheta}$ or its conjugate, and c_p carries $w_a\bar{w}_b = e^{2i\vartheta}$, so the Hermitian matrix $M(m, \vartheta)$ of $\mathcal{F}$, $m = \cos\chi$, is affine in m with $M(-1, \vartheta) = M(1, \vartheta + \pi)$, and $m = 1$ suffices. For the fixed link e_2, in the variables (z_1, z_3, z_4) and with $c = \cos\vartheta$,

$$M = \begin{pmatrix} \frac{21}{5} + 2c & -2e^{i\vartheta} & -\frac{9}{5} - 2e^{i\vartheta} \\ -2e^{-i\vartheta} & \frac{21}{5} + 2c & \frac{9}{5} + 2e^{-i\vartheta} \\ -\frac{9}{5} - 2e^{-i\vartheta} & \frac{9}{5} + 2e^{i\vartheta} & \frac{5}{2} + 2c \end{pmatrix}.$$

With $u = \cos^2(\vartheta/2) \in [0, 1]$ its leading principal minors are

$$\tfrac{20u+11}{5}, \qquad \tfrac{(20u+1)(20u+21)}{25}, \qquad \tfrac{400u(40u^2-23u+4)+21}{250},$$

and $40u^2 - 23u + 4$ has negative discriminant $529 - 640$. The other positions give the same determinant and the leading minors $\frac{20u+11}{5}$ or $\frac{8u+1}{2}$ and $\frac{800u^2-180u+53}{50}$, whose discriminant $32400 - 169600$ is negative. Every block is positive definite.

Appendix C. The single-link law of SU(2)

This appendix proves the sharp quarter, $W_1(\nu_b, \nu_{b'}) \le \frac{1}{4}|b - b'|$ for the one-link law of SU(2) in the chordal metric, which the proof of Theorem A for SU(2), Lemma 7.6(i) and the whole of Sec. 8 use, together with the exact value $\frac{1}{4}$ at $b = 0$, which shows that the single-link argument of Sec. 4 stops at $\beta_W = \frac{1}{3}$. The argument is that of [6]; it is given in full so that the paper is self-contained. The computations it calls on are in Appendix D, and the facts about Bessel functions in Appendix E.

C.1. **The von Mises–Fisher law.** Identify SU(2) with the unit quaternions, that is with the unit sphere $S^3 \subset \mathbb{R}^4$, so that $\operatorname{Re}\operatorname{Tr} U = 2x_0(U)$ and $\operatorname{Re}\operatorname{Tr}(UV) = 2\langle u, \bar v\rangle$ for the Euclidean inner product of $\mathbb{R}^4$ and quaternion conjugation $v \mapsto \bar v$. Fix a link e and the field off e. Every plaquette through e can be rotated cyclically so that its trace reads $\operatorname{Re}\operatorname{Tr}(Q_e A_p)$ with $A_p \in \mathrm{SU}(2)$, the staple of e in p. The action therefore depends on Q_e only through

$$N\beta \sum_{p \ni e} \operatorname{Re}\operatorname{Tr}(Q_e A_p) = \langle x, b\rangle, \qquad b = 4\beta \sum_{p \ni e} \bar A_p \in \mathbb{R}^4,$$

with $x = Q_e \in S^3$. Since there are six staples, each of norm one,

$$\kappa := |b| \le y := 24|\beta|. \tag{C.1}$$

The conditional law of Q_e is the von Mises–Fisher law[10]

$$\nu_b(dx) = Z(\kappa)^{-1} e^{\langle x, b\rangle}\, \sigma(dx) \tag{C.2}$$

on the unit sphere, with σ the normalised surface measure. We use coordinates in which $b = \kappa e_1$. The density of $t = x_1$ is proportional to $(1-t^2)^{1/2} e^{\kappa t}$ on $[-1, 1]$, so $Z(\kappa) = 2I_1(\kappa)/\kappa$ with the modified Bessel function I_1, normalised so that $Z(0) = 1$.

Lemma C.1. *For every $\kappa \ge 0$ the law ν_b on the unit sphere S^3 satisfies the Poincaré inequality $\operatorname{Var}_{\nu_b}(g) \le \frac{1}{3}\int |\nabla g|^2 d\nu_b$, and the constant $\frac{1}{3}$ is attained at $\kappa = 0$.*

Proof. In the polar angle $\theta \in (0, \pi)$ from e_1 the generator of the reversible diffusion of ν_b is $L = \partial_\theta^2 + b(\theta)\partial_\theta + \sin^{-2}\theta\, \Delta_{S^2}$ with $b(\theta) = 2\cot\theta - \kappa\sin\theta$. On spherical harmonics of degree ℓ on S^2 it acts as a Sturm–Liouville operator L_ℓ with potential $\ell(\ell+1)\sin^{-2}\theta$. A ground state comparison of Barta's type [5] with the trial function $\psi = \sin\theta\, e^{a\kappa\cos\theta}$ gives $-L_1\psi \ge 3\psi$ for $a = -\frac{1}{5}$, and the same trial function with $a = -\frac{2}{5}$ gives the bound 3 for the derivatives of zonal eigenfunctions; Appendix D carries out both computations. Hence every nonzero eigenvalue of $-L$ is at least 3. At $\kappa = 0$ the first eigenvalue of the Laplacian of the unit S^3 is 3. □

C.2. **Moments.** Let $A(\kappa) = \mathbb{E}_{\nu_b} x_1 = Z'(\kappa)/Z(\kappa) = I_2(\kappa)/I_1(\kappa)$.

Lemma C.2. *For every $\kappa > 0$, in coordinates with $b = \kappa e_1$,*

(i) $\mathbb{E}x_2^2 = \mathbb{E}x_3^2 = \mathbb{E}x_4^2 = A/\kappa$, $\mathbb{E}(1 - x_1^2) = 3A/\kappa$, *the covariance matrix of x is* $\operatorname{diag}(A', A/\kappa, A/\kappa, A/\kappa)$, *and* $A' = 1 - A^2 - 3A/\kappa$;

(ii) $\dfrac{\kappa}{2 + \sqrt{\kappa^2 + 4}} \le A(\kappa) \le \dfrac{\kappa}{1 + \sqrt{\kappa^2 + 9}}$;

(iii) $A' \le A/\kappa \le \big(1 + \sqrt{\kappa^2 + 9}\big)^{-1} \le \frac{1}{4}$;

(iv) $A(\kappa) \ge \kappa/\big(\frac{3}{2} + \sqrt{\kappa^2 + 25/4}\big)$.

[10] On the sphere $S^{p-1} \subset \mathbb{R}^p$ the von Mises–Fisher law with mean direction μ and concentration κ has density proportional to $e^{\kappa\langle x, \mu\rangle}$ with respect to the surface measure, and its normalising constant is $(2\pi)^{p/2} I_{p/2-1}(\kappa)/\kappa^{p/2-1}$; it is the maximum entropy law with prescribed mean resultant, and it plays on the sphere the role of the Gaussian in $\mathbb{R}^p$. For $p = 4$ the Bessel index is 1. The mean resultant length $\mathbb{E}\langle x, \mu\rangle = I_{p/2}(\kappa)/I_{p/2-1}(\kappa)$ is the function A of Lemma C.2. See Mardia and Jupp [38], Ch. 9, for these facts and for the tangent-normal decomposition used in Appendix D.

Proof. (i) Integrating by parts against $\frac{d}{dt}(1-t^2)^{3/2} = -3t(1-t^2)^{1/2}$ gives $\kappa\,\mathbb{E}(1-x_1^2) = 3\mathbb{E}x_1$, and the three coordinates x_2, x_3, x_4 are exchangeable under ν_b, with $\sum_i x_i^2 = 1$. The off-diagonal covariances vanish by the reflections $x_i \mapsto -x_i$, $i \geq 2$, and $A' = \operatorname{Var} x_1$ because $A = (\log Z)'$. (ii) By (i), A solves the Riccati equation $A' = 1 - A^2 - 3A/\kappa$ with $A(\kappa) = \kappa/4 + O(\kappa^3)$. Appendix D shows that the left side of (ii) is a subsolution and the right side a supersolution of this equation, with the same behaviour at 0, and concludes by comparison. (iii) The middle inequality is (ii). For the first, put $s = \sqrt{\kappa^2+4}$. By (ii), $4A/\kappa \geq 4/(2+s)$ and $1 - A^2 \leq 1 - \kappa^2/(2+s)^2$, and

$$1 - \frac{\kappa^2}{(2+s)^2} \leq \frac{4}{2+s} \iff (2+s)^2 - \kappa^2 \leq 4(2+s)$$
$$\iff s^2 - \kappa^2 \leq 4,$$

which holds with equality. So $A' = 1 - A^2 - 3A/\kappa \leq A/\kappa$. (iv) is the comparison of (ii) with a third barrier, which improves the lower bound of (ii); it is carried out in the same appendix. □

C.3. **The Lipschitz rate of the family.** Let δ be the chordal metric $\delta(x,x') = |x - x'|$ on S^3 and W_1 the Kantorovich distance for δ. For $c \in \mathbb{R}^4$ and a unit vector $h \in \mathbb{R}^4$ put

$$L(c,h) = \sup_g \operatorname{Cov}_{\nu_c}(g, \langle h, x\rangle), \tag{C.3}$$

the supremum over 1-Lipschitz functions g for δ. Since $\frac{d}{dt}\nu_{c+th}(g) = \operatorname{Cov}_{\nu_{c+th}}(g, \langle h, x\rangle)$, the Kantorovich–Rubinstein duality gives, for all b, b',

$$W_1(\nu_b, \nu_{b'}) \leq |b - b'| \sup_{c \in [b,b']} L\Big(c, \frac{b'-b}{|b'-b|}\Big). \tag{C.4}$$

By rotation invariance $L(c,h)$ depends only on $\kappa = |c|$ and on the angle φ between h and $c^\perp$, and we write $L(\kappa, \varphi)$.

Lemma C.3. *For all c and h, $L(c,h) \leq 1/(2\sqrt{3})$. Moreover:*

(i) *at $c = 0$, $L(0,h) = \frac{1}{4}$ for every h;*
(ii) *if $h \perp c$, then $L(c,h) = \mathbb{E}_{\nu_c} x_2^2 = A(\kappa)/\kappa \leq \frac{1}{4}$;*
(iii) *if $\kappa \geq \sqrt{88}/3 \approx 3.127$, then $L(c,h) \leq \frac{1}{4}$ for every h.*

Proof. A 1-Lipschitz function for δ is 1-Lipschitz for the geodesic distance, which is larger, so $|\nabla g| \leq 1$ almost everywhere, and Lemma C.1 gives $\operatorname{Var} g \leq \frac{1}{3}$. By Lemma C.2(i), (iii) the largest eigenvalue of the covariance of x is $A/\kappa \leq \frac{1}{4}$, so $\operatorname{Var}\langle h, x\rangle \leq A/\kappa$. The Cauchy–Schwarz inequality gives

$$L(c,h) \leq \big(\tfrac{1}{3}A/\kappa\big)^{1/2} \leq \Big(3\big(1 + \sqrt{\kappa^2+9}\big)\Big)^{-1/2} \leq \frac{1}{2\sqrt{3}}. \tag{C.5}$$

(i) and (ii) are reflection arguments. Let R be the reflection in the hyperplane $h^\perp$, and suppose ν_c is R-invariant, which is the case when $c = 0$ or $h \perp c$. Then $\mathbb{E}\langle h, x\rangle = 0$ and

$$\operatorname{Cov}(g, \langle h, x\rangle) = \mathbb{E}\big[(g(x) - g(Rx))\langle h, x\rangle;\ \langle h, x\rangle > 0\big]$$
$$\leq \mathbb{E}\big[2\langle h, x\rangle^2;\ \langle h, x\rangle > 0\big] = \mathbb{E}\langle h, x\rangle^2,$$

since $\delta(x, Rx) = 2|\langle h, x\rangle|$. Equality holds for $g = \langle h, x\rangle$, which is 1-Lipschitz. At $c = 0$, $\mathbb{E}\langle h, x\rangle^2 = \frac{1}{4}$; for $h \perp c$, $\mathbb{E}\langle h, x\rangle^2 = A/\kappa$. (iii) By (C.5), $L \leq \frac{1}{4}$ as soon as $1 + \sqrt{\kappa^2+9} \geq \frac{16}{3}$, that is $\kappa^2 \geq \frac{88}{9}$. □

The reflection settles the directions $h \perp c$ and the point $c = 0$; for an oblique h the reflection no longer preserves ν_c, and the Cauchy–Schwarz bound (C.5) is too weak for $\kappa < \sqrt{88}/3$. The next theorem closes this case.

Theorem C.4 (The sharp quarter)**.** *For every $c \in \mathbb{R}^4$ and every unit vector $h \in \mathbb{R}^4$,*

$$L(c,h) \le \tfrac{1}{4},$$

with equality at $c = 0$. Equivalently, $W_1(\nu_b, \nu_{b'}) \le \frac{1}{4}|b-b'|$ for all $b, b' \in \mathbb{R}^4$, and the constant $\frac{1}{4}$ is the best possible.

The equivalence is (C.4) in one direction and the derivative formula above it in the other; the sharpness is Lemma C.3(i). The metric matters.[11] The proof has four steps. A Lipschitz function on S^3 extends to the unit ball $B \subset \mathbb{R}^4$, and Green's formula in B bounds the covariance by a negative Sobolev norm of the density (Lemma C.5). This norm splits into a zonal and a transverse part, both explicit series in modified Bessel functions (Lemma C.6). The transverse part is at most $\frac{1}{4}$ because a certain power series has nonnegative coefficients (Proposition C.8), and the zonal part is at most $\frac{1}{4}$ by a Riccati comparison (Proposition C.9).

C.4. **Harmonic extension.** Let σ be the normalised surface measure of S^3 and H_k the space of spherical harmonics of degree k, the restrictions to S^3 of harmonic homogeneous polynomials of degree k on $\mathbb{R}^4$. The spaces H_k are orthogonal in $L^2(\sigma)$ and span it; we write u_k for the component of u in H_k, and for u with $\int u\, d\sigma = 0$ we put

$$\mathcal{Q}(u) = \sum_{k \ge 1} \frac{1}{k}\, \|u_k\|^2_{L^2(\sigma)}. \tag{C.6}$$

This is $\langle u, \Lambda^{-1} u\rangle$ for the Dirichlet-to-Neumann operator Λ of B, which acts on H_k as multiplication by k.

Lemma C.5. *Let u be a real-analytic function on S^3 with $\int u\, d\sigma = 0$ and let g be 1-Lipschitz for the chordal metric. Then*

$$\int_{S^3} g\, u\, d\sigma \le \tfrac{1}{2}\, \mathcal{Q}(u)^{1/2}.$$

Proof. The chordal metric is the Euclidean metric of $\mathbb{R}^4$ restricted to S^3, so McShane's extension $G(y) = \inf_{x \in S^3}(g(x) + |x - y|)$ is 1-Lipschitz on $\mathbb{R}^4$ and equals g on S^3 [36]; by Rademacher's theorem $|\nabla G| \le 1$ almost everywhere. Let $\psi = \sum_{k \ge 1} k^{-1} r^k u_k(\omega)$ in polar coordinates $y = r\omega$. Each term is a harmonic polynomial, the series converges with all derivatives on the closed ball because u is real-analytic,[12] and $\partial_r \psi = u$ on S^3. Green's formula for the Lipschitz function G and the smooth harmonic function ψ reads $\int_{S^3} G\, \partial_r \psi\, dS = \int_B \nabla G \cdot \nabla \psi\, dy$. With $|S^3| = 2\pi^2$ and $|B| = \pi^2/2$ the Cauchy–Schwarz inequality gives

$$\int g\, u\, d\sigma = \frac{1}{2\pi^2} \int_B \nabla G \cdot \nabla \psi\, dy \le \frac{|B|^{1/2}}{2\pi^2} \Big(\int_B |\nabla \psi|^2 dy \Big)^{1/2}.$$

Green's formula once more gives $\int_B |\nabla \psi|^2 = \int_{S^3} \psi\, \partial_r \psi\, dS = 2\pi^2 \sum_k k^{-1} \|u_k\|^2 = 2\pi^2 \mathcal{Q}(u)$, and the right side becomes $\frac{1}{2}\mathcal{Q}(u)^{1/2}$. □

The lemma is a form of Thomson's principle. By Kantorovich–Rubinstein duality, $\sup_g \int g u\, d\sigma$ is the least cost of a flow in $\mathbb{R}^4$ that carries the positive part of $u\, d\sigma$ to its

[11] In the geodesic metric of S^3 the statement is false already at $c = 0$: the function $\arcsin x_2$ is geodesically 1-Lipschitz and its covariance with x_2 under the uniform law is $\frac{8}{9\pi} \approx 0.283 > \frac{1}{4}$. A proof therefore has to use that the chordal distance between two points is shorter than their geodesic distance, that is, it has to see the sphere from the ball it bounds. The proof below does exactly this: it extends a chordally Lipschitz function to the ball and integrates by parts there.

[12] For real-analytic u the norms $\|u_k\|$ decay geometrically in k, and the supremum of a function in H_k is at most $(\dim H_k)^{1/2} = k + 1$ times its $L^2(\sigma)$ norm, so the series for ψ and for each of its derivatives converge uniformly on $\bar{B}$.

negative part, the cost being the total length travelled by the mass.[13] The harmonic field $\nabla\psi$ is one such flow, through the interior of the ball, and the lemma computes its cost. At $c = 0$ the flow is optimal: for $u = \langle h, x\rangle$ one has $\psi = \langle h, y\rangle$, the field $\nabla\psi = h$ is constant, the Cauchy–Schwarz inequality is an equality, and $\mathcal{Q}(u) = \|u\|^2 = \frac{1}{4}$.

C.5. **Spectral decomposition.** Fix $c = \kappa e_1$ with $\kappa > 0$ and write $h = \cos\varphi\, e_1 + \sin\varphi\, e_2$, which is no loss of generality by rotation invariance. Let $\rho = d\nu_c/d\sigma = e^{\kappa x_1}/Z(\kappa)$, $Z(\kappa) = \int e^{\kappa x_1} d\sigma = 2I_1(\kappa)/\kappa$, and

$$u = \big(\langle h, x\rangle - \mathbb{E}_{\nu_c}\langle h, x\rangle\big)\rho,$$

so that $\mathrm{Cov}_{\nu_c}(g, \langle h, x\rangle) = \int gu\, d\sigma$ and $\int u\, d\sigma = 0$. Let U_n be the Chebyshev polynomials of the second kind and put

$$d_n(\kappa) = \frac{(n+1)\, I_{n+1}(\kappa)}{I_1(\kappa)}, \qquad n \geq 0. \tag{C.7}$$

Lemma C.6. *With this notation*

$$\mathcal{Q}(u) = \cos^2\varphi\, \mathcal{Q}_z(\kappa) + \sin^2\varphi\, \mathcal{Q}_x(\kappa),$$

where

$$\mathcal{Q}_z = \sum_{n\geq 1} \frac{d_n'(\kappa)^2}{n}, \qquad \mathcal{Q}_x = \frac{1}{3\kappa^2} \sum_{n\geq 1} (n+2)\, d_n(\kappa)^2.$$

Proof. The functions $U_n(x_1)$, $n \geq 0$, are the zonal harmonics: $U_n(x_1) \in H_n$, and they are orthonormal in $L^2(\sigma)$ (Appendix E). The expansion $e^{\kappa t} = \frac{2}{\kappa}\sum_n (n+1) I_{n+1}(\kappa) U_n(t)$ of the same appendix gives $\rho = \sum_n d_n(\kappa) U_n(x_1)$, with $d_0 = 1$.

Write $u = \cos\varphi\, u_z + \sin\varphi\, u_x$ with $u_z = (x_1 - A)\rho$ and $u_x = x_2\rho$; here $\mathbb{E}_{\nu_c} x_1 = A$ and $\mathbb{E}_{\nu_c} x_2 = 0$. The reflection $x_2 \mapsto -x_2$ maps each H_k to itself, fixes u_z and reverses u_x, so the components of u_z and u_x in each H_k are orthogonal and the cross term in $\mathcal{Q}(u)$ vanishes.

Since $A = Z'/Z$, we have $u_z = \partial_\kappa \rho = \sum_n d_n'(\kappa) U_n(x_1)$, and the component of u_z in H_n is $d_n' U_n(x_1)$, of norm $|d_n'|$. This gives $\mathcal{Q}_z$.

For u_x, let $y(t) = \cos t\, e_1 + \sin t\, e_2$. The law $\nu_{\kappa y(t)}$ has density $\rho_t = e^{\kappa\langle y(t), x\rangle}/Z(\kappa) = \sum_n d_n(\kappa) U_n(\langle y(t), x\rangle)$, and $\partial_t \rho_t|_{t=0} = \kappa x_2 \rho$. The function $U_n(\langle y(t), x\rangle)$ lies in H_n for every t, hence so does its derivative $x_2 U_n'(x_1)$ at $t = 0$, and the component of u_x in H_n is $\kappa^{-1} d_n x_2 U_n'(x_1)$. Its norm follows from the eigenvalue $n(n+2)$ of the Laplacian of S^3 on H_n. For $f = U_n(x_1)$,

$$\begin{aligned} n(n+2) = n(n+2)\|f\|^2 &= \int |\nabla_{S^3} f|^2 d\sigma \\ &= \int U_n'(x_1)^2 (1 - x_1^2)\, d\sigma, \end{aligned}$$

since the tangential gradient of x_1 has squared length $1 - x_1^2$. Given x_1, the point (x_2, x_3, x_4) is uniform on the sphere of radius $(1 - x_1^2)^{1/2}$, so $\mathbb{E}[x_2^2 \mid x_1] = \frac{1}{3}(1 - x_1^2)$ and $\|x_2 U_n'(x_1)\|^2 = \frac{1}{3} n(n+2)$. Summing $n^{-1}\kappa^{-2} d_n^2 \cdot \frac{1}{3} n(n+2)$ over $n \geq 1$ gives $\mathcal{Q}_x$. □

By Lemmas C.5 and C.6,

$$L(\kappa, \varphi) \leq \tfrac{1}{2} \max\big(\mathcal{Q}_z(\kappa), \mathcal{Q}_x(\kappa)\big)^{1/2}, \tag{C.8}$$

and Theorem C.4 follows once both parts are at most $\frac{1}{4}$. Both are $\frac{1}{4}$ at $\kappa = 0$, by continuity from the values $d_1 \sim \kappa/2$, $d_1' \to \frac{1}{2}$ and $d_n = O(\kappa^n)$.

[13] This is the Beckmann formulation of the transport problem with cost $|x - y|$: the optimal cost equals the least $\int |V|$ over vector measures V on $\mathbb{R}^4$ with divergence $u\, d\sigma$. Restricting the flow to the closed ball does not change the infimum, since projecting onto $\bar{B}$ shortens paths. The gradient $\nabla\psi$ on B is one admissible flow. It minimises the energy $\int |V|^2$ rather than the cost $\int |V|$, and the Cauchy–Schwarz inequality converts one into the other at the price of the factor $|B|^{1/2}$.

C.6. **Three Bessel sums.** The series of Lemma C.6 reduce to closed forms. Write I_j for $I_j(\kappa)$.

Lemma C.7. *For every $\kappa > 0$,*

$$\sum_{j\ge1} j^2 I_j^2 = \frac{\kappa}{4}\, I_1(2\kappa), \qquad \sum_{j\ge1} j^3 I_j^2 = \frac{\kappa^2}{4}(I_0^2 + I_1^2),$$
$$I_1(2\kappa) = 2\sum_{j\ge1} I_{j-1} I_j.$$

The first and third identities follow from Parseval's formula for $e^{\kappa\cos\theta} = \sum_{j\in\mathbb{Z}} I_j e^{ij\theta}$, and the second from the recurrences of I_j by telescoping; Appendix E gives the proofs.

C.7. **The transverse part.**

Proposition C.8. $\mathcal{Q}_x(\kappa) \le \frac14$ *for every $\kappa \ge 0$.*

Proof. By (C.7), $\mathcal{Q}_x = (3\kappa^2 I_1^2)^{-1}\sum_{j\ge2} j^2(j+1)I_j^2$, and the sum equals $\sum_{j\ge1}(j^3+j^2)I_j^2 - 2I_1^2$. By Lemma C.7, $\mathcal{Q}_x \le \frac14$ is equivalent to

$$\mathcal{D}(\kappa) := (8+2\kappa^2)I_1^2 - \kappa^2 I_0^2 - \kappa I_1(2\kappa) \ge 0. \tag{C.9}$$

Each term is an entire function of κ that is even after the factor κ^2 is removed, so $\mathcal{D}(\kappa) = \sum_{n\ge0} \mathcal{D}_n \kappa^{2n}$. Lemma E.2 of Appendix E computes the coefficients: $\mathcal{D}_0 = \mathcal{D}_1 = 0$, and for $n = m+1 \ge 2$

$$\mathcal{D}_n = \frac{N_m}{4^m\, m!\,(m+1)!\,(m+1)(m+2)},$$

with

$$N_m = \binom{2m}{m}\big(4(2m+1) + (m-1)(m+1)(m+2)\big) - 4^m(m+1)(m+2).$$

The numerator N_m vanishes for $m = 0, 1, 2$ and is positive for $m \ge 3$ (Lemma E.2). So every coefficient of $\mathcal{D}$ is nonnegative and $\mathcal{D} \ge 0$. □

The first nonzero coefficient is $\mathcal{D}_4 = \frac{1}{2304}$, so $\frac14 - \mathcal{Q}_x = \mathcal{D}/(12\kappa^2 I_1^2)$ is of order κ^4 at 0: the transverse part leaves the value $\frac14$ very slowly, which is why no estimate that loses a constant can prove the proposition.

C.8. **The zonal part.** Let $A = I_2/I_1$ as before, so that $d_1 = 2A$ and, by Lemma C.2(i), $A' = 1 - A^2 - 3A/\kappa$.

Proposition C.9. *For $0 < \kappa \le \sqrt{88}/3$,*

$$\mathcal{Q}_z(\kappa) \le \frac14 + 4A'^2 + \frac{A^2}{4} - \frac{A}{2\kappa} - \frac{2A^2}{\kappa^2} < \frac14.$$

Proof. The higher modes. With $r_\nu = I_{\nu+1}/I_\nu$ the recurrence $I_\nu' = I_{\nu+1} + \frac{\nu}{\kappa} I_\nu$ gives

$$\frac{d_n'}{d_n} = \frac{I_{n+1}'}{I_{n+1}} - \frac{I_1'}{I_1} = \frac{n}{\kappa} + r_{n+1} - r_1.$$

By Lemma E.3 of Appendix E, $r_{n+1} \le r_1$, and $r_1 = A \le \kappa/4$ by Lemma C.2(iii). Hence $-n/\kappa \le n/\kappa - \kappa/4 \le d_n'/d_n \le n/\kappa$ for $n \ge 2$ and $\kappa^2 \le 16$, so $d_n'^2 \le n^2 d_n^2/\kappa^2$. With $d_1' = 2A'$,

$$\mathcal{Q}_z \le 4A'^2 + \frac{1}{\kappa^2}\sum_{n\ge2} n\, d_n^2 = 4A'^2 + \frac{S_3 - S_2 - 4I_2^2}{\kappa^2 I_1^2},$$

where $S_2 = \sum_{j\ge1} j^2 I_j^2$ and $S_3 = \sum_{j\ge1} j^3 I_j^2$; the second form uses $n\, d_n^2 I_1^2 = (j-1)j^2 I_j^2$ with $j = n+1$.

The closed form. The recurrences $I_0 - I_2 = \frac{2}{\kappa} I_1$ and $I_1 - I_3 = \frac{4}{\kappa} I_2$ give $I_0/I_1 = 2/\kappa + A$ and $I_2 I_3 / I_1^2 = A - 4A^2/\kappa$. By Lemma C.7, $S_3/(\kappa^2 I_1^2) = \frac{1}{4}(2/\kappa + A)^2 + \frac{1}{4}$, and, keeping three terms of the third identity,

$$\frac{S_2}{\kappa^2 I_1^2} \geq \frac{I_0 I_1 + I_1 I_2 + I_2 I_3}{2\kappa I_1^2} = \frac{1}{2\kappa}\Big(\frac{2}{\kappa} + 3A - \frac{4A^2}{\kappa}\Big).$$

Collecting terms, $\mathcal{Q}_z \leq \frac{1}{4} + 4A'^2 + T(A)$ with $T(a) = \frac{a^2}{4} - \frac{a}{2\kappa} - \frac{2a^2}{\kappa^2}$, which is the first inequality.

The barrier. Let $s = \sqrt{\kappa^2 + 25/4}$ and $A_\flat = \kappa/(\frac{3}{2} + s)$. By Lemma C.2(iv), $A_\flat \leq A \leq \kappa/4$. The function $a \mapsto 1 - a^2 - 3a/\kappa$ decreases on $a \geq 0$, and $A' \geq 0$ because $A' = \operatorname{Var} x_1$, so $0 \leq A' \leq \alpha := 1 - A_\flat^2 - 3A_\flat/\kappa$. For $\kappa^2 \leq 12$ the quadratic T decreases on $[0, \kappa/4]$, since

$$T(a) - T(a') = (a' - a)\,\frac{2\kappa - (a + a')(\kappa^2 - 8)}{4\kappa^2}$$

and $(a + a')(\kappa^2 - 8) \leq \frac{\kappa}{2}(\kappa^2 - 8) \leq 2\kappa$ there. So $\mathcal{Q}_z \leq \frac{1}{4} + 4\alpha^2 + T(A_\flat)$. With $\kappa^2 = s^2 - \frac{25}{4}$ and $A_\flat/\kappa = 2/(2s+3)$ both terms are rational in s: $\alpha = 16/(2s+3)^2$, and

$$4\alpha^2 + T(A_\flat) = \frac{(2s-5)\,p(s)}{4(2s+3)^4}, \qquad p(s) = 8s^3 + 28s^2 - 98s - 695. \tag{C.10}$$

For $0 < \kappa \leq \sqrt{88}/3$ we have $\frac{5}{2} < s \leq \sqrt{577}/6 < 4.004$. On $s \geq \frac{5}{2}$ the cubic p increases, since $p'(s) = 24s^2 + 56s - 98 > 0$, and $p(4.004) < -120$. So the right side of (C.10) is negative. □

C.9. **Proof of Theorem C.4.** At $c = 0$ the theorem is Lemma C.3(i), and for $|c| \geq \sqrt{88}/3$ it is Lemma C.3(iii). For $0 < |c| < \sqrt{88}/3$, Propositions C.8 and C.9 give $\mathcal{Q}_x, \mathcal{Q}_z \leq \frac{1}{4}$, and (C.8) gives $L \leq \frac{1}{4}$. □

Appendix D. Computations for the single-link law

D.1. **Barta's inequality.** Let $\rho(\theta) = \sin^2\theta\, e^{\kappa\cos\theta}$ on $(0, \pi)$, let V be a continuous potential, and let

$$T = -\rho^{-1}\partial_\theta\, \rho\, \partial_\theta + V.$$

If $\psi > 0$ is smooth on $(0, \pi)$ with $\rho\psi^2 = O(\sin^4\theta)$ at the end points and $T\psi \geq \lambda_0\psi$, then

$$\int (|h'|^2 + Vh^2)\rho\, d\theta \geq \lambda_0 \int h^2 \rho\, d\theta$$

for every smooth h vanishing to first order at 0 and π. Indeed, writing $h = \psi u$ and integrating by parts, the boundary terms $\rho\psi\psi' u^2$ vanish and

$$\int (|h'|^2 + Vh^2)\rho = \int \rho\psi^2 |u'|^2 + \int \frac{T\psi}{\psi}\, h^2\rho \;\geq\; \lambda_0 \int h^2\rho.$$

This is the inequality of Barta [5] in the form of a ground state transformation.

D.2. **The computations of Lemma C.1.** The generator of the reversible diffusion of ν_b on S^3 is $L = \Delta_{S^3} + \kappa \nabla x_1 \cdot \nabla$. In the polar angle θ from e_1 the round metric is $d\theta^2 + \sin^2\theta\, g_{S^2}$, so $\Delta_{S^3} = \partial_\theta^2 + 2\cot\theta\,\partial_\theta + \sin^{-2}\theta\,\Delta_{S^2}$, and $\nabla x_1 \cdot \nabla = -\sin\theta\,\partial_\theta$. On $h(\theta)Y_\ell$ with a spherical harmonic Y_ℓ of degree ℓ on S^2 the operator acts as $L_\ell = \partial_\theta^2 + b\,\partial_\theta - \ell(\ell+1)\sin^{-2}\theta$, $b = 2\cot\theta - \kappa\sin\theta = \rho'/\rho$, so $-L_\ell = T$ with $V = \ell(\ell+1)\sin^{-2}\theta$.

Degree $\ell \geq 1$. Take $\psi = \sin\theta\, e^{a\kappa\cos\theta}$. Then $\psi'/\psi = \cot\theta - a\kappa\sin\theta$ and $\psi''/\psi = (\psi'/\psi)^2 + (\psi'/\psi)'$, and a short computation gives

$$\frac{-L_1\psi}{\psi} = 3 + (1 + 5a)\kappa\cos\theta - a(1+a)\kappa^2\sin^2\theta.$$

For $a = -\frac{1}{5}$ this is $3 + \frac{4}{25}\kappa^2 \sin^2\theta \geq 3$. Since $V = \ell(\ell+1)\sin^{-2}\theta \geq 2\sin^{-2}\theta$ for $\ell \geq 1$, Barta's inequality gives $-L_\ell \geq 3$ on every sector with $\ell \geq 1$.

Degree $\ell = 0$. Let $g(\theta)$ be a zonal eigenfunction, $-L_0 g = \lambda g$, g not constant, so that $g'(0) = g'(\pi) = 0$. Differentiating $g'' + bg' = -\lambda g$ gives, for $h = g'$,

$$-(h'' + bh') - b'h = \lambda h, \qquad -b' = \frac{2}{\sin^2\theta} + \kappa\cos\theta,$$

so h is an eigenfunction of T with $V = 2\sin^{-2}\theta + \kappa\cos\theta$, with eigenvalue λ, and $h \not\equiv 0$ vanishes at both ends. With the same ψ,

$$\frac{T\psi}{\psi} = 3 + (2+5a)\kappa\cos\theta - a(1+a)\kappa^2\sin^2\theta,$$

which for $a = -\frac{2}{5}$ is $3 + \frac{6}{25}\kappa^2\sin^2\theta \geq 3$. Barta's inequality applied to h gives $\lambda \geq 3$. Together with the case $\ell \geq 1$ every nonzero eigenvalue of $-L$ is at least 3, which is Lemma C.1; the spectrum is discrete because S^3 is compact and the drift is smooth.

D.3. **Moments.** The density of $t = x_1$ under ν_b is proportional to $(1-t^2)^{1/2}e^{\kappa t}$. For every smooth ϕ on $[-1,1]$,

$$\frac{d}{dt}\Big[(1-t^2)^{3/2}e^{\kappa t}\Big] = (1-t^2)^{1/2}e^{\kappa t}(\kappa(1-t^2) - 3t),$$

and integration by parts gives $\mathbb{E}[\phi'(t)(1-t^2)] = \mathbb{E}\big[\phi(t)\big(3t - \kappa(1-t^2)\big)\big]$. With $\phi = 1$ this is $\kappa\mathbb{E}(1-x_1^2) = 3\mathbb{E}x_1$, the identity of Lemma C.2(i). With $\phi = t^k$ it gives the recursion

$$m_{k+2} = m_k - \frac{k+3}{\kappa}\,m_{k+1} + \frac{k}{\kappa}\,m_{k-1}, \qquad m_k = \mathbb{E}x_1^k,$$

from $m_0 = 1$ and $m_1 = A$; all polynomial moments of ν_b are rational in κ and $A(\kappa)$.[14]

D.4. **The Riccati barriers.** By Lemma C.2(i), A solves $A' = \mathcal{F}(\kappa, A) := 1 - A^2 - 3A/\kappa$ on $(0,\infty)$, and $A(\kappa) = \frac{\kappa}{4} - \frac{\kappa^3}{96} + O(\kappa^5)$ from the series of I_1 and I_2. Let $s = \sqrt{\kappa^2+4}$, $u = \sqrt{\kappa^2+9}$, $\phi_- = \kappa/(2+s)$ and $\phi_+ = \kappa/(1+u)$. Direct differentiation gives

$$\phi_-' - \mathcal{F}(\kappa,\phi_-) = -\frac{\kappa^2}{s(s+2)^2} \leq 0,$$

$$\phi_+' - \mathcal{F}(\kappa,\phi_+) = \frac{(u-3)^2}{u(1+u)^2} \geq 0,$$

and $\phi_\pm = \frac{\kappa}{4} + O(\kappa^3)$.[15] The comparison is singular at 0 because of the term $3A/\kappa$, and we carry it out with an integrating factor. Let $D = A - \phi_-$. Then

$$\begin{aligned} D' &= \mathcal{F}(\kappa,A) - \mathcal{F}(\kappa,\phi_-) - (\phi_-' - \mathcal{F}(\kappa,\phi_-)) \\ &= -\Big(A + \phi_- + \frac{3}{\kappa}\Big)D + R, \end{aligned}$$

with $R = \kappa^2/(s(s+2)^2) \geq 0$. Put $\mu(\kappa) = \kappa^3 \exp\int_0^\kappa (A+\phi_-)$. Then $(\mu D)' = \mu R \geq 0$, and $\mu D = O(\kappa^6) \to 0$ as $\kappa \to 0$, so $\mu D \geq 0$ and $A \geq \phi_-$. The same argument with $D = \phi_+ - A$ and $R = (u-3)^2/(u(1+u)^2)$ gives $A \leq \phi_+$. This is Lemma C.2(ii).

[14] For instance $m_2 = 1 - 3A/\kappa$, $m_3 = A - 3/\kappa + 12A/\kappa^2$ and $m_4 = m_2 - 5m_3/\kappa + 2A/\kappa$. The joint moments with x_2, x_3, x_4 follow from exchangeability: $\mathbb{E}[\phi(x_1)x_2^2] = \frac{1}{3}\mathbb{E}[\phi(x_1)(1-x_1^2)]$.

[15] For the first identity write $\phi_- = (s-2)/\kappa$, which follows from $s^2 - 4 = \kappa^2$; then $\phi_-' = (\kappa^2/s - s + 2)/\kappa^2$ and $1 - \phi_-^2 - 3\phi_-/\kappa = (\kappa^2 - (s-2)^2 - 3(s-2))/\kappa^2$, and the difference simplifies with $s^2 = \kappa^2 + 4$. The second is the same computation with $\phi_+ = \kappa(u-1)/(u^2-1) = \kappa(u-1)/(\kappa^2+8)$ in place of ϕ_- and $u^2 = \kappa^2 + 9$ in place of $s^2 = \kappa^2 + 4$. After clearing the denominator $\kappa^2(1+u)^2$ the numerator becomes $\kappa^2(u-3)^2/u$, which is the stated residual.

For (iv) let $s = \sqrt{\kappa^2 + 25/4}$ and $\phi_\flat = \kappa/(\frac{3}{2} + s)$. Since $s^2 - \frac{9}{4} = \kappa^2 + 4$, the same differentiation gives

$$\phi_\flat' - \mathcal{F}(\kappa, \phi_\flat) = -\frac{5\,(s - \frac{5}{2})}{2\,s\,(\frac{3}{2} + s)^2} \le 0,$$

so $\phi_\flat$ is a subsolution, and $\phi_\flat = \frac{\kappa}{4} - \frac{\kappa^3}{80} + O(\kappa^5)$.[16] Hence $A - \phi_\flat = \frac{\kappa^3}{480} + O(\kappa^5)$, and the integrating-factor argument above, with $D = A - \phi_\flat$, $R = \frac{5(s-5/2)}{2s(3/2+s)^2} \ge 0$ and $\mu = \kappa^3 \exp\int_0^\kappa (A + \phi_\flat)$, gives $\mu D = O(\kappa^6)$ at 0, $(\mu D)' = \mu R \ge 0$, and so $A \ge \phi_\flat$. Since $\frac{3}{2} + \sqrt{\kappa^2 + 25/4} < 2 + \sqrt{\kappa^2 + 4}$ for $\kappa > 0$, this improves the lower bound of (ii).

Appendix E. Bessel functions

This appendix proves the auxiliary statements used in Appendix C for Theorem C.4. Throughout, $I_j = I_j(\kappa)$ with $\kappa > 0$, and $I_{-j} = I_j$ for integers j. We use the series

$$I_\nu(\kappa) = \sum_{i \ge 0} \frac{(\kappa/2)^{2i+\nu}}{i!\,(i+\nu)!},$$

the integral $I_j(\kappa) = \pi^{-1} \int_0^\pi e^{\kappa \cos\theta} \cos j\theta \, d\theta$, the Fourier series $e^{\kappa\cos\theta} = \sum_{j\in\mathbb{Z}} I_j e^{ij\theta}$ and the recurrences

$$I_{j-1} - I_{j+1} = \frac{2j}{\kappa} I_j, \quad I_j' = I_{j+1} + \frac{j}{\kappa} I_j, \tag{E.1}$$

all of which are in Watson's treatise [53], §§3.7, 3.71, 2.22. All series below converge absolutely, because $0 < I_j \le (\kappa/2)^j e^\kappa / j!$.

E.1. **Zonal harmonics.** Under σ the coordinate x_1 has the density $\frac{2}{\pi}(1-t^2)^{1/2}$ on $[-1,1]$, the area of the two-sphere $\{x_1 = t\}$ being proportional to $1 - t^2$ and its normal speed to $(1-t^2)^{-1/2}$. The Chebyshev polynomials $U_n(\cos\theta) = \sin((n+1)\theta)/\sin\theta$ satisfy

$$\int U_n U_m \, d\sigma = \frac{2}{\pi} \int_0^\pi \sin((n+1)\theta) \sin((m+1)\theta)\, d\theta = \delta_{nm},$$

and $U_n = C_n^{(1)}$ is the Gegenbauer polynomial of index $\frac{4-2}{2} = 1$, so $U_n(\langle y, x\rangle)$ is the zonal harmonic of degree n on S^3 with pole y: $|x|^n U_n(\langle y, x\rangle/|x|)$ is a harmonic polynomial on $\mathbb{R}^4$ [48], Ch. IV, §2.

Lemma E.1. *$e^{\kappa t} = \frac{2}{\kappa} \sum_{n\ge 0} (n+1) I_{n+1}(\kappa)\, U_n(t)$ in $L^2(\frac{2}{\pi}(1-t^2)^{1/2} dt)$, and uniformly on $[-1,1]$.*

Proof. With $t = \cos\theta$ and $2\sin\theta\sin((n+1)\theta) = \cos n\theta - \cos(n+2)\theta$, the n-th coefficient is

$$\begin{aligned} \frac{2}{\pi} \int_0^\pi e^{\kappa\cos\theta} \sin((n+1)\theta) \sin\theta \, d\theta &= I_n - I_{n+2} \\ &= \frac{2(n+1)}{\kappa} I_{n+1}, \end{aligned}$$

by (E.1). Since $|U_n| \le n+1$ and I_{n+1} decays factorially, the series converges uniformly. □

[16] With $\phi_\flat = \kappa(s - \frac{3}{2})/(\kappa^2 + 4)$ one has $\phi_\flat' = (\frac{3}{2}s + \frac{25}{4})/(s(\frac{3}{2}+s)^2)$, using $ds/d\kappa = \kappa/s$; and $\mathcal{F}(\kappa, \phi_\flat) = 1 - \phi_\flat^2 - 3/(\frac{3}{2} + s)$. Clearing the denominator $s(\frac{3}{2}+s)^2$ and replacing κ^2 by $s^2 - \frac{25}{4}$ leaves $-\frac{5}{2}s + \frac{25}{4}$ in the numerator, which is the stated residual. The expansion follows from $s = \frac{5}{2} + \frac{\kappa^2}{5} + O(\kappa^4)$.

E.2. **Proof of Lemma C.7.** Let $F(\theta) = e^{\kappa\cos\theta} = \sum_j I_j e^{ij\theta}$. Then

$$F'(\theta) = -\kappa\sin\theta\, F(\theta) = \sum_j ijI_j e^{ij\theta},$$

and Parseval's formula gives

$$\begin{aligned}\sum_{j\in\mathbb{Z}} j^2 I_j^2 &= \frac{\kappa^2}{2\pi}\int_0^{2\pi} \sin^2\theta\, e^{2\kappa\cos\theta} d\theta \\ &= \frac{\kappa^2}{2}\big(I_0(2\kappa) - I_2(2\kappa)\big) = \frac{\kappa}{2} I_1(2\kappa),\end{aligned}$$

the last step by (E.1) at the argument 2κ. Halving gives the first identity. The coefficient of $e^{i\theta}$ in $F^2 = e^{2\kappa\cos\theta}$ is $I_1(2\kappa) = \sum_{j\in\mathbb{Z}} I_j I_{1-j}$, and the terms with $j \le 0$ repeat those with $j \ge 1$; this is the third identity.

For the second, $jI_j = \frac{\kappa}{2}(I_{j-1} - I_{j+1})$ by (E.1), so

$$\sum_{j\ge1} j^3 I_j^2 = \frac{\kappa^2}{4}\sum_{j\ge1} j\big(I_{j-1}^2 + I_{j+1}^2 - 2I_{j-1}I_{j+1}\big).$$

Shifting indices gives

$$\sum_{j\ge1} j\big(I_{j-1}^2 + I_{j+1}^2\big) = I_0^2 + \sum_{m\ge1} 2mI_m^2, \qquad \sum_{m\ge1} 2mI_m^2 = \kappa\sum_{m\ge1}\big(I_{m-1}I_m - I_m I_{m+1}\big) = \kappa I_0 I_1,$$

the last by telescoping. For the mixed term write $j = \frac12\big((j-1)+(j+1)\big)$ and apply (E.1) to $(j-1)I_{j-1}$ and to $(j+1)I_{j+1}$, using $I_{-1} = I_1$ at $j = 1$:

$$\begin{aligned}jI_{j-1}I_{j+1} = \frac{\kappa}{4}\big[&(I_{j-1}I_j - I_j I_{j+1}) \\ &+ (I_{j-2}I_{j+1} - I_{j-1}I_{j+2})\big].\end{aligned}$$

Both brackets telescope, to $I_0 I_1$ and to $I_{-1}I_2 = I_1 I_2$, so $\sum_{j\ge1} jI_{j-1}I_{j+1} = \frac{\kappa}{4}(I_0 I_1 + I_1 I_2)$. Altogether

$$\sum_{j\ge1} j^3 I_j^2 = \frac{\kappa^2}{4}\Big(I_0^2 + \frac{\kappa}{2} I_1(I_0 - I_2)\Big) = \frac{\kappa^2}{4}\big(I_0^2 + I_1^2\big). \qquad \square$$

E.3. **The coefficients of $\mathcal{D}$.**

Lemma E.2. *Write $\mathcal{D}(\kappa) = \sum_{n\ge0}\mathcal{D}_n\kappa^{2n}$ for the function of (C.9). Then $\mathcal{D}_0 = \mathcal{D}_1 = 0$, and for $m \ge 1$*

$$\mathcal{D}_{m+1} = \frac{N_m}{4^m\, m!\,(m+1)!\,(m+1)(m+2)},$$

with $N_m = \binom{2m}{m}\big(4(2m+1) + (m-1)(m+1)(m+2)\big) - 4^m(m+1)(m+2)$. Moreover $N_m = 0$ for $m \le 2$ and $N_m > 0$ for $m \ge 3$.

Proof. The coefficients. Multiplying the series of I_0 and I_1 and collecting powers,

$$I_0^2 = \sum_{m\ge0}\alpha_m\kappa^{2m}, \qquad I_1^2 = \sum_{m\ge0}\beta_m\kappa^{2m+2},$$

$$\kappa I_1(2\kappa) = \sum_{m\ge0}\frac{\kappa^{2m+2}}{m!\,(m+1)!},$$

where

$$\alpha_m = \frac{1}{4^m}\sum_{i=0}^{m}\frac{1}{i!^2(m-i)!^2},$$

$$\beta_m = \frac{1}{4^{m+1}}\sum_{i=0}^{m}\frac{1}{i!\,(i+1)!\,(m-i)!\,(m-i+1)!}.$$

By Vandermonde's identity, $\sum_i \binom{m}{i}^2 = \binom{2m}{m}$ and $\sum_i \binom{m}{i}\binom{m+2}{i+1} = \binom{2m+2}{m+1}$, so

$$\alpha_m = \frac{\binom{2m}{m}}{4^m\, m!^2}, \qquad \beta_m = \frac{\binom{2m+2}{m+1}}{4^{m+1} m!\,(m+2)!}.$$

The coefficient of κ^{2m+2} in $\mathcal{D}$ is $\mathcal{D}_{m+1} = 8\beta_m + 2\beta_{m-1} - \alpha_m - 1/(m!(m+1)!)$, with $\beta_{-1} = 0$; for $m = 0$ it is $\frac{8}{4} - 1 - 1 = 0$, and $\mathcal{D}_0 = 0$ because every term of $\mathcal{D}$ is $O(\kappa^2)$. For $m \ge 1$ put $\Delta = 4^m\, m!\,(m+1)!\,(m+1)(m+2)$ and $b = \binom{2m}{m}$. With $\binom{2m+2}{m+1} = \frac{2(2m+1)}{m+1} b$,

$$8\beta_m = \frac{4(2m+1)\,b}{\Delta}, \qquad \alpha_m = \frac{(m+1)^2(m+2)\,b}{\Delta},$$
$$2\beta_{m-1} = \frac{2m(m+1)(m+2)\,b}{\Delta},$$
$$\frac{1}{m!(m+1)!} = \frac{4^m(m+1)(m+2)}{\Delta},$$

and $2m - (m+1) = m - 1$ gives $\mathcal{D}_{m+1} = N_m/\Delta$.

The sign. Directly, $N_0 = N_1 = N_2 = 0$ $(2-2,\ 24-24,\ 192-192)$, $N_3 = 80$, $N_4 = 1140$ and $N_5 = 10416$. For $m \ge 6$ we first show $16^m \le 4m\binom{2m}{m}^2$ for $m \ge 1$ by induction: it is an equality at $m = 1$, and $(m+1)\binom{2m+2}{m+1} = 2(2m+1)\binom{2m}{m}$ gives

$$4(m+1)\binom{2m+2}{m+1}^2 = \frac{16(2m+1)^2}{m+1}\binom{2m}{m}^2$$
$$\ge \frac{16(2m+1)^2}{4m(m+1)} 16^m \ge 16^{m+1}.$$

Since $(m-1)^2 \ge 4m$ for $m \ge 6$, it follows that $\binom{2m}{m}^2 (m-1)^2 \ge 16^m$, that is $\binom{2m}{m}(m-1) \ge 4^m$, and

$$N_m = \Big(\binom{2m}{m}(m-1) - 4^m\Big)(m+1)(m+2)$$
$$+ 4(2m+1)\binom{2m}{m} > 0.$$

□

E.4. **Monotonicity of Bessel ratios.**

Lemma E.3. *For $n \ge 1$ and $\kappa > 0$, $I_{n+2}(\kappa)/I_{n+1}(\kappa) \le I_2(\kappa)/I_1(\kappa)$.*

Proof. We use a lemma of Biernacki and Krzyż [11]: if $f = \sum a_j x^j$ and $g = \sum b_j x^j$ have positive coefficients and a_j/b_j is nonincreasing in j, then f/g is nonincreasing on $x > 0$. Indeed

$$f'g - fg' = \sum_{i,j} (i-j) a_i b_j x^{i+j-1}$$
$$= \frac{1}{2}\sum_{i,j} (i-j) b_i b_j \Big(\frac{a_i}{b_i} - \frac{a_j}{b_j}\Big) x^{i+j-1} \le 0.$$

Take $x = \kappa^2$, $f = I_{n+1}(\kappa)/\kappa^{n+1}$ and $g = I_1(\kappa)/\kappa$. Then $a_j/b_j = 2^{-n}(j+1)!/(j+n+1)!$ decreases in j, so $I_{n+1}/(\kappa^n I_1)$ is nonincreasing in κ. Its logarithmic derivative is, by the second recurrence of (E.1),

$$\frac{n+1}{\kappa} + \frac{I_{n+2}}{I_{n+1}} - \frac{n}{\kappa} - \frac{1}{\kappa} - \frac{I_2}{I_1}$$
$$= \frac{I_{n+2}}{I_{n+1}} - \frac{I_2}{I_1} \le 0.$$

□

Appendix F. A finite-size criterion for the cube condition

This appendix proves the finite-size criterion used in Sec. 10: a gap for the dynamics that resamples the blocks of one patch of m^4 blocks, uniform in the configuration outside the patch, gives the cube condition on every large torus. Blocks that share no plaquette give commuting projections, and this replaces the disjointness used in the finite-size criteria of Knabe [34] and of Gosset and Mozgunov [26] for quantum spin systems.[17]

Setting. The notation is that of Sec. 6 and of the setting of Sec. 10. Fix integers $d \geq 2$ and $m \geq 4$ with $d \mid L$, put $K = L/d$, $n = 2d-1$, and fix a shift $a \in [0,d)^4$. For j in the coarse torus $T_K = (\mathbb{Z}/K\mathbb{Z})^4$ let $r_j = a + dj$, let $B_j = V_{r_j}$ be the set of links with base point in $r_j + [0,n)^4$, and write $\mathcal{P}_j = \mathcal{P}_{r_j}$. Displacements $v = j' - j$ in T_K are represented in $(-K/2, K/2]^4$. Two sets A, A' of links are *non-interacting* if no plaquette contains a link of each; for a set A of links, P_A is the conditional expectation under μ given the links outside A.

Lemma F.1. (i) *For non-interacting sets A, A' of links, $P_A P_{A'} = P_{A'} P_A = P_{A\cup A'}$, and $(I - P_A)(I - P_{A'})$ is an orthogonal projection.*
(ii) *If $K \geq 4$ and $|j' - j|_\infty \geq 2$, then B_j and $B_{j'}$ are non-interacting.*
(iii) *The blocks B_j, $j \in T_K$, cover E.*
(iv) *If $f \in L^2(\mu)$ and $\mathcal{P}_j f = f$ for every $j \in T_K$, then f is almost surely constant.*

Proof. (i) Every link lies on a plaquette, so $A \cap A' = \emptyset$. Given the links outside $A \cup A'$, the law of $(U_A, U_{A'})$ has density proportional to $\prod_{p \cap (A\cup A') \neq \emptyset} \varphi(U_p)$, and each factor involves the links of exactly one of A, A'. The density is therefore a product of a function of U_A and a function of $U_{A'}$, with coefficients depending on the links outside $A \cup A'$. Hence U_A and $U_{A'}$ are conditionally independent, the conditional law γ_A of U_A given the links outside A does not depend on $U_{A'}$, and likewise for A'. So $P_A P_{A'} f$ and $P_{A'} P_A f$ are both the integral of f against $\gamma_A \otimes \gamma_{A'}$, which is $P_{A\cup A'} f$, and $(I - P_A)(I - P_{A'}) = I - P_A - P_{A'} + P_{A\cup A'}$ is self-adjoint and idempotent.

(ii) Let $|v_i| \geq 2$ for $v = j' - j$. In the coordinate i the base points of the links of B_j and of $B_{j'}$ lie in $dj_i + a_i + [0, 2d-2]$ and in $dj_i + a_i + dv_i + [0, 2d-2]$ modulo L. If $v_i \geq 2$, their distance in $\mathbb{Z}/L\mathbb{Z}$ is at least $\min\{(v_i - 2)d + 2, (K - v_i - 2)d + 2\} \geq 2$, since $v_i \leq K/2$ and $K \geq 4$; the case $v_i \leq -2$ is symmetric. A plaquette meeting both blocks would contain two links whose base points differ by at least two in the coordinate i, while the base points of the links of a plaquette differ by at most one in each coordinate.

(iii) The sets $r_j + [0,d)^4$ partition Λ_L, and $[0,d) \subset [0,n)$.

(iv) The density of μ with respect to the Haar measure h of G^E is bounded above and below by positive constants, so μ and h have the same null sets and $L^2(\mu) = L^2(h)$. If $\mathcal{P}_j f = f$, then f is almost surely a function of $U_{E\setminus B_j}$, and it is fixed by the conditional expectation Q_j of h given $U_{E\setminus B_j}$, which integrates out U_{B_j}. By Fubini's theorem the Q_j commute and their product integrates out the links of $\bigcup_j B_j = E$, so $f = \int f \, dh$ almost surely. □

In four dimensions the displacements v with $|v|_\infty = 1$ are 80 in number, and those with $|v|_\infty \geq 2$ give commuting projections. For $s \in T_K$ let $S_s = s + [0, m-1]^4$ be the *patch* at s and $S_s^\circ = s + [0, m-2]^4$ the set of its $(m-1)^4$ block indices, and put

$$H = \sum_{j \in T_K} (I - \mathcal{P}_j), \qquad H_s = \sum_{j \in S_s^\circ} (I - \mathcal{P}_j).$$

These are bounded nonnegative operators on $L^2(\mu)$ and preserve the subspace $\mathcal{A}$ of gauge-invariant functions.

[17] In those criteria the local terms of a Hamiltonian act on overlapping sets of sites, and the gap of the whole system is bounded below by the gap of a finite patch minus a correction that decays with the size of the patch. Here the local terms are the complements $I - \mathcal{P}_j$ of conditional expectations, and two of them commute as soon as the blocks are at distance at least two.

Lemma F.2. *Let $m \geq 4$, $p = m-1$, $K \geq 2m$, let $\mathcal{K}$ be $L^2(\mu)$ or $\mathcal{A}$, and suppose that $H_s^2 \geq \gamma H_s$ on $\mathcal{K}$ for every $s \in T_K$, with $\gamma > 0$. Then $H^2 \geq \varepsilon H$ on $\mathcal{K}$ with*

$$\varepsilon = 1 - \frac{(1-\gamma)p^4 + 72p^3 - 168p^2 + 96p - 16}{p^3(p-2)} \geq \frac{\gamma p - 74}{p-2}.$$

In particular $\varepsilon \geq \gamma/2$ if $m - 1 \geq 148/\gamma$.

Proof. All operators below preserve $\mathcal{K}$, and a restriction of an orthogonal projection to an invariant closed subspace is an orthogonal projection; so it suffices to argue on $L^2(\mu)$ with inequalities read on $\mathcal{K}$. Write $\mathbf{P}_j = I - \mathcal{P}_j$ and $A_v = \sum_j \mathbf{P}_j \mathbf{P}_{j+v}$. Since $K \geq 2m$, two indices j and $j+v$ lie in a common S_s° only if $|v_i| \leq m-2$ for all i, and then for exactly $c_v = \prod_{i=1}^4 (p - |v_i|)$ points s. Let O be the set of the 80 displacements with $|v|_\infty = 1$, D the set of those with $|v|_\infty \geq 2$, and D' the set of $v \in D$ with all $|v_i| \leq m-2$. For $v \in D$, $\mathbf{P}_j \mathbf{P}_{j+v}$ is an orthogonal projection by Lemma F.1(i),(ii), so $A_v \geq 0$. Summing the hypothesis over s,

$$\sum_s H_s^2 = c_0 H + \sum_{v \in O} c_v A_v + \sum_{v \in D'} c_v A_v \geq \gamma \sum_s H_s = \gamma c_0 H.$$

Put $c_* = p^3(p-2)$; then $c_v \leq c_*$ for $v \in D'$, since one factor of c_v is at most $p-2$ and the others are at most p. For orthogonal projections $\mathbf{P}, \mathbf{P}'$, $(\mathbf{P} \pm \mathbf{P}')^2 \geq 0$ gives $-(\mathbf{P} + \mathbf{P}') \leq \mathbf{P}\mathbf{P}' + \mathbf{P}'\mathbf{P} \leq \mathbf{P} + \mathbf{P}'$, so $-2H \leq A_v + A_{-v} \leq 2H$. As O is symmetric and $c_{-v} = c_v$,

$$\sum_{v \in O} (c_* - c_v) A_v \geq -\Sigma H, \qquad \Sigma = \sum_{v \in O} |c_* - c_v|,$$

and therefore

$$c_* \sum_{v \in O} A_v = \sum_{v \in O} c_v A_v + \sum_{v \in O} (c_* - c_v) A_v \geq -\big((1-\gamma)c_0 + \Sigma\big) H - \sum_{v \in D'} c_v A_v.$$

Since $H^2 = \sum_v A_v = H + \sum_{v \in O} A_v + \sum_{v \in D} A_v$,

$$H^2 \geq \Big(1 - \frac{(1-\gamma)c_0 + \Sigma}{c_*}\Big) H + \sum_{v \in D'} \Big(1 - \frac{c_v}{c_*}\Big) A_v + \sum_{v \in D \setminus D'} A_v,$$

and the last two sums are nonnegative. It remains to compute Σ. A displacement $v \in O$ with k nonzero coordinates has $c_v = p^{4-k}(p-1)^k$, and there are 8, 24, 32 and 16 of them for $k = 1, 2, 3, 4$. The differences $c_v - c_*$ are

$$p^3, \qquad p^2, \qquad -p(p^2 - 3p + 1), \qquad -2p(p-1)(p-2) + 1,$$

with the signs shown since $p \geq 3$. Hence $\Sigma = 8p^3 + 24p^2 + 32p(p^2 - 3p + 1) + 16(2p(p-1)(p-2) - 1) = 72p^3 - 168p^2 + 96p - 16$, and with $c_0 = p^4$ this is the formula for ε. As $-168p^2 + 96p - 16 < 0$, $\varepsilon \geq 1 - ((1-\gamma)p + 72)/(p-2) = (\gamma p - 74)/(p-2)$, and if $\gamma p \geq 148$ this is at least $(\gamma p - 74)/p \geq \gamma/2$. □

For a patch S_s let $\Pi_s = \bigcup_{j \in S_s^\circ} B_j$, the set of links with base point in $r_s + [0, md-1)^4$. For $L \geq md + 2$ the patch and the plaquettes meeting it embed in $\mathbb{Z}^4$, and the conditional law of U_{Π_s} under μ given the links outside Π_s is the measure ν_Q^ω of Sec. 10 for the box $Q = r_s + [0, md-1)^4$, with ω the configuration outside. For $j \in S_s^\circ$ write $\mathcal{P}_j^\omega$ for the conditional expectation under ν_Q^ω given the links of $\Pi_s \setminus B_j$.

Definition F.3. The *gauge-invariant patch gap* $\gamma_{d,m}$ is the largest $\gamma \geq 0$ such that

$$\gamma \operatorname{Var}_{\nu_Q^\omega}(F) \leq \sum_{j \in S_s^\circ} \nu_Q^\omega\big((F - \mathcal{P}_j^\omega F)^2\big)$$

for every patch box $Q = r + [0, md-1)^4$ of $\mathbb{Z}^4$, every exterior configuration ω and every $F \in \mathcal{A}_Q^\omega$.

By translation invariance $\gamma_{d,m}$ depends only on G, φ, d and m. It is the gap, on gauge-invariant functions, of the dynamics that resamples the $(m-1)^4$ blocks of one patch, uniform in the configuration outside the patch.

Theorem F.4 (finite-size criterion)**.** *Let $d \geq 2$, $m \geq 4$, $d \mid L$, L even, $L \geq 2md$, and suppose that $\gamma_{d,m} \geq \gamma > 0$ and that the number ε of Lemma F.2 is positive. Then*

$$\frac{\varepsilon}{16}\operatorname{Var}_\mu(f) \leq \frac{1}{n^4}\sum_{x\in\Lambda_L}\mu((f-\mathcal{P}_x f)^2) \qquad \text{for every real } f \in \mathcal{A},$$

that is, $\mathrm{BP}(2d-1,\varepsilon/16)$*. If $m-1 \geq 148/\gamma$, then $\varepsilon/16 \geq \gamma/32$.*

Proof. Fix a. Since $L \geq 2md$, $K \geq 2m \geq 8$.

The patch inequality on the torus. Let $F \in \mathcal{A}$; replacing F by $\int F(gU)\,dg$, the integral over all gauge transformations of Λ_L with Haar measure, we may take F invariant at every point. Write $U = (\sigma,\omega)$ with σ the links of Π_s and ω the others. Under μ the conditional law of σ given ω is ν_Q^ω, and for $j \in S_s^\circ$ the conditional law of U_{B_j} given the other links is the same under μ and under ν_Q^ω, so $(\mathcal{P}_j F)(\cdot,\omega) = \mathcal{P}_j^\omega(F(\cdot,\omega))$ for almost every ω and H_s acts on each fibre as $H_s^\omega = \sum_{j\in S_s^\circ}(I-\mathcal{P}_j^\omega)$. A gauge transformation of Λ_L that is trivial at the vertices not interior to Q fixes ω, so each fibre $F(\cdot,\omega)$ lies in $\mathcal{A}_Q^\omega$. On $\mathcal{A}_Q^\omega$ the operator H_s^ω is self-adjoint and nonnegative, preserves the constants and their orthogonal complement (Lemma 10.1(iii)), and by Definition F.3 is at least γ on that complement. So its spectrum there lies in $\{0\}\cup[\gamma,\infty)$ and $(H_s^\omega)^2 \geq \gamma H_s^\omega$. Integrating over ω, $\|H_s F\|^2 \geq \gamma\langle F, H_s F\rangle$ for $F \in \mathcal{A}$, that is, $H_s^2 \geq \gamma H_s$ on $\mathcal{A}$.

Conclusion. By Lemma F.2 with $\mathcal{K} = \mathcal{A}$, $H^2 \geq \varepsilon H$ on $\mathcal{A}$, so the spectrum of H on $\mathcal{A}$ lies in $\{0\}\cup[\varepsilon,\infty)$. If $F \in \mathcal{A}$ and $HF = 0$, then $\sum_j \|F-\mathcal{P}_j F\|^2 = \langle F, HF\rangle = 0$, so F is constant by Lemma F.1(iv). Hence for real $f \in \mathcal{A}$ the function $g = f - \mu(f) \in \mathcal{A}$ is orthogonal to the kernel of H on $\mathcal{A}$, and

$$\varepsilon\operatorname{Var}_\mu(f) \leq \langle g, Hg\rangle = \sum_{j\in T_K}\mu((f-\mathcal{P}_{r_j}f)^2).$$

As a runs over $[0,d)^4$ and j over T_K, $r_j = a + dj$ runs over Λ_L once. Averaging over a gives $\varepsilon d^4\operatorname{Var}_\mu(f) \leq \sum_x \mu((f-\mathcal{P}_x f)^2)$, and $d^4 \geq n^4/16$. Since $L \geq 8d \geq 2n+4$ this is $\mathrm{BP}(2d-1,\varepsilon/16)$, and the last claim is Lemma F.2. □

The passage from the patch to the torus costs a factor: at $\varphi \equiv 1$ the links are independent, $\gamma_{d,m} \geq 1$ by the Efron–Stein inequality [22], and the theorem gives $\mathrm{BP}(2d-1,\varepsilon/16)$ with $\varepsilon < 1$, against $\mathrm{BP}(n,1)$ from Proposition 7.1.

Appendix G. The threshold one half, two-dimensional models and limits of methods

This appendix proves the statements about the hypothesis of Section 10 and about other approaches to (D) that are used in the text. Section G.1 shows that the threshold $c < \frac{1}{2}$ of Theorem D is the exact limit of the slab mechanism, the boundary value being attained by a massless Gaussian chain. Sections G.2 and G.3 treat two dimensions: the slab hypothesis for fat boxes holds along the weak-coupling tuning, uniformly in the boundary condition, while the same condition for strips of width one fails. Sections G.4 to G.6 prove that four other routes cannot give (D) without an input of the same strength: curvature of the Wilson measure, entropy factorisation, tensorisation over blocks of fixed lattice size, and the transfer matrix taken alone.

G.1. **The Gaussian chain.** For a probability measure and σ-algebras $\mathcal{F}_1, \mathcal{F}_2$ let $\rho(\mathcal{F}_1,\mathcal{F}_2) = \sup|E[XY]|$, the supremum over centred $X \in L^2(\mathcal{F}_1)$ and $Y \in L^2(\mathcal{F}_2)$ with $E[X^2] = E[Y^2] = 1$; this is the maximal correlation of Definition 10.2, without the constraint of gauge invariance.

Lemma G.1. *Let (U,V,M,W,X) be random variables with values in standard Borel spaces whose joint law has the form $\psi_1(u,v)\psi_2(v,m,w)\psi_3(w,x)\,d\lambda$ for a product measure λ and nonnegative functions ψ_i. Then $\rho(\sigma(U,V),\sigma(W,X)) \le \rho(\sigma(V),\sigma(W))$.*

Proof. By the factorisation, the conditional law of U given (V,M,W,X) has density proportional to $\psi_1(\cdot,V)$ and that of X given (U,V,M,W) has density proportional to $\psi_3(W,\cdot)$, and given (V,M,W) the variables U and X are conditionally independent. Let $f=f(U,V)$ and $g=g(W,X)$ be centred and square integrable. Then $E[f \mid V,M,W,X] = \tilde f(V)$ with $\tilde f = E[f \mid V]$, and $E[g \mid U,V,M,W] = \tilde g(W)$ with $\tilde g = E[g \mid W]$. Hence $E[fg] = E[E[fg \mid V,M,W]] = E[\tilde f(V)\tilde g(W)]$, where $\tilde f$ and $\tilde g$ are centred with $\|\tilde f\| \le \|f\|$ and $\|\tilde g\| \le \|g\|$. □

Lemma G.2 (Gebelein)**.** *If (V,W) is a nondegenerate Gaussian vector in $\mathbb{R}^2$ with correlation coefficient r, then $\rho(\sigma(V),\sigma(W)) = |r|$.*

Proof. The choice $X=V$, $Y=\pm W$, standardised, gives $\rho \ge |r|$. For the converse standardise V and W. Let H_n be the Hermite polynomials with generating function $\sum_n H_n(x)s^n/n! = e^{sx-s^2/2}$; then $h_n = H_n/\sqrt{n!}$ is a complete orthonormal system of $L^2(\mathbb{R},\gamma)$, γ the standard Gaussian law [40, Sec. 1.1.1]. From $E[e^{sV-s^2/2}e^{uW-u^2/2}] = e^{rsu}$ and a comparison of coefficients one gets $E[h_j(V)h_k(W)] = \delta_{jk}r^j$; the double series converges absolutely, since $H_n(x) = E(x+iZ)^n$ with $Z\sim\gamma$ gives $|H_n(x)| \le E(|x|+|Z|)^n$ and $Ee^{t(|V|+|W|+2|Z|)} < \infty$ for every t. For centred $f=\sum_{j\ge1} a_j h_j$ and $g=\sum_{j\ge1} b_j h_j$ in $L^2(\gamma)$ the form $(f,g)\mapsto E[f(V)g(W)]$ is bounded by $\|f\|\|g\|$, so passing to the limit in the partial sums, $E[f(V)g(W)] = \sum_{j\ge1} a_j b_j r^j$, whose modulus is at most $|r|\sum_j |a_jb_j| \le |r|\|f\|\|g\|$. □

The inequality $\rho \le |r|$ is Gebelein's theorem [24].

The chain. Let $\mu \ge 0$ and define $\kappa \ge 0$ by $2\cosh\kappa = 2+\mu^2$. For integers $p<q$ with $q-p\ge 2$ and real boundary values z_p, z_q let ν be the Gaussian law of $(z_{p+1},\dots,z_{q-1})$ with density proportional to

$$\exp\Big(-\tfrac12\sum_{j=p}^{q-1}(z_{j+1}-z_j)^2 - \tfrac12\mu^2\sum_{j=p+1}^{q-1} z_j^2\Big),$$

a nearest-neighbour Gaussian chain, massless for $\mu=0$. Its precision matrix on the $q-p-1$ interior sites is $\mathsf{L} = \operatorname{tridiag}(-1, 2+\mu^2, -1)$, independent of z_p and z_q, which only shift the mean, and its conditional laws on subintervals are chains of the same form.

Lemma G.3. *Put $T=q-p$ and, for $0\le i\le T$, $u(i)=\sinh(\kappa i)$ and $v(i)=\sinh(\kappa(T-i))$ if $\mu>0$, $u(i)=i$ and $v(i)=T-i$ if $\mu=0$. For $0<i\le j<T$, $\operatorname{Cov}_\nu(z_{p+i},z_{p+j}) = A\,u(i)v(j)$ with $A>0$ independent of i and j; for $\mu=0$, $A=1/T$. Hence the correlation of z_{p+i} and z_{p+j} is $(u(i)v(j)/(v(i)u(j)))^{1/2}$.*

Proof. Both u and v satisfy $(2+\mu^2)f(l) = f(l-1)+f(l+1)$ (for $\mu>0$ because $\sinh(x-\kappa)+\sinh(x+\kappa) = 2\cosh\kappa\sinh x$), $u(0)=v(T)=0$, and both are positive on $(0,T)$. The covariance matrix is $G=\mathsf{L}^{-1}$, which is symmetric. For fixed j the column $G(\cdot,j)$, extended by 0 at 0 and T, satisfies the recursion at every $l\ne j$, so $G(i,j)=\alpha_j u(i)$ for $i\le j$ and $G(i,j)=\beta_j v(i)$ for $i\ge j$: a solution of the second-order recursion on $[0,j]$ that vanishes at 0 is a multiple of u, and similarly for v. For $i<j$, $\alpha_j u(i) = G(i,j) = G(j,i) = \beta_i v(j)$, so $\alpha_j/v(j) = \beta_i/u(i)$ for all $i<j$, a constant A; and $\alpha_j u(j) = \beta_j v(j)$ gives $G(j,j) = Au(j)v(j)$, so $A>0$. For $\mu=0$ the j-th entry of $\mathsf{L}G(\cdot,j)$ equals one, that is $A[2j(T-j)-(j-1)(T-j)-j(T-j-1)] = AT = 1$. □

Slab geometry. Let $s\ge1$, $s\le w\le 2s-1$ and $M\ge1$. Take the sites $I=[0,sM+w)$ with the two exterior sites -1 and $sM+w$, the blocks $R_t=[st,st+w)$ for $0\le t\le M$, and, for $t<M$, $Q_t = R_t\cup R_{t+1}$ split into $Q_t^- = R_t\setminus R_{t+1} = [st,st+s)$, $Q_t^0 = R_t\cap R_{t+1} = [st+s,st+w)$ and $Q_t^+ = R_{t+1}\setminus R_t = [st+w,st+s+w)$. The condition $w\le 2s-1$ says that R_t and R_{t+2} contain no adjacent sites, so that, exactly as in Proposition 10.7, the resampling moves of

non-adjacent blocks commute; Section 10 uses $s = d$ and $w = 2d - 1$. Let ν be the chain on I with any exterior values, P_t the conditional expectation given the sites outside R_t, and ρ_t the supremum, over all values outside Q_t, of $\rho(\sigma(z_{Q_t^-}), \sigma(z_{Q_t^+}))$ under the conditional law of z_{Q_t}.

Proposition G.4 (the Gaussian chain)**.**

(i) *For every $t < M$, $\rho_t = \sinh(\kappa s)/\sinh(\kappa(w+1))$ if $\mu > 0$, and $\rho_t = s/(w+1)$ if $\mu = 0$. For $w = 2s - 1$ this is $1/(2\cosh(\kappa s))$, and $\frac{1}{2}$ for $\mu = 0$. For $\mu = 0$, $\rho_t \geq \frac{1}{2}$ in every admissible geometry, with equality only for $w = 2s - 1$.*

(ii) *For $\mu = 0$ the best constant γ_M in the inequality $\gamma_M \operatorname{Var}_\nu(F) \leq \sum_{t=0}^{M} \nu((F - P_t F)^2)$ for all $F \in L^2(\nu)$ satisfies $\gamma_M \leq 16/(3M)$.*

(iii) *For a subinterval split into consecutive nonempty intervals of A', r and C' sites, with the sites outside it conditioned, the maximal correlation of the two outer intervals is*

$$\Big(\frac{\sinh(\kappa A')\sinh(\kappa C')}{\sinh(\kappa(A'+r+1))\sinh(\kappa(C'+r+1))}\Big)^{1/2} \leq e^{-\kappa(r+1)} \quad \text{if } \mu > 0,$$

with supremum $e^{-\kappa(r+1)}$ over A' and C', and $(A'C'/((A'+r+1)(C'+r+1)))^{1/2}$ if $\mu = 0$, with supremum 1.

Proof. (i) Given the values outside Q_t, the law of z_{Q_t} is the chain with $p = st-1$, $q = st+s+w$, $T = s + w + 1$. Its density factorises as in Lemma G.1 with $U = z_{[st,st+s-1)}$, $V = z_{st+s-1}$, $M = z_{Q_t^0}$, $W = z_{st+w}$ and $X = z_{(st+w,st+s+w)}$, the interactions being between nearest neighbours; so $\rho_t \leq \rho(\sigma(V), \sigma(W))$, with equality since V and W are functions of the two outer parts. By Lemma G.2 this is the modulus of the correlation of V and W, which by Lemma G.3 with $i = s$, $j = w+1$ and $T = s+w+1$ equals $(u(s)v(w+1)/(v(s)u(w+1)))^{1/2}$. Here $v(w+1) = u(s)$ and $v(s) = u(w+1)$, which gives the formulas; they do not depend on the conditioning values. For $w = 2s - 1$, $\sinh(\kappa s)/\sinh(2\kappa s) = 1/(2\cosh \kappa s)$. For $\mu = 0$, $w + 1 \leq 2s$ gives $s/(w+1) \geq \frac{1}{2}$, with equality only when $w = 2s - 1$.

(ii) Let $T = sM+w+1 \geq sM \geq 2$ and take $F = z_j$ with $j = \lfloor T/2 \rfloor - 1 \in I$. By Lemma G.3 with $p = -1$, $\operatorname{Var}_\nu(F) = i(T-i)/T$ with $i = \lfloor T/2 \rfloor$, so $\operatorname{Var}_\nu(F) \geq (T^2-1)/(4T) \geq 3T/16$. If $j \notin R_t$, then $P_t F = F$. If $j \in R_t$, then $\nu((F - P_t F)^2)$ is the mean conditional variance of z_j given z_{st-1} and z_{st+w}, the variance of a chain of $w+1$ steps at an interior site, which is at most $(w+1)/4 \leq s/2$ by Lemma G.3. The t with $j \in R_t$ satisfy $(j-w)/s < t \leq j/s$, an interval of length $w/s < 2$, so there are at most two. Hence $\sum_t \nu((F - P_t F)^2) \leq s$, and $\gamma_M \leq 16s/(3T) \leq 16/(3M)$.

(iii) Put the left conditioned site at 0. The outer parts are $[1, A']$ and $[A'+r+1, A'+r+C']$, and the right conditioned site is $T = A' + r + C' + 1$. As in (i), by Lemmas G.1 and G.2, the maximal correlation is the modulus of the correlation of the inner ends $z_{A'}$ and $z_{A'+r+1}$, which by Lemma G.3 is $(u(A')v(A'+r+1)/(v(A')u(A'+r+1)))^{1/2}$. With $u_0(x) = \sinh \kappa x$, respectively $u_0(x) = x$, one has $u(i) = u_0(i)$ and $v(i) = u_0(T-i)$, so $v(A'+r+1) = u_0(C')$ and $v(A') = u_0(C'+r+1)$, which gives the formulas. For $\mu > 0$ the inequality $\sinh x/\sinh(x+y) \leq e^{-y}$ for $x, y \geq 0$, which is equivalent to $e^{-x-y} \leq e^{-x+y}$, bounds both factors by $e^{-\kappa(r+1)}$, and both tend to $e^{-\kappa(r+1)}$ as $A', C' \to \infty$. For $\mu = 0$ the expression tends to 1. □

Three consequences explain the role of the chain in Section 10. First, for $w = 2s - 1$ the massless chain satisfies the hypothesis of Proposition 10.7 with $c = \frac{1}{2}$ for every exterior condition, its non-adjacent moves commute, and yet by (ii) no inequality $\gamma \operatorname{Var}_\nu(F) \leq \sum_t \nu((F - P_t F)^2)$ holds with $\gamma > 0$ independent of M; in any geometry with commuting non-adjacent moves the same holds for every $c \geq s/(w+1) \geq \frac{1}{2}$. So the threshold $\frac{1}{2}$ of Lemma 10.5 cannot be raised by an argument that uses only these two properties, and no slab geometry of this kind improves it. Second, for $w = 2s - 1$ the slab constant of the chain is $1/(2\cosh \kappa s)$, which is below $\frac{1}{2}$ if and only if $\mu > 0$: along a tuning $s = \ell/(2a)$ with lattice mass $\mu = \mu_{\text{phys}} a$ one has $\kappa/\mu \to 1$, so the constant tends to $1/(2\cosh(\mu_{\text{phys}}\ell/2)) < \frac{1}{2}$.

In the chain the hypothesis is exactly a mass at the scale $\ell/2$. Third, the condition $(\mathrm{M}_r^*(c))$ with a middle of width r has in the chain the best constant $e^{-\kappa(r+1)}$ by (iii), and for $\mu = 0$ it fails for every $c < 1$ and every r; the iteration of Proposition 10.10 reproduces the true decay up to the integer part of the ratio of widths, so it loses nothing, but it only transports a mass present at the starting scale.

G.2. **Two dimensions: fat boxes.** In this section and the next, G is a compact group with normalised Haar measure λ, also on products of copies of G, and φ is a positive continuous class function on G with $\varphi(g^{-1}) = \varphi(g)$; for $G = \mathrm{SU}(2)$ the last condition holds for every class function, since every element of $\mathrm{SU}(2)$ is conjugate to its inverse. Multiplying φ by a constant changes no measure, so we normalise $\int \varphi \, d\lambda = 1$. For class functions put $(\alpha * \beta)(g) = \int \alpha(gy^{-1})\beta(y)\, dy$ and let α^{*k} be the k-fold convolution power.

Links of $\mathbb{Z}^2$ are (y, ν), $\nu \in \{1, 2\}$, from y to $y + e_\nu$, and the plaquette with base point z has holonomy $U_p = U_{(z,1)} U_{(z+e_1,2)} U_{(z+e_2,1)}^{-1} U_{(z,2)}^{-1}$. Boxes $Q = ([\alpha_1, \alpha_1 + s_1) \times [\alpha_2, \alpha_2 + s_2)) \cap \mathbb{Z}^2$, the sets $V(Q)$, the measures ν_Q^ω, the groups $\mathcal{G}_Q$, the spaces $\mathcal{A}_Q^\omega$, splittings and the maximal correlation $\rho^\omega(Q_-, Q_+)$ are defined as in Section 10 and Definition 10.2, with $\mathbb{Z}^2$ in place of $\mathbb{Z}^4$; a vertex v is interior to Q when its four incident links lie in $V(Q)$, that is, when v, $v - e_1$ and $v - e_2$ lie in Q. The *two-dimensional form* of $(\mathrm{M}_{d,m}(c))$ is the condition $\rho^\omega(Q_-, Q_+) \le c$ for every box of $\mathbb{Z}^2$ whose two widths lie in $[2d - 1, md - 1]$, every coordinate in which it has width $3d - 1$, the splitting into widths d, $d - 1$, d in that coordinate, and every exterior configuration ω.

Lemma G.5 (merging)**.** *Let α, β be continuous class functions on G.*

(i) *$\alpha * \beta$ is a continuous class function, positive if α and β are, and $\int \alpha(Ax)\beta(x^{-1}B)\, dx = (\alpha * \beta)(AB)$ for all $A, B \in G$.*

(ii) *Let $\alpha_0, \dots, \alpha_M$ be continuous class functions, $M \ge 1$, and $g_j, c_j, V_0, V_{M+1} \in G$. Then*

$$\int \prod_{j=0}^{M} \alpha_j(V_{j+1} g_j V_j^{-1} c_j)\, dV_1 \cdots dV_M = (\alpha_0 * \cdots * \alpha_M)(V_{M+1}\, g_M \cdots g_0\, V_0^{-1}\, c_0 \cdots c_M).$$

Proof. (i) The substitution $y = x^{-1}B$ preserves Haar measure and turns the integral into $\int \alpha(ABy^{-1})\beta(y)\, dy$. For $h \in G$ the substitution $y = hy'h^{-1}$ gives $(\alpha * \beta)(hgh^{-1}) = \int \alpha(hgy'^{-1}h^{-1})\beta(hy'h^{-1})\, dy' = (\alpha * \beta)(g)$. Continuity and positivity are clear.

(ii) For $M = 1$: since α_0 and α_1 are class functions, $\alpha_0(V_1 g_0 V_0^{-1} c_0) = \alpha_0(g_0 V_0^{-1} c_0 V_1)$ and $\alpha_1(V_2 g_1 V_1^{-1} c_1) = \alpha_1(V_1^{-1} c_1 V_2 g_1)$, so by (i) the integral over V_1 is $(\alpha_0 * \alpha_1)(g_0 V_0^{-1} c_0 c_1 V_2 g_1) = (\alpha_0 * \alpha_1)(V_2 g_1 g_0 V_0^{-1} c_0 c_1)$. For $M \ge 2$ integrate $V_1, \dots, V_{M-1}$ first; by induction the result is $\beta(V_M g_{M-1} \cdots g_0 V_0^{-1} c_0 \cdots c_{M-1})$ with $\beta = \alpha_0 * \cdots * \alpha_{M-1}$, and the case $M = 1$ with $g_{M-1} \cdots g_0$ and $c_0 \cdots c_{M-1}$ in place of g_0 and c_0 gives the claim. □

Fix a box, after translation $Q = [0, s_1) \times [0, s)$ with $s_1 = 3d - 1$, $d \ge 2$, $s \ge 1$, split in the first coordinate into $Q_- = [0, d) \times [0, s)$, $Q_0 = [d, 2d - 1) \times [0, s)$ and $Q_+ = [2d - 1, 3d - 1) \times [0, s)$. Write $h(x, y) = \sigma_{((x,y),1)}$ and $v(x, y) = \sigma_{((x,y),2)}$. The interior vertices are $I = [1, s_1 - 1] \times [1, s - 1]$. Let

$$F' = \{((x, y), 2) : 1 \le x \le s_1 - 1,\ 0 \le y \le s - 2\},$$

the vertical links entering the interior vertices from below ($F' = \emptyset$ if $s = 1$), and $E' = V(Q) \setminus F'$: all $h(x, y)$, all $v(0, y)$, and $v(x, s - 1)$ for $x \ge 1$.

Lemma G.6 (axial gauge)**.** *For $\sigma \in G^{V(Q)}$ define $H(\sigma) \in \mathcal{G}_Q$ recursively in the second coordinate by $H_u = \sigma_{(u-e_2,2)}^{-1} H_{u-e_2}$ for $u \in I$, with $H_w = 1$ for $w \notin I$, and put $\tau(\sigma) = (H(\sigma)^{-1}\sigma)|_{E'}$.*

(i) *$(H(\sigma)^{-1}\sigma)_e = 1$ for $e \in F'$, and $H(\sigma)$ depends only on $\sigma_{F'}$.*

(ii) *Under λ on $G^{V(Q)}$, $\tau(\sigma)$ has law λ on $G^{E'}$.*

(iii) *Under $\nu = \nu_Q^\omega$, $\tau(\sigma)$ has density proportional to $\mathcal{W}(\tau) = \prod_{p \cap V(Q) \neq \emptyset} \varphi((\tau, 1_{F'}, \omega)_p)$.*

(iv) *If $X \in \mathcal{A}_Q^\omega$ is measurable with respect to $\sigma_{V(Q_-)}$, there is a Borel function $\bar X$ with $X = \bar X(\tau(\sigma)_{E'\cap V(Q_-)})$ ν-almost surely; likewise for Q_+.*

Proof. (i) For $u \in I$, $(H^{-1}\sigma)_{(u-e_2,2)} = H_{u-e_2}^{-1}\sigma_{(u-e_2,2)}H_u = 1$ by the definition of H_u. The recursion terminates, since $u - e_2$ has a smaller second coordinate, and it uses only the links $(u - je_2, 2)$, $1 \le j \le u_2$, which lie in F'.

(ii) Under λ the families $\sigma_{F'}$ and $\sigma_{E'}$ are independent with laws λ. Given $\sigma_{F'}$, H is fixed, and for a link e of E' from x to y, $\tau_e = H_x^{-1}\sigma_e H_y$ is a two-sided translate of σ_e. Translations preserve λ, so the conditional law of τ given $\sigma_{F'}$ is λ, and so is its law.

(iii) Let $w(\sigma) = \prod_{p\cap V(Q)\neq\emptyset}\varphi((\sigma,\omega)_p)$. Every $h \in \mathcal{G}_Q$ fixes the links outside $V(Q)$, whose end points are not interior, so $(h\sigma,\omega) = h\cdot(\sigma,\omega)$ and each plaquette holonomy is conjugated; as φ is a class function, $w(h\sigma) = w(\sigma)$. Hence $w(\sigma) = w(H(\sigma)^{-1}\sigma) = \mathcal{W}(\tau(\sigma))$ by (i), and for bounded Borel Ψ, by (ii), $\int \Psi(\tau(\sigma))w(\sigma)\,d\lambda(\sigma) = \int \Psi(\tau)\mathcal{W}(\tau)\,d\lambda(\tau)$.

(iv) The group $\mathcal{G}_Q \cong G^I$ acts on $G^{V(Q_-)}$ link by link. Let $X = f(\sigma_{V(Q_-)})$ almost surely with f Borel, and put $\bar f(\eta) = \int f(h\eta)\,dh$ where the integral converges absolutely and $\bar f = 0$ elsewhere; by invariance of Haar measure both the set of convergence and $\bar f$ are invariant under every $k \in \mathcal{G}_Q$. Since ν and λ have the same null sets, the density being bounded above and below, and $f(h\eta) = f(\eta)$ for ν-almost every configuration for each h, Fubini's theorem gives $\bar f = f$ almost surely. Then $X = \bar f(\sigma_{V(Q_-)}) = \bar f((H(\sigma)^{-1}\sigma)_{V(Q_-)})$ almost surely, and by (i) the right side is a function of $\tau(\sigma)_{E'\cap V(Q_-)}$, the coordinates in F' being 1. □

Lemma G.7 (near-product densities)**.** *Let (S_1,μ_1) and (S_2,μ_2) be probability spaces and π a probability measure on $S_1\times S_2$ with density r with respect to $\mu_1\otimes\mu_2$, $|r-1| \le \delta < 1$. If $X \in L^2(\pi)$ depends only on s_1, $Y \in L^2(\pi)$ only on s_2, and $\pi(X) = \pi(Y) = 0$, then $|\pi(XY)| \le \frac{\delta+\delta^2}{1-\delta}\|X\|_{L^2(\pi)}\|Y\|_{L^2(\pi)}$. If $r = k/Z'$ with $1-\varepsilon \le k \le 1+\varepsilon$ and $Z' = \int k\,d(\mu_1\otimes\mu_2)$, then $\delta = 2\varepsilon/(1-\varepsilon)$ is admissible, and for $\varepsilon \le \frac{1}{10}$ the bound is at most 4ε.*

Proof. The marginal density $m_1(s_1) = \int r(s_1,\cdot)\,d\mu_2$ lies in $[1-\delta, 1+\delta]$, so $\int X^2 d\mu_1 \le (1-\delta)^{-1}\|X\|_{L^2(\pi)}^2$; and $\int X\,d\mu_1 = \int X(1-m_1)\,d\mu_1$ since $\int X m_1\,d\mu_1 = \pi(X) = 0$, so $|\int X\,d\mu_1| \le \delta\int|X|\,d\mu_1$. The same holds for Y. Now $\pi(XY) = \iint XY(r-1)\,d\mu_1 d\mu_2 + \int X\,d\mu_1\int Y\,d\mu_2$, and by the Cauchy–Schwarz inequality the two terms together are at most $(\delta+\delta^2)(\int X^2 d\mu_1\int Y^2 d\mu_2)^{1/2}$, which gives the first claim. If $r = k/Z'$, then $Z' \in [1-\varepsilon,1+\varepsilon]$ and $r \in [\frac{1-\varepsilon}{1+\varepsilon}, \frac{1+\varepsilon}{1-\varepsilon}]$, so $|r-1| \le 2\varepsilon/(1-\varepsilon)$. For $\varepsilon \le \frac{1}{10}$, $\delta \le \frac{20}{9}\varepsilon \le \frac{2}{9}$ and $\frac{\delta(1+\delta)}{1-\delta} \le \frac{11}{7}\delta \le \frac{220}{63}\varepsilon < 4\varepsilon$. □

For $G = \mathrm{SU}(2)$ write χ_j for the character of spin $j \in \frac12\mathbb{Z}_{\ge0}$, $d_j = 2j+1$ and $c_j = j(j+1)$, and let

$$p_t = \sum_j d_j e^{-tc_j}\chi_j \qquad (t > 0)$$

be the heat kernel. It is a uniformly convergent series of real class functions, $\int p_t = 1$, $p_t * p_{t'} = p_{t+t'}$ because $\chi_i * \chi_j = \delta_{ij}\chi_j/d_j$, and $p_t > 0$, as the heat kernel of a compact connected Riemannian manifold [27, Chs. 7–8]. It is the weight of the *heat-kernel action*, for which the lattice theory in two dimensions reproduces the continuum Yang–Mills measure on every lattice [25, 21, 35], and the two-dimensional tuning is $t = g^2a^2$ with a coupling g of the dimension of a mass. With $n = 2j+1$, $c_j = (n^2-1)/4$ and $|\chi_j| \le d_j$,

$$\|p_\theta - 1\|_\infty \le \sum_{n\ge2} n^2 e^{-\theta(n^2-1)/4} \le \sum_{k\ge1}(k+1)^2 q^k \le \frac{4q}{(1-q)^3}, \qquad q = e^{-3\theta/4}, \tag{G.1}$$

using $n^2 - 1 \ge 3(n-1)$ and $\sum_{k\ge1}(k+1)^2q^k = (4q - 3q^2 + q^3)/(1-q)^3$. For $\theta \ge 6$, $q \le e^{-4.5} < \frac{1}{20}$ and the right side is at most $5q \le 0.06$.

Theorem G.8 (two dimensions, fat boxes)**.**

(i) *Let $d \geq 2$, $s \geq 1$, let Q be a box of $\mathbb{Z}^2$ of widths $3d-1$ and s, in either order, split in the coordinate of width $3d-1$ into widths d, $d-1$, d, and put $N = 1+(d-1)(s+1)$ and $\varepsilon_N = \|\varphi^{*N} - 1\|_\infty$. If $\varepsilon_N \leq \frac{1}{10}$, then $\rho^\omega(Q_-, Q_+) \leq 4\varepsilon_N$ for every exterior configuration ω.*

(ii) *Let $G = \mathrm{SU}(2)$, $\varphi = p_t$ with $t = g^2a^2$, $d \geq 2$ and $gda \geq \sqrt{6}$. For every box of widths $3d-1$ and $s \geq 2d-1$ and every ω, $\rho^\omega(Q_-, Q_+) \leq 20e^{-3Nt/4}$ with $N = 1+(d-1)(s+1)$. In particular the two-dimensional form of $(\mathrm{M}_{d,m}(c))$ holds for every $m \geq 4$ with*

$$c = 20\, e^{-3g^2(da)^2/4}.$$

Proof. (i) *Reduction to the split in the first coordinate.* The exchange of the two coordinates maps links to links, boxes to boxes, interior vertices to interior vertices and $\mathcal{G}_Q$ to $\mathcal{G}_{Q'}$, and it maps the holonomy of the plaquette at z to the inverse of the holonomy of the plaquette at the image of z; since $\varphi(g^{-1}) = \varphi(g)$, it maps ν_Q^ω to $\nu_{Q'}^{\omega'}$. So we may take Q as above.

Reduction to the axial model. By Lemma G.6(iii) and (iv), $\rho^\omega(Q_-, Q_+)$ is at most the supremum of $|\pi(\bar{X}\bar{Y})|$ over centred unit vectors $\bar{X} \in L^2(\pi)$ depending on $\tau_- = \tau_{E' \cap V(Q_-)}$ and $\bar{Y}$ depending on $\tau_+ = \tau_{E' \cap V(Q_+)}$, where π is the law on $G^{E'}$ with density proportional to $\mathcal{W}$.

Which plaquettes couple the three parts. Every plaquette meeting $V(Q)$ contains a link of E': the plaquette at $z \in Q$ contains $h(z)$, the plaquette at $(-1, y)$ contains $v(0, y)$, and the plaquette at $(x, -1)$ contains $h(x, 0)$. The plaquette at (x, y) contains $h(x, y)$, $h(x, y+1)$ and $v(x, y)$, which lie in the part containing the column x, and $v(x+1, y)$, which lies in the part containing the column $x+1$ and belongs to F', so is fixed to 1, unless $y = s-1$. Hence the only plaquettes that contain free links of two different parts are the two seam plaquettes at $(d-1, s-1)$, joining Q_- and Q_0, and at $(2d-2, s-1)$, joining Q_0 and Q_+, and no plaquette contains free links of both Q_- and Q_+. So $\mathcal{W} = \mathcal{W}_-(\tau_-)\mathcal{W}_+(\tau_+)\widehat{\mathcal{W}}$, where $\widehat{\mathcal{W}}$ is the product of the factors of the plaquettes that contain a free link of Q_0.

Integration of the middle. Write $H_x = h(x, s-1)$, $V_x = v(x, s-1)$ and $e_x = \sigma_{((x,s),1)}$, an exterior link. For a column $x \in [d, 2d-2]$ the factors of $\widehat{\mathcal{W}}$ are: the bottom plaquette at $(x, -1)$, which equals $\varphi(h(x,0)^{-1}c_x)$ with c_x a product of exterior links, by the cyclic invariance of φ; the plaquettes at (x, y), $0 \leq y \leq s-2$, which equal $\varphi(h(x,y)h(x,y+1)^{-1})$, both vertical links being in F'; and the top plaquette $\varphi(H_xV_{x+1}e_x^{-1}V_x^{-1})$. Integrating $h(x,0), \dots, h(x, s-2)$ successively with Lemma G.5(i), each step having the form $\int \varphi(h_{y+1}^{-1}h_y)\psi(h_y^{-1}c_x)\, dh_y$, gives $\varphi^{*s}(H_x^{-1}c_x)$, and then $\int \varphi(V_{x+1}e_x^{-1}V_x^{-1}H_x)\varphi^{*s}(H_x^{-1}c_x)\, dH_x = \varphi^{*(s+1)}(V_{x+1}e_x^{-1}V_x^{-1}c_x)$, again by Lemma G.5(i). The seam plaquette at $(d-1, s-1)$ is $\varphi(H_{d-1}V_de_{d-1}^{-1}V_{d-1}^{-1}) = \varphi(V_de_{d-1}^{-1}V_{d-1}^{-1}H_{d-1})$, with H_{d-1} and V_{d-1} free links of Q_-, and the seam plaquette at $(2d-2, s-1)$ is the top plaquette of the column $2d-2$, which contains V_{2d-1}, a free link of Q_+. Lemma G.5(ii) with $M = d-1$, $\alpha_0 = \varphi$, $\alpha_j = \varphi^{*(s+1)}$ for $j \geq 1$, $V_0 = V_{d-1}$, $c_0 = H_{d-1}$, $g_j = e_{d-1+j}^{-1}$ and $c_j = c_{d-1+j}$ gives, after a cyclic rearrangement,

$$\int \widehat{\mathcal{W}}\, d\lambda(\tau_0) = \varphi^{*N}(\Phi_-\Phi_+), \qquad \Phi_- = V_{d-1}^{-1}H_{d-1}C, \quad \Phi_+ = V_{2d-1}\Gamma,$$

with $C = c_d \cdots c_{2d-2}$ and $\Gamma = e_{2d-2}^{-1} \cdots e_{d-1}^{-1}$ exterior, and $N = 1+(d-1)(s+1)$.

Conclusion. Under π the law of the pair (τ_-, τ_+) has a density proportional to

$$\mathcal{W}_-(\tau_-)\, \mathcal{W}_+(\tau_+)\, \varphi^{*N}(\Phi_-\Phi_+).$$

Let μ_1 and μ_2 be the probability measures proportional to $\mathcal{W}_-\lambda$ and to $\mathcal{W}_+\lambda$. Then the law has density k/Z' with respect to $\mu_1 \otimes \mu_2$, where $k = \varphi^{*N}(\Phi_-\Phi_+)$ takes values in $[1-\varepsilon_N, 1+\varepsilon_N]$, and Lemma G.7 gives the bound $4\varepsilon_N$.

(ii) For $s \geq 2d-1$, $N \geq 1+2d(d-1) \geq d^2$, so $Nt \geq g^2d^2a^2 \geq 6$. Since $\varphi^{*N} = p_{Nt}$, (G.1) gives $\varepsilon_N \leq 5e^{-3Nt/4} \leq 0.06$, and (i) gives $\rho^\omega \leq 20e^{-3Nt/4} \leq 20e^{-3g^2d^2a^2/4}$. □

The heat-kernel weight has string tension $\frac{3}{4}g^2$, and Nt is g^2 times the area of the middle slab up to boundary terms, so the bound reads $\rho \le 20e^{-\frac{3}{4}g^2\cdot\mathrm{area}}$: in two dimensions the slab decorrelation of Section 10 is the confinement of the middle slab, and it holds uniformly in the boundary condition with $c \to 0$ as the physical size da of the box grows. The proof rests on one fact. After the axial gauge the plaquettes straddling each seam contain a single free link, so the middle slab integrates out exactly to a function $\varphi^{*N}(\Phi_-\Phi_+)$ of one group element, uniformly close to 1 once the middle slab is confined. In four dimensions the same gauge leaves on each seam the free links of a three-dimensional layer, and integrating out the middle produces the $(d-1)$-step transfer operator of the box, a kernel on of order $(md)^3$ copies of G; the hypothesis $(\mathrm{M}_{d,m}(c))$ is a bound on its second singular value on gauge-invariant functions, uniform in the boundary condition, and the argument above has no four-dimensional counterpart.

G.3. **Two dimensions: thin strips.** The restriction of $(\mathrm{M}_{d,m}(c))$ to boxes all of whose widths are at least $2d-1$ is needed. We show that for strips of width one the same bound fails along the two-dimensional tuning for every $c < 1$.

Identify SU(2) with the unit quaternions $g = g_0 + g_1\mathbf{i} + g_2\mathbf{j} + g_3\mathbf{k}$; then $\chi := \chi_{1/2}$ satisfies $\chi(g) = \operatorname{Tr} g = 2g_0$, $\chi(g^{-1}) = \chi(g)$, and $\langle g, h\rangle_{\mathbb{R}^4} = \frac{1}{2}\chi(g^{-1}h)$. For $\epsilon \in \{1, -1\}$ and $a, b, c > 0$ with $T = a + b + c$ let $\mu^{\epsilon}_{a,b,c}$ be the probability measure on $\mathrm{SU}(2)^2$ with density

$$p_a(x)\, p_b(x^{-1}y)\, p_c(\epsilon y)/p_T(\epsilon)$$

with respect to $\lambda\otimes\lambda$: a Brownian bridge on SU(2) from 1 to ϵ observed at the times a and $a+b$. The normalisation follows from $\int p_a(x)p_b(x^{-1}y)\,dx = p_{a+b}(y)$ and $\int p_{a+b}(y)p_c(\epsilon y)\,dy = p_T(\epsilon)$, both instances of Lemma G.5(i).

Lemma G.9 (the strip is a bridge)**.** *Let $Q = [0, 3d-1) \times \{0\}$, $d \ge 2$, split in the first coordinate, let $\varphi = p_t$, and let ω_ϵ be the exterior configuration with all links equal to 1 except $\sigma_{((3d-1,0),2)} = \epsilon$. Then $\mathcal{G}_Q$ is trivial, so $\mathcal{A}_Q^{\omega_\epsilon} = L^2(\nu_Q^{\omega_\epsilon})$, and with $h_x = h(x,0)$ and $v_x = v(x,0)$ the pair $(x, y) = (v_{d-1}^{-1}h_{d-1}, v_{2d-1}^{-1})$, formed by a function of $\sigma_{V(Q_-)}$ and a function of $\sigma_{V(Q_+)}$, has law $\mu^{\epsilon}_{a,b,c}$ with $a = c = 2dt$ and $b = (2d-1)t$.*

Proof. No vertex v of Q has $v - e_2 \in Q$, so there are no interior vertices. The plaquettes meeting $V(Q)$ are those at $(-1, 0)$, with holonomy v_0; at $(x, -1)$, with holonomy h_x^{-1}; and at $(x, 0)$, $0 \le x \le 3d-2$, with holonomy $h_x v_{x+1} v_x^{-1}$, where $v_{3d-1} := \epsilon$. So the density of ν is proportional to $p_t(v_0)\prod_x p_t(h_x)\prod_x p_t(h_x v_{x+1}v_x^{-1})$. For $x \ne d-1$, $\int p_t(h)p_t(hg)\,dh = (p_t * p_t)(g) = p_{2t}(g)$ by Lemma G.5(i) and $p_t(h^{-1}) = p_t(h)$. Integrating then $v_0, \dots, v_{d-2}$, $v_{2d}, \dots, v_{3d-2}$ and $v_d, \dots, v_{2d-2}$ with Lemma G.5(i), in the form $\int \alpha(gy^{-1})\beta(yh)\,dy = (\alpha * \beta)(gh)$, leaves the density of $(v_{d-1}, h_{d-1}, v_{2d-1})$ proportional to

$$p_{(2d-1)t}(v_{d-1})\, p_t(h_{d-1})\, p_{(2d-1)t}(v_{2d-1}v_{d-1}^{-1}h_{d-1})\, p_{2dt}(\epsilon v_{2d-1}^{-1}).$$

Substituting $x = v_{d-1}^{-1}h_{d-1}$ for h_{d-1}, which preserves Haar measure for fixed v_{d-1}, and integrating v_{d-1} gives $(p_{(2d-1)t} * p_t)(x) = p_{2dt}(x)$. With $y = v_{2d-1}^{-1}$ the third factor is $p_{(2d-1)t}(y^{-1}x) = p_b(x^{-1}y)$, and the last is $p_{2dt}(\epsilon y)$. □

Lemma G.10 (moments of the bridge)**.** *Under $\mu^{\epsilon}_{a,b,c}$, $E[x_1] = E[y_1] = 0$ and*

$$E[x_1y_1] = \tfrac{1}{3}(\tfrac{1}{2}D - \tfrac{1}{4}C), \quad E[x_1^2] = \tfrac{1}{3}(1 - \tfrac{1}{4}A), \quad E[y_1^2] = \tfrac{1}{3}(1 - \tfrac{1}{4}B),$$

where $D = E\chi(x^{-1}y)$, $C = E[\chi(x)\chi(y)]$, $A = E\chi(x)^2$ *and* $B = E\chi(y)^2$ *are given, with* $q_\theta(n) = e^{-\theta(n^2-1)/4}$ *and* $Z = p_T(\epsilon)$, *by*

$$\begin{aligned} Z &= \tfrac{1}{2}\textstyle\sum_{n\in\mathbb{Z}} \epsilon^{n-1} n^2 q_T(n), \qquad DZ = \sum_{n\in\mathbb{Z}} \epsilon^{n-1} n(n+1) q_{a+c}(n) q_b(n+1), \\ CZ &= \textstyle\sum_{n\in\mathbb{Z}} \epsilon^n \big[(n+1)^2 q_a(n+1) q_c(n+1) + (n^2-1) q_a(n+1) q_c(n-1)\big] q_b(n), \\ AZ &= 2Z + \textstyle\sum_{n\in\mathbb{Z}} \epsilon^{n-1} n(n+2) q_a(n+2) q_{b+c}(n), \\ BZ &= 2Z + \textstyle\sum_{n\in\mathbb{Z}} \epsilon^{n-1} n(n+2) q_c(n+2) q_{a+b}(n). \end{aligned}$$

Proof. Symmetry. Conjugation $(x,y) \mapsto (gxg^{-1}, gyg^{-1})$ preserves μ^ϵ, since ϵ is central and the p_θ are class functions. It fixes x_0 and y_0 and rotates (x_1,x_2,x_3) and (y_1,y_2,y_3) by the same element of SO(3). Conjugation by $\mathbf{j}$ maps $\mathbf{i}$ to $-\mathbf{i}$, so $E[x_1] = E[y_1] = 0$; conjugation by $\frac{1}{2}(1+\mathbf{i}+\mathbf{j}+\mathbf{k})$ permutes $\mathbf{i},\mathbf{j},\mathbf{k}$ cyclically, so $E[x_1y_1] = \frac{1}{3}E\sum_k x_k y_k$ and $E[x_1^2] = \frac{1}{3}E\sum_k x_k^2$. Finally $\sum_k x_k y_k = \langle x,y\rangle - x_0y_0 = \frac{1}{2}\chi(x^{-1}y) - \frac{1}{4}\chi(x)\chi(y)$ and $\sum_k x_k^2 = 1 - \frac{1}{4}\chi(x)^2$.

Reduction to character sums. Write class functions as $\sum_{n\ge1} F_n \chi_{(n-1)/2}$. Then p_θ has coefficients $nq_\theta(n)$, and $p_\theta(\epsilon\,\cdot)$ has coefficients $\epsilon^{n-1} n q_\theta(n)$, since $\chi_j(\epsilon g) = \epsilon^{2j}\chi_j(g)$; moreover $\int(\sum F_n\chi)(\sum G_n\chi) = \sum F_nG_n$, and $(\alpha * p_\theta * \gamma)(1) = \sum_n \alpha_n\gamma_n q_\theta(n)$ because $\chi_i * \chi_j = \delta_{ij}\chi_j/d_j$ and $\chi_j(1) = d_j$. By the Clebsch–Gordan rule $\chi\chi_j = \chi_{j-1/2} + \chi_{j+1/2}$ for $j \ge \frac{1}{2}$ and $\chi\chi_0 = \chi$, multiplication by χ acts on coefficients by $F_n \mapsto F_{n+1} + F_{n-1}$ with $F_0 := 0$. The marginal of x is $p_a(x)p_{b+c}(\epsilon x)/Z$ and that of y is $p_{a+b}(y)p_c(\epsilon y)/Z$; substituting $z = x^{-1}y$, the law of z is $p_b(z)p_{a+c}(\epsilon z)/Z$; and $C = ((\chi p_a) * p_b * \gamma)(1)/Z$ with $\gamma(w) = \chi(w)p_c(\epsilon w)$. All series converge absolutely. Hence, with $P_n^{(\theta)} = nq_\theta(n)$ and $\alpha_n^{(\theta)} = P_{n+1}^{(\theta)} + P_{n-1}^{(\theta)}$,

$$Z = \sum_{n\ge1} \epsilon^{n-1} n^2 q_T(n), \quad DZ = \sum_{n\ge1} \epsilon^{n-1} \alpha_n^{(b)}\, n q_{a+c}(n), \quad CZ = \sum_{n\ge1} \epsilon^n \alpha_n^{(a)} \alpha_n^{(c)} q_b(n),$$

$$AZ = \sum_{n\ge1} \epsilon^{n-1} (\alpha_{n+1}^{(a)} + \alpha_{n-1}^{(a)})\, n q_{b+c}(n), \quad BZ = \sum_{n\ge1} \epsilon^{n-1} (\alpha_{n+1}^{(c)} + \alpha_{n-1}^{(c)})\, n q_{a+b}(n),$$

where for CZ we used that the coefficients of γ are $\sum_\pm \epsilon^{n\pm1-1} P_{n\pm1}^{(c)} = \epsilon^n \alpha_n^{(c)}$, and $\alpha_0 := 0$.

Symmetrisation. Extend $P_n^{(\theta)} = nq_\theta(n)$ to $n \in \mathbb{Z}$; it is odd, so $\alpha^{(\theta)}$, now defined for all n, is odd with $\alpha_0^{(\theta)} = 0$ and agrees with the above for $n \ge 1$, and so is $\alpha_{n+1} + \alpha_{n-1}$. Each summand above is then an even function of n vanishing at 0, as $\epsilon^{n\pm1}$ is even in n, so $\sum_{n\ge1} = \frac{1}{2}\sum_{n\in\mathbb{Z}}$. Expanding the products and using the substitution $n \mapsto -n$, under which $P_{n-1} \mapsto -P_{n+1}$ and $P_{n-2} \mapsto -P_{n+2}$, to identify the terms in pairs gives the stated formulas; for AZ the middle term $2P_n^{(a)} \cdot nq_{b+c}(n)$ gives $2Z$, since $q_a q_{b+c} = q_T$. □

Lemma G.11 (Poisson summation)**.** *Let* $0 < T \le 1$, $\lambda = T/4$, $\gamma = \sqrt{4\pi/T}$, $\mu \in \mathbb{R}$, $\widetilde{Q}(m) = q_2m^2 + q_1m + q_0$ *real, and* $f(x) = \widetilde{Q}(x+\mu)e^{-\lambda(x+\mu)^2}$.

(i) $|\sum_{n\in\mathbb{Z}} f(n) - \gamma(q_0 + 2q_2/T)| \le 30\gamma(|q_0| + 4|q_1|/T + 16|q_2|/T^2)e^{-4\pi^2/T}$.

(ii) *If* $q_2 = 1$, $|q_1| \le 2$ *and* $|q_0| \le 1$, *then* $\sum_{n\in\mathbb{Z}}(-1)^{n-1} f(n) = \Lambda_0(\cos\pi\mu - \mathcal{E})$ *with* $\Lambda_0 = \frac{8\pi^2}{T^2}\gamma e^{-\pi^2/T}$ *and* $|\mathcal{E}| \le 0.22\,T$.

Proof. f is a Schwartz function. With $\widehat{f}(\xi) = \int f(x)e^{-2\pi i\xi x}dx$, the substitution $y = x + \mu$ and the formulas $\int e^{-\lambda y^2 - i\omega y}dy = \gamma e^{-\omega^2/(4\lambda)} =: G(\omega)$, $\int y e^{-\lambda y^2 - i\omega y}dy = iG'(\omega)$, $\int y^2 e^{-\lambda y^2 - i\omega y}dy = -G''(\omega)$, $G' = -\frac{\omega}{2\lambda}G$ and $G'' = (\frac{\omega^2}{4\lambda^2} - \frac{1}{2\lambda})G$ give

$$\widehat{f}(\xi) = e^{2\pi i\xi\mu}\gamma e^{-\pi^2\xi^2/\lambda}\Big[q_0 - iq_1\frac{\pi\xi}{\lambda} + q_2\Big(\frac{1}{2\lambda} - \frac{\pi^2\xi^2}{\lambda^2}\Big)\Big].$$

(i) By the Poisson summation formula [47, Ch. 5], $\sum_n f(n) = \sum_k \widehat{f}(k)$, and $\widehat{f}(0) = \gamma(q_0 + 2q_2/T)$. For $k \ne 0$, since $\lambda \le \frac{1}{4}$, each of 1, $\pi|k|/\lambda$ and $\frac{1}{2\lambda} + \frac{\pi^2k^2}{\lambda^2}$ is at most $M_k\lambda^{-e}$ with $e = 0,1,2$ respectively and $M_k = 1 + \pi|k| + \pi^2k^2$; so $|\widehat{f}(k)| \le \gamma e^{-\pi^2k^2/\lambda} M_k(|q_0| + |q_1|/\lambda + |q_2|/\lambda^2)$, and $\sum_{k\ne0} e^{-\pi^2k^2/\lambda}M_k \le 2e^{-\pi^2/\lambda}(M_1 + \sum_{k\ge2} M_k e^{-12\pi^2(k-1)}) \le 30e^{-\pi^2/\lambda}$, using $k^2 - 1 \ge 3(k-1)$ and $M_1 < 14.1$. Put $\lambda = T/4$.

(ii) Poisson summation for $f(x)e^{i\pi x}$ gives $\sum_n (-1)^n f(n) = \sum_{\xi\in\frac12+\mathbb{Z}} \widehat{f}(\xi)$. The terms $\xi = \pm\frac12$ are complex conjugate and add up to $2\gamma e^{-\pi^2/(4\lambda)}[\cos(\pi\mu)(q_0+\frac{1}{2\lambda}-\frac{\pi^2}{4\lambda^2})+\sin(\pi\mu)\frac{q_1\pi}{2\lambda}]$. Hence $\sum(-1)^{n-1}f(n) = \Lambda_0(\cos\pi\mu - \mathcal{E})$ with $\Lambda_0 = \frac{\pi^2}{2\lambda^2}\gamma e^{-\pi^2/(4\lambda)}$, the stated value, and $\mathcal{E} = \mathcal{E}_1 + \mathcal{E}_2$, where $\mathcal{E}_1 = \frac{2\lambda^2}{\pi^2}[\cos(\pi\mu)(q_0+\frac{1}{2\lambda}) + \sin(\pi\mu)\frac{q_1\pi}{2\lambda}]$ and $\mathcal{E}_2 = \Lambda_0^{-1}\sum_{|\xi|\ge 3/2}\widehat{f}(\xi)$. First, $|\mathcal{E}_1| \le \frac{2}{\pi^2}(\lambda^2+\frac{\lambda}{2}+\pi\lambda) \le \frac{2}{\pi^2}(\frac14+\frac12+\pi)\lambda < 0.79\lambda$. Second, for $|\xi|\ge\frac32$, $|\widehat{f}(\xi)|/\Lambda_0 \le e^{-\pi^2(\xi^2-1/4)/\lambda}(\frac{2\lambda^2}{\pi^2}+\frac{4\lambda|\xi|}{\pi}+\frac{\lambda}{\pi^2}+2\xi^2) \le 3\xi^2 e^{-\pi^2(\xi^2-1/4)/\lambda}$, and with $\xi = \pm(k+\frac12)$, $\xi^2-\frac14 = k^2+k \ge 2k$, so $|\mathcal{E}_2| \le 6\sum_{k\ge1}(k+\frac12)^2 e^{-2\pi^2 k/\lambda} \le 14e^{-2\pi^2/\lambda} \le 56e^{-8\pi^2}\lambda$, as $e^{-2\pi^2/\lambda}/\lambda$ increases on $(0,\frac14]$. So $|\mathcal{E}| \le 0.8\lambda \le 0.22\,T$. □

Lemma G.12 (the bridge from 1 to 1)**.** *Let $0 < T \le 1$ and suppose that $\kappa := ac/T$, $\tau_x := a(b+c)/T$ and $\tau_y := c(a+b)/T$ are all at least $T/10$. Under $\mu^1_{a,b,c}$,*

$$\mathrm{Corr}(x_1,y_1) \ge r_G(1-T), \qquad r_G = \Big(\frac{ac}{(a+b)(b+c)}\Big)^{1/2}.$$

Here r_G is the correlation of a Euclidean Brownian bridge on $[0,T]$ at the times a and $a+b$.

Proof. Main terms. Complete the squares in the exponents of Lemma G.10 with $\epsilon = 1$: $q_T(n) = e^{T/4}e^{-Tn^2/4}$; $q_{a+c}(n)q_b(n+1) = e^{E_D}e^{-T(n+\mu_D)^2/4}$ with $\mu_D = b/T$ and $E_D = \frac{b^2}{4T}+\frac{a+c}{4}$; $q_a(n+1)q_c(n+1)q_b(n) = e^{E_1}e^{-T(n+\mu_1)^2/4}$ with $\mu_1 = \frac{a+c}{T}$ and $E_1 = \frac{(a+c)^2}{4T}+\frac{b}{4}$; $q_a(n+1)q_c(n-1)q_b(n) = e^{E_2}e^{-T(n+\mu_2)^2/4}$ with $\mu_2 = \frac{a-c}{T}$ and $E_2 = \frac{(a-c)^2}{4T}+\frac{b}{4}$; and $q_a(n+2)q_{b+c}(n) = e^{E_A}e^{-T(n+\mu_A)^2/4}$ with $\mu_A = \frac{2a}{T}$ and $E_A = \frac{a^2}{T} - \frac{3a-b-c}{4}$. In the variable $m = n+\mu$ the polynomial factors become m^2 for Z, $m^2+(1-2\mu_D)m+\mu_D^2-\mu_D$ for D, $(m+1-\mu_1)^2$ with $1-\mu_1 = b/T$ and $(m-\mu_2)^2-1$ for C, and $(m-\mu_A)(m-\mu_A+2)$ for A. The main terms $\gamma(q_0+2q_2/T)$ of Lemma G.11(i), divided by the main term $\gamma e^{T/4}/T$ of Z, give, after the simplifications $T-b = a+c$, $(a-c)^2 - T(a+c) = -4ac - b(a+c)$ and $a^2/T - a = -\tau_x$,

$$D_0 = e^{-u}(2-4u), \qquad C_0 = e^{-u}\Big(2+\frac{b^2}{T}\Big) + e^{-u-\kappa}\Big(2+\frac{(a-c)^2}{T}-T\Big),$$
$$A_0 = 2+e^{-\tau_x}(2-4\tau_x), \qquad B_0 = 2+e^{-\tau_y}(2-4\tau_y), \qquad u = \frac{b(a+c)}{4T}.$$

Put $P = \frac{b(T+a+c)}{4T}$. Using $\frac{(a-c)^2-T^2}{4T} = -\frac{(2a+b)(2c+b)}{4T}$ and $\frac{(2a+b)(2c+b)}{4T} = P+\kappa$, one finds

$$X_0 := \tfrac12 D_0 - \tfrac14 C_0 = e^{-u}[(\tfrac12 - P)(1-e^{-\kappa}) + \kappa e^{-\kappa}], \qquad V_0^x := 1 - \tfrac14 A_0 = \tfrac12(1-e^{-\tau_x}) + \tau_x e^{-\tau_x}.$$

Since $ac \le T^2/4$ and $b(a+c) \le T^2/4$, we have $\kappa \le T/4$, $u \le T/16$ and $P \le b/2 \le T/2 \le \frac12$. With $1-e^{-\kappa} \ge \kappa - \kappa^2/2$, $e^{-\kappa} \ge 1-\kappa$ and $e^{-u} \ge 1-u$,

$$X_0 \ge (1-u)\kappa(\tfrac32 - P - \tfrac54\kappa) \ge \tfrac32\kappa(1-\tfrac{T}{16})(1-\tfrac{13}{24}T) \ge \tfrac32\kappa(1-\tfrac{29}{48}T).$$

Also $V_0^x \le \frac32\tau_x$ and $V_0^y := 1 - \frac14 B_0 \le \frac32\tau_y$.

Errors. Each of the five sums is its main term times $1+\theta$ with $|\theta| \le \eta := 750e^{-4\pi^2/T}/T$. This follows from Lemma G.11(i), with $T \le 1$, the coefficient bounds $q_2 = 1$, $|q_1| \le 2$ and $|q_0| \le 1$ read off above, and the lower bounds $2/T - 1/4$, $2/T$, $2/T-1$ and $2/T-1 \ge 1/T$ for the brackets $q_0 + 2q_2/T$ of the main terms: the relative errors are at most $240e^{-4\pi^2/T}/T$ for Z and $750e^{-4\pi^2/T}/T$ for the others. As $e^{-4\pi^2/T}/T^3$ increases on $(0,1]$, $\eta \le 750e^{-4\pi^2}T^2 < 10^{-14}T^2$. Since $|D_0| \le 2$, both parts of C_0 are at most 3 in absolute value, and $0 \le A_0 - 2 \le 2$, we get $|D-D_0| \le 6\eta$, $|C-C_0| \le 18\eta$ and $|A-A_0|, |B-B_0| \le 6\eta$, hence $|X-X_0| \le 7.5\eta$ and $|V^x - V_0^x|, |V^y - V_0^y| \le 1.5\eta$, where $X = \frac12 D - \frac14 C$, $V^x = 1 - \frac14 A$ and $V^y = 1 - \frac14 B$.

Conclusion. By Lemma G.10, $\mathrm{Corr}(x_1,y_1) = X/(V^xV^y)^{1/2}$. Using $\kappa,\tau_x,\tau_y \ge T/10$,

$$\mathrm{Corr}(x_1,y_1) \ge \frac{\frac{3}{2}\kappa(1-\frac{29}{48}T-50\eta/T)}{\frac{3}{2}(\tau_x\tau_y)^{1/2}(1+10\eta/T)} \ge r_G\Big(1-\frac{29}{48}T-60\frac{\eta}{T}\Big) \ge r_G(1-T),$$

since $\kappa/(\tau_x\tau_y)^{1/2} = r_G$ and $60\eta/T < 10^{-12}T$. □

Lemma G.13 (the bridge from 1 to -1)**.** *Let $s_0 > 0$ and $0 < T \le \min(1, s_0^2/4)$, and suppose $\sin(\pi a/T) \ge s_0$ and $\sin(\pi c/T) \ge s_0$. Under $\mu^{-1}_{a,b,c}$,*

$$\mathrm{Corr}(x_1,y_1) \ge 1 - \frac{5T}{2s_0^2}.$$

Proof. Write $r_a = a/T$, $r_b = b/T$, $r_c = c/T$, $s_a = \sin\pi r_a$ and $s_c = \sin\pi r_c$. The sums of Lemma G.10 with $\epsilon = -1$ have the exponents and polynomial factors of the proof of Lemma G.12, all with $q_2 = 1$, $|q_1| \le 2$ and $|q_0| \le 1$, and carry the sign $(-1)^{n-1}$, except CZ, which carries $(-1)^n$. Lemma G.11(ii) gives

$$Z = \tfrac{1}{2}e^{T/4}\Lambda_0(1-\mathcal{E}_Z), \qquad D = 2e^{-u}\frac{\cos\pi r_b - \mathcal{E}_D}{1-\mathcal{E}_Z},$$
$$C = -2\,\frac{e^{-u}(\cos\pi(r_a+r_c)-\mathcal{E}_1) + e^{-u-\kappa}(\cos\pi(r_a-r_c)-\mathcal{E}_2)}{1-\mathcal{E}_Z},$$
$$A = 2 + 2e^{-\tau_x}\frac{\cos 2\pi r_a - \mathcal{E}_A}{1-\mathcal{E}_Z}, \qquad B = 2 + 2e^{-\tau_y}\frac{\cos 2\pi r_c - \mathcal{E}_B}{1-\mathcal{E}_Z},$$

with all $|\mathcal{E}_\bullet| \le 0.22T$ and u,κ,τ_x,τ_y as before; the exponential prefactors are those of the main terms in the proof of Lemma G.12. Each quotient has the form $w(\cos\theta - \mathcal{E})/(1-\mathcal{E}_Z)$ with $w \in [1-T/4, 1]$, as $u+\kappa, \tau_x, \tau_y \le T/4$; since $|1/(1-\mathcal{E}_Z)| \le 1.29$ and $|1/(1-\mathcal{E}_Z)-1| \le 0.29T$, it differs from $\cos\theta$ by at most $0.25T\cdot 1.29 + 0.29T + 0.22T\cdot 1.29 < T$. Hence

$$\big|\tfrac{D}{2} - \cos\pi r_b\big| \le T, \quad \big|\tfrac{C}{4} + \cos\pi r_a\cos\pi r_c\big| \le T, \quad \big|\tfrac{A}{4} - \cos^2\pi r_a\big| \le \tfrac{T}{2}, \quad \big|\tfrac{B}{4} - \cos^2\pi r_c\big| \le \tfrac{T}{2},$$

using $\frac{1}{2}(\cos\pi(r_a+r_c) + \cos\pi(r_a-r_c)) = \cos\pi r_a\cos\pi r_c$. Since $r_b = 1 - r_a - r_c$, $\cos\pi r_b = -\cos\pi(r_a+r_c)$, so $\cos\pi r_b + \cos\pi r_a\cos\pi r_c = s_as_c$. Thus $X = \frac{1}{2}D - \frac{1}{4}C \ge s_as_c - 2T > 0$, and $0 < V^x = 1 - \frac{1}{4}A \le s_a^2 + \frac{T}{2}$ and $0 < V^y \le s_c^2 + \frac{T}{2}$, positivity coming from Lemma G.10 and $T \le s_0^2/4$. By Lemma G.10 and $\sqrt{(1+x)(1+y)} \le 1 + \frac{x+y}{2}$,

$$\mathrm{Corr}(x_1,y_1) \ge \frac{s_as_c - 2T}{s_as_c(1+\frac{T}{4s_a^2}+\frac{T}{4s_c^2})} \ge \frac{1-2T/s_0^2}{1+T/(2s_0^2)} \ge 1 - \frac{5T}{2s_0^2}.$$ □

Proposition G.14 (thin strips)**.** *Let $G = \mathrm{SU}(2)$, $\varphi = p_t$, $d \ge 2$, let $Q = [0, 3d-1)\times\{0\}$ be split in the first coordinate into widths d, $d-1$, d, put $T = (6d-1)t$, and let $\omega_{\pm1}$ be as in Lemma G.9.*

(i) *If $T \le 1$, then $\rho^{\omega_1}(Q_-,Q_+) \ge \frac{2d}{4d-1}(1-T)$.*

(ii) *If $T \le \frac{3}{16}$, then $\rho^{\omega_{-1}}(Q_-,Q_+) \ge 1 - \frac{10}{3}T$.*

(iii) *Along the tuning $t_k = g^2a_k^2$, $a_k \to 0$, with integers $d_k \ge 2$ and $d_ka_k \le \ell$, for every $m \ge 4$ and every $c < 1$ the bound $\rho^{\omega}(Q_-,Q_+) \le c$ fails for all large k for the strip of widths $3d_k - 1$ and 1, both at most $md_k - 1$; already the trivial boundary condition gives $\rho \ge \frac{1}{2}(1 - 6g^2\ell a_k)$.*

Proof. By Lemma G.9 the functions $X = x_1$ and $Y = y_1$ of $\sigma_{V(Q_-)}$ and $\sigma_{V(Q_+)}$ lie in $\mathcal{A}_Q^{\omega_\epsilon}$, and their joint law is that of $\mu^{\epsilon}_{a,b,c}$ with $a = c = 2dt$ and $b = (2d-1)t$; so $\rho^{\omega_\epsilon} \ge |\mathrm{Corr}(x_1,y_1)|$. (i) Here $\kappa/T = 4d^2/(6d-1)^2 \ge \frac{1}{9}$, $\tau_x/T = \tau_y/T = 2d(4d-1)/(6d-1)^2 \ge \frac{1}{5}$ for $d \ge 2$, and $r_G = 2d/(4d-1)$; apply Lemma G.12. (ii) $a/T = c/T = 2d/(6d-1) \in (\frac{1}{3},\frac{1}{2})$, so $\sin(\pi a/T) \ge \sin\frac{\pi}{3} = \frac{\sqrt{3}}{2} =: s_0$ with $s_0^2/4 = \frac{3}{16}$; apply Lemma G.13. (iii) $T_k = (6d_k-1)t_k \le 6g^2(d_ka_k)a_k \le 6g^2\ell a_k \to 0$, so for large k, by (ii), $\rho^{\omega_{-1}} \ge 1 - \frac{10}{3}T_k > c$, and by (i), $\rho^{\omega_1} \ge \frac{2d_k}{4d_k-1}(1-T_k) \ge \frac{1}{2}(1-6g^2\ell a_k)$. □

The mechanism is a bridge. The fixed boundary pins the holonomy of the loop surrounding the strip, and both outer parts read off the same collective variable. For the condition ω_{-1} the measure is invariant under the global conjugation $\sigma \mapsto g\sigma g^{-1}$ of all links of $V(Q)$, the exterior links being central, and part (ii) shows that this global SU(2) symmetry is broken at small T: the bridge from 1 to -1 concentrates near the two-sphere of minimising geodesics, whose point is read off by both outer parts, and the slab correlation tends to one. Theorem G.8 and Proposition G.14 together show that in two dimensions, along the weak-coupling tuning of a non-abelian theory, the slab hypothesis holds for fat boxes uniformly in the boundary condition with $c \to 0$, and fails for strips of width one for every $c < 1$. In four dimensions the boundary conditions with values in the centre are 't Hooft's twisted ones [50], and for fat boxes they are the natural test of $(\mathrm{M}_{d,m}(c))$; we have no estimate for them along the four-dimensional tuning.

G.4. **Curvature.** Theorem A is a curvature argument on the gauge-fixed measure, and its range is limited by the Hessian bound of Theorem 5.3. We show that no curvature argument applied to the Wilson measure itself, with or without the gauge, can be uniform along the weak-coupling tuning. We use the notation of Section 5: $G = \mathrm{SU}(N)$ with the metric $\langle X, Y\rangle = -\operatorname{Tr}(XY)$, tangent vectors $v = (X_e Q_e)$, the transported vectors $Y_1, \dots, Y_4$ of a plaquette, and $S_0 = \sum_p \operatorname{Re}\operatorname{Tr} Q_p$. For a probability measure with density proportional to e^{-V} on a compact Riemannian manifold, the *Bakry–Émery constant* is $\inf(\mathrm{Ric} + \mathrm{Hess}\, V)(v,v)/|v|^2$ over all points and nonzero tangent vectors; by Bochner's formula it is the largest K with $\Gamma_2(f) \geq K\Gamma(f)$ pointwise for all smooth f, and it controls the Poincaré and logarithmic Sobolev constants of the Langevin dynamics [10]. For a plaquette put $Z_p = Y_1 + Y_2 - Y_3 - Y_4$.

Lemma G.15 (central plaquettes)**.** *Let Q be a field and p a plaquette with $Q_p = wI$ central. Along the geodesics $Q_e(t) = e^{tX_e} Q_e$,*

$$\frac{d^2}{dt^2}\Big|_{t=0} \operatorname{Re}\operatorname{Tr} Q_p(t) = -\operatorname{Re}(w)\, |Z_p|^2.$$

If φ is a smooth class function on G, then $\frac{d^2}{dt^2}|_0 \varphi(Q_p(t)) = H_w(Z_p, Z_p)$ for an Ad*-invariant quadratic form H_w on $\mathfrak{g}$; for $G = \mathrm{SU}(2)$, $H_{\pm 1} = h_\pm |\cdot|^2$ with real constants $h_\pm$.*

Proof. As in Section 5, $Q_p(t) = e^{tY_1} e^{tY_2} W e^{-tY_3} e^{-tY_4}$ with $W = Q_p$. Since $W = wI$ is central, $Q_p(t) = w\, c(t)$ with $c(t) = e^{tY_1} e^{tY_2} e^{-tY_3} e^{-tY_4}$. With $a = (1, 1, -1, -1)$, $c''(0) = \sum_i a_i^2 Y_i^2 + 2\sum_{i<j} a_i a_j Y_i Y_j$, and as $\operatorname{Tr}(Y_i Y_j) = \operatorname{Tr}(Y_j Y_i)$, $\operatorname{Tr} c''(0) = \operatorname{Tr}(Z_p^2) = -|Z_p|^2$. Hence $\frac{d^2}{dt^2} \operatorname{Re}\operatorname{Tr} Q_p = \operatorname{Re}(w \operatorname{Tr} Z_p^2) = -\operatorname{Re}(w)|Z_p|^2$, the trace of a product of two skew-Hermitian matrices being real. For φ, the curve $\gamma(t) = Q_p(t)$ has $\gamma(0) = wI$ and $\gamma'(0) = Z_p wI$, and $(\varphi \circ \gamma)''(0) = \mathrm{Hess}\, \varphi(\gamma'(0), \gamma'(0)) + d\varphi(\nabla_t \gamma'(0))$. The gradient of φ at wI is YwI with Y fixed by every Ad_g, because conjugation by g is an isometry that fixes wI and preserves φ; the centre of $\mathfrak{su}(N)$ is trivial, so $Y = 0$ and the second term vanishes. The same isometries show that $H_w(Z, Z) = \mathrm{Hess}\, \varphi_{wI}(ZwI, ZwI)$ is Ad-invariant, and Ad is irreducible on $\mathfrak{su}(2)$, so $H_{\pm 1}$ is a multiple of the metric. □

The configuration

$$U^{\pi}_{x,\mu} = (-1)^{\sum_{\nu<\mu} x_\nu} I \tag{G.2}$$

has every plaquette equal to $-I$: for $\mu < \nu$ the link $(x + e_\mu, \nu)$ sees the coordinate x_μ raised by one and the link $(x + e_\nu, \mu)$ does not, so the four signs of the plaquette at x multiply to -1. It is defined on every box and on Λ_L for even L, and lies in $\mathrm{SU}(N)$ for even N. Put $s(x) = (-1)^{|x|}$ with $|x| = \sum_\mu x_\mu$.

Theorem G.16 (full space)**.** *Let N be even, $L \geq 4$ even, and μ_L the* SU(N) *Wilson measure on Λ_L with density proportional to $e^{N\beta S_0}$. Its Bakry–Émery constant satisfies*

$$K_{\text{full}} \leq \frac{N}{2} - 16N|\beta| = \frac{N}{2} - \frac{16|\beta_W|}{N}.$$

For SU(2)*, $K_{\text{full}} \leq 1 - 8|\beta_W|$.*

Proof. By Lemma 5.4 and Table 1, $\mathrm{Ric} = \frac{N}{2}\langle\cdot,\cdot\rangle$. Let $c \in \mathbb{R}^4$ with $\sum_\mu c_\mu = 0$ and $c \neq 0$, let $E \in \mathfrak{su}(N)$ with $|E| = 1$, and put $X_{x,\mu} = c_\mu s(x)E$. For $\beta \geq 0$ take the field U^π, at which all transports are $\mathrm{Ad}_{\pm I} = \mathrm{id}$. For the plaquette at x in the plane $\mu < \nu$, $Y_1 = c_\mu s(x)E$, $Y_2 = -c_\nu s(x)E$, $Y_3 = -c_\mu s(x)E$ and $Y_4 = c_\nu s(x)E$, so $Z_p = 2s(x)(c_\mu - c_\nu)E$. By Lemma G.15 with $w = -1$,

$$\mathrm{Hess}\, S_0(v,v) = \sum_p |Z_p|^2 = 4L^4 \sum_{\mu<\nu} (c_\mu - c_\nu)^2 = 4L^4\Big(4\sum_\mu c_\mu^2 - \Big(\sum_\mu c_\mu\Big)^2\Big) = 16|v|^2.$$

With $V = -N\beta S_0$ this gives $(\mathrm{Ric} + \mathrm{Hess}\, V)(v,v) = (\frac{N}{2} - 16N\beta)|v|^2$. For $\beta < 0$ take the field $Q \equiv I$, where Lemma G.15 with $w = 1$ gives $\mathrm{Hess}\, S_0(v,v) = -16|v|^2$ for the same v, hence $(\mathrm{Ric} + \mathrm{Hess}\, V)(v,v) = (\frac{N}{2} - 16N|\beta|)|v|^2$. Finally $\beta_W = N^2\beta$, and $\beta_W = 4\beta$ for SU(2). □

Without the gauge the bound $|\mathrm{Hess}\, S_0(v,v)| \leq 16|v|^2$ holds for every field and vector [6], so Theorem G.16 is an equality, $K_{\text{full}} = \frac{N}{2} - 16N|\beta|$. The Ricci curvature grows with N, but so does the Hessian, and the criterion $K_{\text{full}} > 0$ holds exactly for $|\beta_W| < N^2/32$; at fixed N along the tuning $\beta_W \to \infty$ the constant tends to $-\infty$ linearly. The Bakry–Émery constant is also the best rate of contraction of the Langevin dynamics in the quadratic Kantorovich distance [52], so no synchronous coupling of the dynamics does better. The gauge of Section 3 improves the constant 16 to $\frac{29}{2}$, and the next theorem shows that it cannot improve it below $\frac{38}{3}$.

Theorem G.17 (the gauge slice)**.** *Let $G = \mathrm{SU}(2)$, $m \geq 1$, $\Lambda_m = [0, 2m]^4 \cap \mathbb{Z}^4$ with free boundary, and let $K_F(\Lambda_m)$ be the Bakry–Émery constant of the gauge-fixed measure $\mu^F_{\Lambda_m}$ of Lemma 4.1. Then*

$$K_F(\Lambda_m) \leq 1 - 2|\beta|\rho_m, \qquad \rho_m = \frac{304\, m^4}{24\, m^4 + 8(2m+1)^3} \xrightarrow[m\to\infty]{} \frac{38}{3}.$$

Hence the constant $\frac{29}{2}$ of Theorem 5.3 cannot be replaced by any number below $\frac{38}{3}$, and asymptotically in m

$$1 - \tfrac{29}{4}|\beta_W| \leq K_F \leq 1 - \tfrac{19}{3}|\beta_W|,$$

so no Bakry–Émery argument for the gauge-fixed measure reaches $|\beta_W| = \frac{3}{19}$.

Proof. The field. For $\beta \geq 0$ start from U^π on Λ_m and apply the gauge transformation of the proof of Lemma 4.1 along the forest $F \cap E(\Lambda_m)$; the recursion $g_{x_{i+1}} = g_{x_i} U_{(x_i,x_{i+1})}$ produces $g_x \in \{\pm I\}$, so the result Q takes values in $\{\pm I\}$, equals I on F, and has all plaquettes conjugate to those of U^π, that is $-I$. For $\beta < 0$ take $Q \equiv I$. For v vanishing on F the Hessian of the gauge-fixed action is $\mathrm{Hess}\, S_{\Lambda_m}(v,v)$, which by Lemma G.15 equals $\pm\sum_{p\in P(\Lambda_m)} |Z_p|^2$, with $+$ for $\beta \geq 0$ and $-$ for $\beta < 0$; all transports are trivial.

The vector. Let $c = (1, -1, 0, 0)$, $|E| = 1$, and put $X_{x,\mu} = c_\mu s(x)E$ on the active links of $E(\Lambda_m)$ and $X = 0$ on F. Let p be a plaquette at x in the plane $\mu < \nu$ with all its links in $E(\Lambda_m)$; it contains exactly one link of F. Without that link the computation of the proof of Theorem G.16 gives $Z_p = 2s(x)(c_\mu - c_\nu)E$, the links e_1, e_3 of direction μ contributing $s(x)c_\mu E$ each and the links e_2, e_4 of direction ν contributing $-s(x)c_\nu E$ each. Removing the link of F gives $Z_p = s(x)(c_\mu - 2c_\nu)E$ if that link has direction μ, and $Z_p = s(x)(2c_\mu - c_\nu)E$ if it has direction ν.

Counting. Let $B = [0, 2m-1]^4$; every plaquette with base point in B has its links in $E(\Lambda_m)$. Fix $\mu < \nu$. The link of F of the plaquette at x has direction μ exactly when

$\sigma(x)=\mu$ or $\sigma(x+e_\nu)=\mu$, and the two cases exclude each other. By Table 2, σ has period two and takes each value at four of the sixteen residues. B and $B+e_\nu$ are disjoint unions of m^4 translates of $\{0,1\}^4$, each containing every residue once, so each case occurs for $4m^4$ base points in B; of the $16m^4$ plaquettes of the plane with base point in B, exactly $8m^4$ have their link of F in direction μ and $8m^4$ in direction ν. Hence, with $|c|^2=2$ and $\sum_\mu c_\mu=0$,

$$\sum_{p\in P(\Lambda_m)}|Z_p|^2\ge 8m^4\sum_{\mu<\nu}[(c_\mu-2c_\nu)^2+(2c_\mu-c_\nu)^2]=8m^4\Big(19|c|^2-4\Big(\sum_\mu c_\mu\Big)^2\Big)=304m^4,$$

using $\sum_{\mu<\nu}(5c_\mu^2+5c_\nu^2-8c_\mu c_\nu)=15|c|^2-4((\sum_\mu c_\mu)^2-|c|^2)$. On the other side, for $\mu\in\{0,1\}$ the links (x,μ) with $x\in B$ lie in $E(\Lambda_m)$, and exactly $16m^4-4m^4=12m^4$ of them are active; they contribute $24m^4$ to $|v|^2$. The remaining links of $E(\Lambda_m)$ with $X\ne 0$ have direction 0 or 1 and base point in $[0,2m]^4\setminus B$, at most $(2m+1)^4-(2m)^4\le 4(2m+1)^3$ in each direction, and each contributes at most 1. So $|v|^2\le 24m^4+8(2m+1)^3$, and $|\operatorname{Hess}S_{\Lambda_m}(v,v)|\ge\rho_m|v|^2$ with the sign that gives $\operatorname{Hess}V(v,v)\le -2|\beta|\rho_m|v|^2$, $V=-2\beta S_{\Lambda_m}$.

Conclusion. $(\operatorname{Ric}+\operatorname{Hess}V)(v,v)\le(1-2|\beta|\rho_m)|v|^2$. If Theorem 5.3 held with a constant $c'<\frac{38}{3}$, then $\rho_m\le c'$ for all m, contradicting $\rho_m\to\frac{38}{3}$. The lower bound for K_F is Lemma 5.5 with $\lambda=0$, and $\beta_W=4\beta$. □

Proposition G.18 (plaquette actions at a coarse scale)**.** *Let $L\ge 4$ be even and $\nu(dU)\propto\exp(\sum_p\phi(U_p))\prod_e dU_e$ on $\mathrm{SU}(2)^{E(\Lambda_L)}$ with $\phi(U)=g(\frac12\operatorname{Re}\operatorname{Tr}U)$, $g\in C^2([-1,1])$. The Bakry–Émery constant K of ν satisfies*

$$K\le 1-8\max(g'(-1),-g'(1)).$$

Proof. $\frac12\operatorname{Re}\operatorname{Tr}(\pm e^{tZ})=\pm(1-\frac{t^2}{4}|Z|^2)+O(t^3)$, since $\operatorname{Tr}Z^2=-|Z|^2$; so in Lemma G.15 $h_+=-\frac12 g'(1)$ and $h_-=\frac12 g'(-1)$. The potential is $-\sum_p\phi(U_p)$. At U^π, with the vector of the proof of Theorem G.16, $\operatorname{Hess}\sum_p\phi(U_p)(v,v)=h_-\sum_p|Z_p|^2=16h_-|v|^2$, and at $Q\equiv I$ it is $16h_+|v|^2$. Hence $K\le 1-16h_\pm$. □

For the Wilson weight $g(s)=4\beta s$ this is Theorem G.16. For an effective plaquette action at a coarse scale the Bakry–Émery route needs $|g'(\pm1)|<\frac18$, an effective theory deep in strong coupling, and producing such an effective measure from the weak-coupling lattice theory, and transporting the resulting estimate back, are the two missing steps of Section 11.

G.5. **Entropy, link covariances and blocks of fixed size.** In the setting of Section 6, say that μ satisfies *block entropy factorisation* with constant C if $\operatorname{Ent}_\mu(F)\le Cn^{-4}\sum_x\mu(\operatorname{Ent}_{\gamma^0_{V_x}}F)$ for all bounded gauge-invariant $F\ge 0$, where $\operatorname{Ent}_{\gamma^0_{V_x}}F$ is the entropy of F under $\gamma^0_{V_x}(\cdot\mid U)$.

Proposition G.19. *Block entropy factorisation with constant C implies* $\mathrm{BP}(n,1/C)$.

Proof. Let $\Phi(s)=s\log s$. For $|u|\le\frac12$, $\Phi(1+u)=u+\frac{u^2}{2}+r(u)$ with $r(u)=\sum_{k\ge3}(-u)^k/(k(k-1))$ and $|r(u)|\le|u|^3\sum_{k\ge3}2^{3-k}/(k(k-1))\le|u|^3$. Let g be real, bounded and gauge invariant, and $0<\varepsilon\le 1/(2\|g\|_\infty)$. Then $\operatorname{Ent}_\mu(1+\varepsilon g)=\mu\Phi(1+\varepsilon g)-\Phi(1+\varepsilon\mu g)=\frac{\varepsilon^2}{2}\operatorname{Var}_\mu g+R$ with $|R|\le 2\varepsilon^3\|g\|_\infty^3$, and in the same way $\operatorname{Ent}_{\gamma^0_{V_x}}(1+\varepsilon g)=\frac{\varepsilon^2}{2}\operatorname{Var}_{\gamma^0_{V_x}}g+R_x$ with $|R_x|\le 2\varepsilon^3\|g\|_\infty^3$. Inserting $F=1+\varepsilon g$, dividing by $\varepsilon^2/2$ and letting $\varepsilon\to0$ gives $\operatorname{Var}_\mu g\le Cn^{-4}\sum_x\mu(\operatorname{Var}_{\gamma^0_{V_x}}g)=Cn^{-4}\sum_x\mu((g-\mathcal{P}_x g)^2)$. Truncations of $f\in\mathcal{A}$ are bounded and gauge invariant and converge in $L^2(\mu)$, and both sides are continuous on $L^2(\mu)$. □

So entropy factorisation at the physical scale, uniform along the tuning, implies (D) with $\varpi=1/C$, and every obstruction to the cube condition applies to it: it fails for U(1) at weak coupling and at fixed cube size (Section 11). The methods that derive block factorisation from spatial mixing uniformly in the boundary condition [15] need, at the physical scale, an input of the strength of (D) itself.

Proposition G.20 (link covariances)**.** *Let μ be a gauge-invariant probability measure on $\mathrm{SU}(2)^E$, E the links of a box or of Λ_L with $L \geq 3$, and identify U_e with a unit vector $\sigma_e \in \mathbb{R}^4$. Then $E\sigma_e = 0$, $E\sigma_e\sigma_e^T = \frac{1}{4}I_4$ and $E\sigma_e\sigma_{e'}^T = 0$ for $e \neq e'$.*

Proof. Distinct links have distinct sets of end points, so e has an end point x that is not an end point of e'. The gauge transformation g at x alone maps σ_e to $\ell_g\sigma_e$, where ℓ_g is left multiplication by g or right multiplication by g^{-1}, an orthogonal map, and fixes $\sigma_{e'}$. By invariance $E\sigma_e\sigma_{e'}^T = \int \ell_g \, dg \; E\sigma_e\sigma_{e'}^T = 0$, since ℓ_g is linear in g and $\int g \, dg = 0$. The same argument gives $E\sigma_e = 0$ and shows that the law of σ_e is invariant under multiplication, hence Haar measure, so $E\sigma_e\sigma_e^T = \frac{1}{4}I_4$. □

The influence and covariance matrices that spectral independence and localisation schemes ask to be bounded are therefore exactly $\frac{1}{4}I$ for the Wilson measure at every β, weak coupling included, and carry no information. Those methods need the bound for all tilted measures, and the tilts break gauge invariance. Their output is a tensorisation inequality over blocks of bounded lattice size, uniform in the volume, and the next proposition shows that such an inequality cannot hold uniformly along the tuning when the continuum limit is nontrivial.

Proposition G.21 (fixed blocks force triviality)**.** *Let $a_k = a_0 2^{-k}$ and β_k be given, and suppose that for every k there is $C_k < \infty$ such that the Wilson measure $\mu_{\beta_k,L}$ satisfies*

$$\mathrm{Var}_\mu(f) \leq C_k \sum_{e \in E} \mu(\mathrm{Var}_e f) \qquad (f \in \mathcal{A})$$

for all L in an infinite set $\mathcal{L}_k$ of even integers $L \geq 8$, where $\mathrm{Var}_e f$ is the conditional variance given the links other than e. Then $\mathrm{BP}(2, 1/(64C_k))$ holds on these tori, every torus limit along $\mathcal{L}_k$ has transfer-matrix gap at least $1/(5000e^2C_k)$, and every continuum limit of such torus limits in the sense of Definition 6.5 has $\sigma(H) \subset \{0\} \cup [E_*, \infty)$ with $E_* = \limsup_k (5000e^2C_k a_k)^{-1}$. In particular, if $C_k a_k \to 0$, every continuum limit is trivial.

Proof. For $e \in V_x$ the σ-algebra of the links off V_x is contained in that of the links other than e, so $\mu(\mathrm{Var}_e f) = \|f - E[f \mid e^c]\|^2 \leq \|f - \mathcal{P}_x f\|^2$. With $n = 2$ each link lies in 16 cubes V_x and $|V_x| = 4 \cdot 2^4 = 64$, so

$$\sum_e \mu(\mathrm{Var}_e f) \leq \frac{1}{16} \sum_x \sum_{e \in V_x} \mu((f - \mathcal{P}_x f)^2) = 4 \sum_x \mu((f - \mathcal{P}_x f)^2),$$

and $\mathrm{Var}_\mu f \leq 64C_k \cdot 2^{-4} \sum_x \mu((f - \mathcal{P}_x f)^2)$, which is $\mathrm{BP}(2, 1/(64C_k))$ on tori with $L \geq 8 = 2 \cdot 2 + 4$. Corollary 6.4 gives the gap $m_k = \varpi/(2e^2 q_2)$ with $q_2 = \frac{625}{16}$, that is $m_k = 1/(5000e^2C_k)$. For a continuum limit, as in the proof of Theorem B, for $f \in \mathcal{T}$, $\varphi = \psi(f) - \langle \Omega, \psi(f) \rangle \Omega$ and dyadic $\tau > 0$, $\langle \varphi, e^{-\tau H}\varphi \rangle = \lim_k \langle \varphi_k, \mathbb{T}_k^{\tau/a_k} \varphi_k \rangle \leq \liminf_k e^{-m_k \tau/a_k} \|\varphi_k\|^2 = e^{-E_* \tau} \|\varphi\|^2$, using $\|\varphi_k\|^2 \to \|\varphi\|^2$, the case $\tau = 0$ of the definition. These φ are dense in $\Omega^\perp$, and the spectral argument of the proof of Theorem B gives $\sigma(H|_{\Omega^\perp}) \subset [E_*, \infty)$. If $E_* = \infty$, then $\langle \varphi, e^{-\tau H}\varphi \rangle = 0$ for all dyadic $\tau > 0$, and strong continuity of $e^{-\tau H}$ at $\tau = 0$ gives $\varphi = 0$, so $\Omega^\perp = \{0\}$. □

The same holds for blocks of any fixed lattice size b in place of single links, with $\mathrm{BP}(b, \cdot)$. So the tensorisation constants of single links or of bounded blocks must grow at least like $1/a_k$ along the tuning whenever the continuum limit is nontrivial; a usable version must work with blocks of physical size, and is then a form of (D).

G.6. **The transfer matrix alone.** The next results show what the continuum gap needs and that a lattice gap tending to zero at the right rate is, by itself, a kinematic statement. In this section Definition 6.5 is used with $a_k = a_0 2^{-j_k}$ for a strictly increasing sequence of integers j_k, with τ in $\mathbb{D} = \{a_0 p 2^{-q} : p, q \in \mathbb{Z}_{\geq 0}\}$, and with an arbitrary complex vector space $\mathcal{T}$ and linear maps O_k into the gauge-invariant local observables at nonnegative times; the proof of Theorem B uses nothing more. The states ω_k are translation invariant and reflection positive, with transfer matrices $0 \leq \mathbb{T}_k \leq 1$ on $\mathcal{H}_k$, vacua Ω_k and maps Ψ_k from

observables at nonnegative times onto dense subspaces, as in Section 2. Put $\gamma_k = \|\mathbb{T}_k|_{\Omega_k^\perp}\|$, $m_k = -\log\gamma_k \in [0,\infty]$, and $\mathrm{C}_k(f;s) = \omega_k(\vartheta(\overline{O_k f})\tau_{s/a_k}O_k f) - |\omega_k(O_k f)|^2$.

Proposition G.22 (the spectral input)**.** *Let $(\mathcal{H},\Omega,H,\psi)$ be a continuum limit of (ω_k,O_k) and $\Delta>0$. The following are equivalent.*

(i) *$\sigma(H|_{\Omega^\perp}) \subset [\Delta,\infty)$.*
(ii) *For every $f\in\mathcal{T}$ there is $C_f<\infty$ with $\lim_k \mathrm{C}_k(f;s) \le C_f e^{-\Delta s}$ for every $s\in\mathbb{D}$.*

Proof. For $f\in\mathcal{T}$ let $\phi_f = \psi f - \langle\Omega,\psi f\rangle\Omega$. Since $e^{-sH}\Omega=\Omega$ and e^{-sH} is self-adjoint, $\langle\phi_f,e^{-sH}\phi_f\rangle = \langle\psi f,e^{-sH}\psi f\rangle - |\langle\Omega,\psi f\rangle|^2$, so $\lim_k \mathrm{C}_k(f;s)$ exists and equals $\langle\phi_f,e^{-sH}\phi_f\rangle$. If (i) holds, the spectral theorem gives (ii) with $C_f=\|\phi_f\|^2$. Conversely, let μ_f be the spectral measure of H at ϕ_f. If $\mu_f([0,\Delta-\eta]) = c > 0$ for some $\eta>0$, then $\langle\phi_f,e^{-sH}\phi_f\rangle \ge ce^{-(\Delta-\eta)s} > C_f e^{-\Delta s}$ for large $s\in\mathbb{D}$, which contradicts (ii); so the spectral projection P of H for $[0,\Delta)$ satisfies $P\phi_f=0$. The ϕ_f are dense in $\Omega^\perp$, so $P=0$ on $\Omega^\perp$. □

So the continuum gap needs exponential decay, at physical times and only asymptotically in k, of the autocorrelations of the observables that generate the limit; it does not need a lattice gap, since lattice states of physical energy below Δ are allowed if they decouple from every $O_k(f)$.

Lemma G.23. *Let $(\mathcal{H},\Omega,H,\psi)$ be a continuum limit of (ω_k,O_k) along a subsequence Λ and $\bar E = \limsup_{k\in\Lambda} m_k/a_k \in [0,\infty]$. Then $\langle\phi,e^{-sH}\phi\rangle \le e^{-\bar E s}\|\phi\|^2$ for all $\phi\in\Omega^\perp$ and $s\ge 0$; so $\sigma(H|_{\Omega^\perp})\subset[\bar E,\infty)$, and $\Omega^\perp=\{0\}$ if $\bar E=\infty$.*

Proof. For $f\in\mathcal{T}$ and $k\in\Lambda$ put $O=O_k f$ and $\phi_k = \Psi_k(O) - \omega_k(O)\Omega_k \perp \Omega_k$. Since $\mathbb{T}_k\ge 0$ is self-adjoint and fixes Ω_k, it preserves $\Omega_k^\perp$, and for $s\in\mathbb{D}$ and $t = s/a_k \in \mathbb{Z}$, $\mathrm{C}_k(f;s) = \langle\phi_k,\mathbb{T}_k^t\phi_k\rangle \le \gamma_k^t\|\phi_k\|^2 = e^{-(m_k/a_k)s}\|\phi_k\|^2$. Along a further subsequence $m_k/a_k\to\bar E$, $\mathrm{C}_k(f;s)\to\langle\phi_f,e^{-sH}\phi_f\rangle$ and $\|\phi_k\|^2 = \mathrm{C}_k(f;0)\to\|\phi_f\|^2$, with $e^{-\infty s}=0$ for $s>0$. The ϕ_f are dense in $\Omega^\perp$, both sides are continuous in ϕ, and $s\mapsto\langle\phi,e^{-sH}\phi\rangle$ is continuous, so the inequality holds on $\Omega^\perp$ for all $s\ge0$. The spectral statement follows as in Proposition G.22; if $\bar E=\infty$, then $\langle\phi,e^{-sH}\phi\rangle=0$ for $s>0$, so $\phi=0$ by strong continuity. □

Theorem G.24 (gap tuning)**.** *Suppose $0<m_k<\infty$ for all k and $m_k\to0$, and let $E>0$ and $a_0>0$. For each k let j_k be the integer with $E\le m_k 2^{j_k}/a_0 < 2E$, pass to a subsequence on which j_k is strictly increasing, and put $a_k = a_0 2^{-j_k}$.*

(a) *Every continuum limit of (ω_k,O_k) along any subsequence, for any $\mathcal{T}$ and O_k, has $\sigma(H|_{\Omega^\perp})\subset[E,\infty)$.*
(b) *There are a subsequence, a space $\mathcal{T}$ with maps O_k, and a nontrivial continuum limit along it with $\inf\sigma(H|_{\Omega^\perp}) = \lim_k m_k/a_k \in [E,2E]$.*

Proof. The integer j_k exists and is unique, because the condition reads

$$2^{j_k} \in [Ea_0/m_k,\ 2Ea_0/m_k),$$

and $j_k\to\infty$ since $m_k\to0$. Part (a) is Lemma G.23, since $m_k/a_k\ge E$.

(b) Pass to a subsequence with $m_k/a_k\to\bar E\in[E,2E]$ and fix k. As $\gamma_k>0$, $\Omega_k^\perp\ne\{0\}$, and γ_k, the norm of the positive operator $\mathbb{T}_k|_{\Omega_k^\perp}$, lies in its spectrum; so the spectral projection P_k of $\mathbb{T}_k|_{\Omega_k^\perp}$ for $[\gamma_k e^{-a_k/k},\gamma_k]$ is nonzero. Choose a unit vector v_k in its range. The vectors $\Psi_k(F-\omega_k(F))$ are the projections of $\Psi_k(F)$ onto $\Omega_k^\perp$, hence dense in it; choose F_k' with $\|u_k'-v_k\|\le\frac{1}{10}$ for $u_k'=\Psi_k(F_k'-\omega_k(F_k'))$, and put $F_k = (F_k'-\omega_k(F_k'))/\|u_k'\|$ and $u_k=\Psi_k(F_k)$. Then $\|u_k\|=1$, $u_k\perp\Omega_k$ and $\|P_k u_k\|\ge(1-\frac{1}{10})/\frac{11}{10} = \frac{9}{11}$. Let $c_k(t)=\langle u_k,\mathbb{T}_k^t u_k\rangle$. Since $\mathbb{T}_k^t\ge0$ commutes with P_k and has norm γ_k^t on $\Omega_k^\perp$,

$$\tfrac{81}{121}e^{-(m_k+a_k/k)t} \le \langle P_k u_k,\mathbb{T}_k^t P_k u_k\rangle \le c_k(t) \le e^{-m_k t}. \tag{G.3}$$

Let σ_k be the spectral measure of $\mathbb{T}_k$ at u_k, a probability measure on $[0,1]$, and ρ_k its image under $x \mapsto -a_k^{-1}\log x$, a probability measure on the compact space $[0,\infty]$, with $0 \mapsto \infty$. Then $c_k(s/a_k) = \int e^{-sx}\,d\rho_k(x)$ for $s \in \mathbb{D}$, with $e^{-s\infty} = 0$ for $s > 0$. Pass to a subsequence with $\rho_k \to \rho$ weakly on $[0,\infty]$. For $s > 0$ the function $x \mapsto e^{-sx}$ is continuous on $[0,\infty]$, so with $\rho' = \rho|_{[0,\infty)}$,

$$B(s) := \lim_k c_k(s/a_k) = \int_{[0,\infty)} e^{-sx}\,d\rho'(x) \qquad (s \in \mathbb{D},\ s > 0).$$

Let $\mathcal{T}$ be the free complex vector space on the symbols $\mathbf{1}$ and j_s, $s \in \mathbb{D}$, $s > 0$; put $O_k(\mathbf{1}) = 1$ and $O_k(j_s) = \tau_{s/a_k}F_k$ for the k with $s/a_k \in \mathbb{Z}$, extended linearly. Let $\mathcal{H} = \mathbb{C}\Omega \oplus L^2([0,\infty),\rho')$ with $\Omega = (1,0)$, $H = 0 \oplus x$, $\psi(\mathbf{1}) = \Omega$ and $\psi(j_s) = (0, e^{-sx})$. Since $\Psi_k(O_k j_s) = \mathbb{T}_k^{s/a_k}u_k$, for $s, s' > 0$ and r in $\mathbb{D}$,

$$\omega_k\big(\vartheta(\overline{O_k j_s})\tau_{r/a_k}O_k j_{s'}\big) = c_k\big((s+s'+r)/a_k\big) \to B(s+s'+r) = \langle \psi j_s, e^{-rH}\psi j_{s'}\rangle;$$

moreover $\omega_k(\tau_{r/a_k}O_k j_s) = \langle \Omega_k, \mathbb{T}_k^{(r+s)/a_k}u_k\rangle = \langle \Omega_k, u_k\rangle = 0 = \langle \Omega, e^{-rH}\psi j_s\rangle$, the same holds with the roles exchanged, and $\omega_k(1) = 1 = \langle \Omega, e^{-rH}\Omega\rangle$. By sesquilinearity the identities of Definition 6.5 hold on $\mathcal{T}$. The span of $\{e^{-sx} : s \in \mathbb{D}, s > 0\}$ is an algebra of functions in $C_0([0,\infty))$ that separates points and vanishes nowhere, hence dense in $C_0([0,\infty))$ by the Stone–Weierstrass theorem and dense in $L^2(\rho')$; so ψ has dense range, and $H \geq 0$ is self-adjoint with $H\Omega = 0$.

By (G.3) with $t = s/a_k$, $\frac{81}{121}e^{-\bar{E}s} \leq B(s) \leq e^{-\bar{E}s}$ for $s \in \mathbb{D}$, $s > 0$. The lower bound gives $\rho' \neq 0$, so $\mathcal{H} \neq \mathbb{C}\Omega$. The upper bound gives $\rho'([0, \bar{E} - \eta]) = 0$ for every $\eta > 0$, and the lower bound gives $\rho'([\bar{E}, \bar{E} + \eta]) > 0$ for every $\eta > 0$. Since $\sigma(H|_{\Omega^\perp})$ is the support of ρ', its infimum is $\bar{E}$. □

Theorem G.24 uses no property of the gauge theory: it applies equally to the free scalar field with lattice masses tending to zero. A nontrivial continuum limit with a gap, in the sense of Definition 6.5, therefore exists as soon as the lattice gaps $m(\beta_k)$ are positive and tend to zero along a sequence $\beta_k \to \infty$, and neither property is known for four-dimensional SU(N). The limit so obtained is generated by one sequence of observables chosen through spectral projections; nothing identifies it with Yang–Mills theory, since it carries no Euclidean invariance, no locality and no Wilson loops. The physical content of (E) is the convergence of fixed physical observables, smeared fields and Wilson loops of fixed physical size, and by Lemma G.23 a gap coming from the transfer matrices requires m_k/a_k to stay bounded below along a subsequence: along the asymptotic-freedom tuning, the lattice gap must decay at exactly the asymptotic-freedom rate.

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The von Mises–Fisher modulus and the Hessian bound without gauge fixing are those of [6]; the line field, the curvature bound in the gauge, the cube condition and its consequences follow the longer account [7, 8]. The constructive framework is that of Osterwalder, Schrader and Seiler, the curvature criterion that of Bakry and Émery, and the uniqueness criterion that of Dobrushin. The maximal correlation of Gaussian variables is Gebelein's, the finite-size criterion follows Knabe and Gosset–Mozgunov, the two-dimensional analysis rests on the heat kernel of the continuum measure of Driver, Gross, King, Sengupta and Lévy, and the twisted boundary conditions are 't Hooft's. The Lean 4 formalisation of the strong-coupling analysis described in the introduction is the work of I. J. Sessford [46].

Declarations

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References

[1] A. Adhikari and S. Cao, Correlation decay for finite lattice gauge theories at weak coupling, Ann. Probab. **53** (2025), 140–174, doi:10.1214/24-AOP1702.

[2] T. Bałaban, Propagators and renormalization transformations for lattice gauge theories. I, Comm. Math. Phys. **95** (1984), 17–40, doi:10.1007/BF01215753.

[3] T. Bałaban, Averaging operations for lattice gauge theories, Comm. Math. Phys. **98** (1985), 17–51, doi:10.1007/BF01211042.

[4] T. Bałaban, Large field renormalization. II. Localization, exponentiation, and bounds for the **R** operation, Comm. Math. Phys. **122** (1989), 355–392, doi:10.1007/BF01238433.

[5] J. Barta, Sur la vibration fondamentale d'une membrane, C. R. Acad. Sci. Paris **204** (1937), 472–473.

[6] D. Bhattacharjee and U. Bhattacharya, Mass gap of lattice Yang–Mills theory up to the Dobrushin ceiling, in *Lattice Yang–Mills theory: spacing-uniform obstructions, the mass gap up to the Dobrushin ceiling, and Lean certificate*, Zenodo, release v2.0.2, 2026, doi:10.5281/zenodo.23449466.

[7] D. Bhattacharjee, U. Bhattacharya, J. M. Lockwood and T. Romanov, One cube estimate implies the Yang–Mills mass gap, Hexagon preprint 2610.00341v7, 2026, `https://hexagonmath.org/2610.00341v7`.

[8] D. Bhattacharjee, U. Bhattacharya, J. M. Lockwood, T. Romanov and S. Singha Roy, Yang–Mills theory and the mass gap, research archive, Zenodo, release v4.44.0, 2026, doi:10.5281/zenodo.24085959.

[9] D. Bakry and M. Émery, Diffusions hypercontractives, in *Séminaire de Probabilités XIX*, Lecture Notes in Math. 1123, Springer, Berlin, 1985, 177–206, doi:10.1007/BFb0075847.

[10] D. Bakry, I. Gentil and M. Ledoux, *Analysis and Geometry of Markov Diffusion Operators*, Grundlehren Math. Wiss. 348, Springer, Cham, 2014, doi:10.1007/978-3-319-00227-9.

[11] M. Biernacki and J. Krzyż, On the monotonity of certain functionals in the theory of analytic functions, Ann. Univ. Mariae Curie-Skłodowska Sect. A **9** (1955), 135–147.

[12] S. Chatterjee, A probabilistic mechanism for quark confinement, Comm. Math. Phys. **385** (2021), 1007–1039, doi:10.1007/s00220-021-04086-y.

[13] S. Chatterjee, Yang–Mills for probabilists, in *Probability and Analysis in Interacting Physical Systems*, Springer Proc. Math. Stat. 283, Springer, Cham, 2019, 1–16, doi:10.1007/978-3-030-15338-0_1.

[14] S. Cao, R. Nissim and S. Sheffield, Expanded regimes of area law for lattice Yang–Mills theories, preprint, 2025, arXiv:2505.16585.

[15] P. Caputo and D. Parisi, Block factorization of the relative entropy via spatial mixing, Comm. Math. Phys. **388** (2021), 793–818, doi:10.1007/s00220-021-04237-1.

[16] M. Creutz, Gauge fixing, the transfer matrix, and confinement on a lattice, Phys. Rev. D **15** (1977), 1128–1136, doi:10.1103/PhysRevD.15.1128.

[17] F. Deutsch, *Best Approximation in Inner Product Spaces*, CMS Books in Mathematics, Springer, New York, 2001, doi:10.1007/978-1-4684-9298-9.

[18] J. Dimock, The renormalization group according to Balaban, I. Small fields, Rev. Math. Phys. **25** (2013), 1330010, doi:10.1142/S0129055X13300100.

[19] R. L. Dobrushin, The description of a random field by means of conditional probabilities and conditions of its regularity, Theory Probab. Appl. **13** (1968), 197–224, doi:10.1137/1113026.

[20] R. L. Dobrushin, Prescribing a system of random variables by conditional distributions, Theory Probab. Appl. **15** (1970), 458–486, doi:10.1137/1115049.

[21] B. K. Driver, YM_2: continuum expectations, lattice convergence, and lassos, Comm. Math. Phys. **123** (1989), 575–616, doi:10.1007/BF01218586.

[22] B. Efron and C. Stein, The jackknife estimate of variance, Ann. Statist. **9** (1981), 586–596, doi:10.1214/aos/1176345462.

[23] J. Fröhlich and T. Spencer, Massless phases and symmetry restoration in abelian gauge theories and spin systems, Comm. Math. Phys. **83** (1982), 411–454, doi:10.1007/BF01213610.

[24] H. Gebelein, Das statistische Problem der Korrelation als Variations- und Eigenwertproblem und sein Zusammenhang mit der Ausgleichsrechnung, Z. Angew. Math. Mech. **21** (1941), 364–379, doi:10.1002/zamm.19410210604.

[25] L. Gross, C. King and A. Sengupta, Two dimensional Yang–Mills theory via stochastic differential equations, Ann. Physics **194** (1989), 65–112, doi:10.1016/0003-4916(89)90032-8.

[26] D. Gosset and E. Mozgunov, Local gap threshold for frustration-free spin systems, J. Math. Phys. **57** (2016), 091901, doi:10.1063/1.4962337.

[27] A. Grigor'yan, *Heat Kernel and Analysis on Manifolds*, AMS/IP Studies in Advanced Mathematics 47, Amer. Math. Soc., Providence, RI, 2009, doi:10.1090/amsip/047.

[28] D. J. Gross and F. Wilczek, Ultraviolet behavior of non-Abelian gauge theories, Phys. Rev. Lett. **30** (1973), 1343–1346, doi:10.1103/PhysRevLett.30.1343.
[29] R. Holley and D. Stroock, Logarithmic Sobolev inequalities and stochastic Ising models, J. Stat. Phys. **46** (1987), 1159–1194, doi:10.1007/BF01011161.
[30] S. Janson, *Gaussian Hilbert Spaces*, Cambridge Tracts in Mathematics **129**, Cambridge University Press, Cambridge, 1997, doi:10.1017/CBO9780511526169.
[31] A. Jaffe and E. Witten, Quantum Yang–Mills theory, in *The Millennium Prize Problems*, Clay Mathematics Institute and American Mathematical Society, Cambridge, MA and Providence, RI, 2006, 129–152, `https://www.claymath.org/wp-content/uploads/2022/06/yangmills.pdf`.
[32] V. G. Kac, *Infinite-Dimensional Lie Algebras*, 3rd ed., Cambridge University Press, Cambridge, 1990, doi:10.1017/CBO9780511626234.
[33] O. Kallenberg, *Foundations of Modern Probability*, 3rd ed., Springer, Cham, 2021, doi:10.1007/978-3-030-61871-1.
[34] S. Knabe, Energy gaps and elementary excitations for certain VBS-quantum antiferromagnets, J. Stat. Phys. **52** (1988), 627–638, doi:10.1007/BF01019721.
[35] T. Lévy, Yang–Mills measure on compact surfaces, Mem. Amer. Math. Soc. **166** (2003), no. 790, doi:10.1090/memo/0790.
[36] E. J. McShane, Extension of range of functions, Bull. Amer. Math. Soc. **40** (1934), 837–842, doi:10.1090/S0002-9904-1934-05978-0.
[37] J. Milnor, Curvatures of left invariant metrics on Lie groups, Adv. Math. **21** (1976), 293–329, doi:10.1016/S0001-8708(76)80002-3.
[38] K. V. Mardia and P. E. Jupp, *Directional Statistics*, Wiley, Chichester, 2000, doi:10.1002/9780470316979.
[39] J. Magnen, V. Rivasseau and R. Sénéor, Construction of YM_4 with an infrared cutoff, Comm. Math. Phys. **155** (1993), 325–383, doi:10.1007/BF02097397.
[40] D. Nualart, *The Malliavin Calculus and Related Topics*, 2nd ed., Probability and its Applications, Springer, Berlin, 2006, doi:10.1007/3-540-28329-3.
[41] K. Osterwalder and R. Schrader, Axioms for Euclidean Green's functions, Comm. Math. Phys. **31** (1973), 83–112, doi:10.1007/BF01645738.
[42] K. Osterwalder and R. Schrader, Axioms for Euclidean Green's functions. II, Comm. Math. Phys. **42** (1975), 281–305, doi:10.1007/BF01608978.
[43] K. Osterwalder and E. Seiler, Gauge field theories on a lattice, Ann. Physics **110** (1978), 440–471, doi:10.1016/0003-4916(78)90039-8.
[44] H. D. Politzer, Reliable perturbative results for strong interactions?, Phys. Rev. Lett. **30** (1973), 1346–1349, doi:10.1103/PhysRevLett.30.1346.
[45] E. Seiler, *Gauge Theories as a Problem of Constructive Quantum Field Theory and Statistical Mechanics*, Lecture Notes in Phys. 159, Springer, Berlin, 1982, doi:10.1007/3-540-11559-5.
[46] I. J. Sessford, The mass gap of SU(N) gauge theory seen by an observer of fixed physical resolution, Lean 4 development and preprint, Zenodo, release v0.3.1, 2026, doi:10.5281/zenodo.24073388; `https://github.com/Agience/entroptics-mass-gap`.
[47] E. M. Stein and R. Shakarchi, *Fourier Analysis: An Introduction*, Princeton Lectures in Analysis I, Princeton University Press, Princeton, NJ, 2003.
[48] E. M. Stein and G. Weiss, *Introduction to Fourier Analysis on Euclidean Spaces*, Princeton University Press, Princeton, NJ, 1971, doi:10.1515/9781400883899.
[49] H. Shen, R. Zhu and X. Zhu, A stochastic analysis approach to lattice Yang–Mills at strong coupling, Comm. Math. Phys. **400** (2023), 805–851, doi:10.1007/s00220-022-04609-1.
[50] G. 't Hooft, A property of electric and magnetic flux in non-Abelian gauge theories, Nuclear Phys. B **153** (1979), 141–160, doi:10.1016/0550-3213(79)90595-9.
[51] C. Villani, *Optimal Transport: Old and New*, Grundlehren Math. Wiss. 338, Springer, Berlin, 2009, doi:10.1007/978-3-540-71050-9.
[52] M.-K. von Renesse and K.-T. Sturm, Transport inequalities, gradient estimates, entropy, and Ricci curvature, Comm. Pure Appl. Math. **58** (2005), 923–940, doi:10.1002/cpa.20060.
[53] G. N. Watson, *A Treatise on the Theory of Bessel Functions*, 2nd ed., Cambridge University Press, Cambridge, 1944.
[54] K. G. Wilson, Confinement of quarks, Phys. Rev. D **10** (1974), no. 8, 2445–2459, doi:10.1103/PhysRevD.10.2445.

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